Data Science Across Domains

From Methods and Models to Business Applications

Rodrigo Kang

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Welcome

Welcome to my portfolio:

Data Science Across Domains
From Methods and Models to Business Applications

This portfolio brings together a collection of technical work developed across a range of domains. Organized as a series of self-contained case studies and technical writings, it reflects an approach to quantitative problem solving grounded in statistical reasoning, computational modeling, and the scientific method.

Although the topics vary, the underlying philosophy remains consistent. Each piece begins with a clearly defined question, develops an appropriate quantitative approach, and evaluates the resulting solution through structured reasoning, careful experimentation, and thoughtful interpretation of the results.

The emphasis throughout is on selecting methods that are appropriate to the problem rather than applying complexity for its own sake. Depending on the objectives of each project, computational artifacts implemented in `Python`, `R`, or both are provided to support reproducibility, illustrate the analytical workflow, and document the implementation behind the results.

Each chapter is intended to stand on its own, while together they present a coherent view of quantitative problem solving across different domains. Whether addressing an applied problem, exploring a computational method, or discussing broader scientific ideas, the common objective remains the same: to develop transparent, reproducible, and well-founded solutions through quantitative reasoning.

Preface

This portfolio reflects the way I naturally approach quantitative problem solving. Throughout my professional experience, I have found that successful analytical work rarely begins with selecting a model or choosing a particular algorithm. It begins with understanding the problem, making assumptions explicit, and identifying the most appropriate methodology for the question being asked.

That perspective has shaped both the projects collected here and the way they are presented. Rather than documenting only the final solution, each chapter follows the reasoning that connects a practical or scientific question to an appropriate quantitative approach. The emphasis is not on methodological complexity, but on developing solutions that are transparent, reproducible, and appropriate for their intended purpose.

The selection of topics is deliberately selective rather than exhaustive. It does not aim to cover every area of data science or follow emerging trends for their own sake. Instead, it reflects problems and ideas that I have found interesting enough to investigate in depth, whether motivated by practical applications, methodological questions, or broader scientific curiosity.

Although the individual projects differ considerably in their subject matter, they are connected by a common way of thinking. Statistical reasoning, mathematical modeling, computational methods, and careful experimentation are treated as complementary tools rather than independent disciplines. The choice of method is always guided by the problem itself, not by the perceived sophistication of a particular technique.

I do not consider this portfolio a finished body of work. Like any long-term technical practice, it continues to evolve through new projects, new ideas, and, occasionally, by revisiting old ones with a better understanding than before. If there is a single theme running throughout these pages, it is the belief that quantitative work is at its strongest when it combines scientific thinking, computational rigor, and a willingness to question its own assumptions.

About

I am a physicist and data scientist with a background in applied mathematics, machine learning, and computational modeling. Throughout my professional experience, I have worked on problems where quantitative reasoning, statistical modeling, and scientific computing come together to support practical decision-making.

My academic training in physics has strongly influenced the way I approach data science. My research focused on **quantum optimal control theory**, particularly on algorithmic methods based on *shortcuts to adiabaticity* and the study of computational complexity in physical systems. Although my work has since expanded into a wide range of applied domains, that experience continues to shape my thinking through an emphasis on clear problem formulation, explicit assumptions, and careful evaluation of model behavior.

Professionally, I have applied quantitative methods, statistical modeling, and machine learning across a variety of application areas, including quantitative finance, retail analytics, energy, and industrial research. My work has involved optimization, simulation, predictive modeling, and decision-support systems, always with the objective of translating complex problems into reliable computational solutions.

What connects these experiences is not the industries themselves, but a consistent way of approaching quantitative problems. Regardless of the domain, I believe that successful solutions begin with understanding the problem, selecting methods that are appropriate to the task, and validating results carefully before drawing conclusions.

This portfolio has grown naturally from that way of thinking. The work presented here brings together ideas from mathematics, statistics, computer science, and physics, but its primary purpose remains the same: to develop solutions that are technically rigorous, transparent, reproducible, and appropriate to the problem they are intended to solve.

Outside my professional work, I enjoy playing the bass guitar and maintain a longstanding interest in theology.

How to Read This Portfolio

This portfolio is organized as a collection of independent chapters. Most are presented as self-contained case studies, while others take the form of technical writings that explore broader methodological or scientific questions. Each chapter is intended to stand on its own, allowing readers to explore topics according to their interests without following the entire sequence.

Although the subject matter varies considerably, most project-based chapters follow a common structure:

1. **Problem Definition**
   Introduces the problem, its context, and the question being addressed.

2. **Modeling Approach**
   Describes the conceptual framework, key assumptions, and methodological reasoning that guide the proposed solution.

3. **Methodology and Implementation**
   Presents the data, computational workflow, experimental design, and implementation details.

4. **Results, Discussion, and Limitations**
   Evaluates the proposed solution, interprets the results, discusses limitations, and identifies possible extensions.

5. **Technical Appendix (Optional)**
   Provides supplementary mathematical derivations, methodological notes, or additional computational details when they contribute to a deeper understanding of the work.

This structure serves as a common framework rather than a fixed template. Individual chapters may place greater emphasis on particular aspects depending on the nature of the problem, the methodology employed, or the purpose of the work.

Where appropriate, computational artifacts implemented in `Python`, `R`, or both accompany the written material. These resources provide additional implementation details, support reproducibility, and allow readers to explore complementary analyses where relevant.

Part I

Applied Data Science

Overview

The projects presented in this section demonstrate how quantitative methods can be applied to practical problems across a wide range of domains. Although the application contexts differ, they are united by a common objective: to develop computational solutions that support understanding, prediction, optimization, or decision-making in real-world settings.

Each project begins with a clearly defined problem and develops an appropriate quantitative approach based on the characteristics of the data, the objectives of the analysis, and the assumptions required by the methodology. Rather than promoting particular algorithms or technologies, the emphasis is placed on selecting methods that are appropriate to the problem and evaluating them critically within their intended context.

The projects have been developed as independent case studies and may therefore be read in any order. Together, they illustrate applied data science as a disciplined process of quantitative problem solving, where careful modeling, reproducible computation, and thoughtful interpretation are considered as important as the final results themselves.

Chapter 1

CAPM and Fama–French Models

This project studies how classical and multi-factor asset pricing models explain the relationship between systematic risk and expected return. Using historical market data, the analysis evaluates the assumptions, explanatory power, and empirical limitations of the CAPM and Fama–French frameworks through regression-based factor modeling.

1.1 Problem Definition

This project evaluates how asset pricing models—from the traditional CAPM to the multi-factor extensions of Fama and French—formalise the relationship between risk and expected return. In practice, an investor faced with a price signal must determine whether the potential return justifies the associated risk. Doing this effectively means discounting uncertain future cash flows back to present value, which requires a reliable discount rate. The core question is straightforward: what minimum return should be demanded from an asset given its specific risk profile?

Getting this right is critical for asset valuation, portfolio construction, and fund performance evaluation. Without a structured framework, investment decisions often default to guesswork or oversimplified rules of thumb—such as chasing past returns or assuming that any form of risk automatically deserves a premium. Modern financial theory offers a more rigorous perspective: the market does not reward idiosyncratic or stock-specific risk, since it can be diversified away.

Only systematic risk warrants a higher expected return. This principle is what separates the CAPM from more naive approaches and serves as the foundation for multi-factor models.

Key decisions—ranging from efficient capital allocation to identifying market mispricings—rely heavily on these models. If an asset offers an expected return above what its systematic risk predicts, it may indicate undervaluation. However, the real-world challenge is methodological: true expected returns are unobservable. Relying on historical data introduces noise, sample-period sensitivity, and specification errors. Furthermore, the CAPM compresses all systematic risk into a single factor (the market portfolio), a simplification that empirical evidence has repeatedly challenged.

The Fama–French models attempt to address this limitation by capturing additional dimensions of systematic risk, including size, value, profitability, and investment patterns. From a data science perspective, this project explores how these models are constructed, tests their underlying assumptions, and evaluates how well they fit empirical data. The final output is a comparative analysis of the models' explanatory power and practical limitations when confronted with real-world financial data that rarely behaves as textbook theory predicts.

1.2 Modeling Approach

To approach the problem of asset valuation, this project is grounded on two fundamental principles of finance. The first is the time value of money: a dollar today is worth more than a dollar tomorrow because it can be invested immediately and start generating returns. The second is risk aversion: an uncertain cash flow is only attractive if it offers additional compensation relative to a risk-free alternative. Together, these principles reflect the two central variables underlying any financial valuation problem: time and risk.

Under this framework, the valuation of an asset can be understood as the present value of an uncertain future cash flow:

$$PV = \left(\frac{1}{1+r_t}\right)^t \mathbb{E}[X_t] \tag{1.1}$$

Here, PV represents the expected present value of a future cash flow X_t received at time t, while r_t denotes the discount rate associated with risk and the opportunity cost of capital. The fundamental challenge for the investor is that this discount rate is not known in advance. In other words, the key question is how much return should be required to compensate for the level of risk being taken. This problem provides the conceptual motivation for the Capital Asset Pricing Model (CAPM) and its multifactor extensions.

Since future returns are uncertain, this project models them as random variables. A practical way to characterize them is through their expected value, representing the anticipated average return, and their volatility, which measures the dispersion of outcomes around that average. This mean-variance framework, originally developed by Harry Markowitz in modern portfolio theory, shows that total portfolio risk can be reduced by combining assets whose returns do not move exactly in the same way [1]. The intuition is straightforward: through diversification, part of the individual fluctuations of each asset tends to offset one another.

However, diversification has an observable practical limit. As portfolios are constructed with an increasing number of assets, total volatility initially falls rapidly but eventually stabilizes around a lower bound. As illustrated in Figure 1.1, this behavior reveals the existence of two distinct components of risk. On the one hand, idiosyncratic or diversifiable risk, which is associated with firm-specific events and can largely be eliminated through diversification. On the other hand, systematic risk, which is linked to broader macroeconomic and market-wide movements and therefore persists even in highly diversified portfolios.

This distinction has a fundamental implication for asset pricing. In a competitive market, rational investors should not expect compensation for taking risks that can be eliminated at virtually no cost through diversification. As a result, the market only rewards exposure to systematic risk. This idea forms the conceptual foundation of the CAPM, which states that the expected risk premium of an asset should be proportional to the expected risk premium of the market [2] [3]:

$$\mathbb{E}[r_i] - r_f = \beta_i \left(\mathbb{E}[r_m] - r_f\right) \tag{1.2}$$

In this expression, $\mathbb{E}[r_i]$ represents the expected return of asset i, r_f is the risk-free rate, and $\mathbb{E}[r_m]$ corresponds to the expected return of the market portfolio. The parameter β_i measures the sensitivity of the asset to market fluctuations and summarizes its exposure to systematic risk.

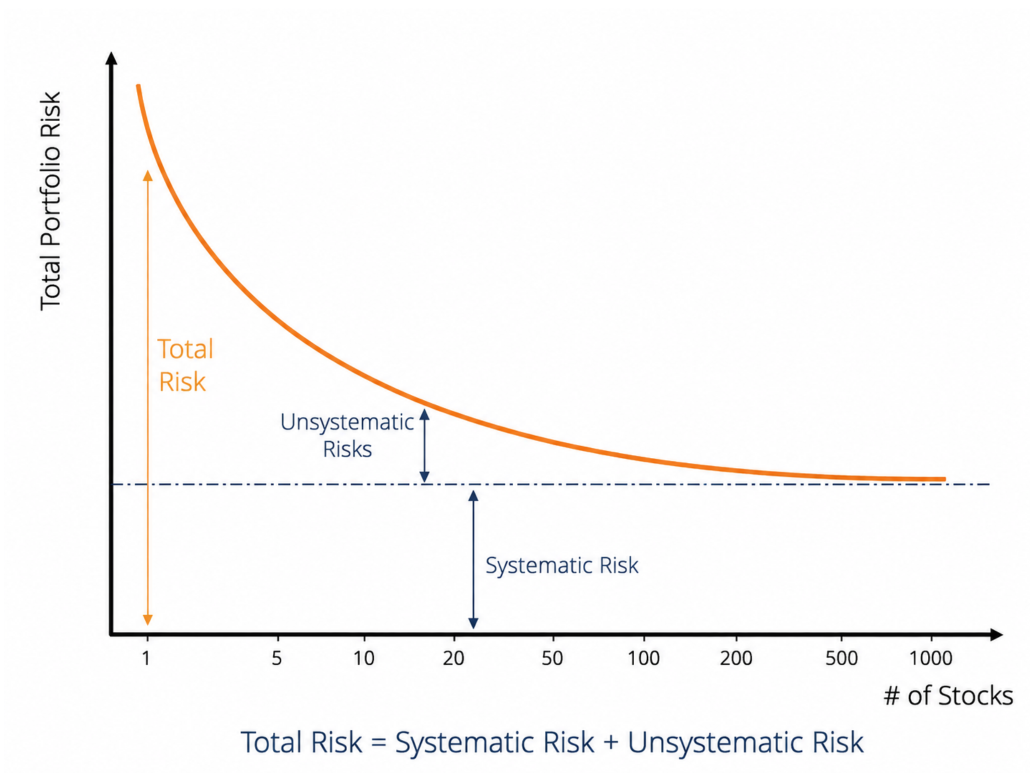

Figure 1.1: Decomposition of total risk into systematic and diversifiable risk as assets are progressively added to a portfolio.

From a modeling perspective, the estimation of beta can be interpreted as a simple linear regression between asset returns and market returns [4]. In this project, the CAPM is implemented precisely under this framework, using simple linear regression to estimate the market exposure of different assets and analyze the relationship between systematic risk and expected return.

Despite its conceptual simplicity and historical importance, the CAPM assumes that the entire structure of systematic risk can be summarized by a single factor: the market itself. Empirical evidence, however, suggests that certain firm characteristics — such as company size or the relationship between book value and market value — capture persistent return patterns that the CAPM does not fully explain. These observations motivated the development of the multifactor models proposed by Fama and French [5] [6].

In this project, these extensions are implemented using multiple linear regression. The three-factor model can be expressed as:

$$\mathbb{E}[r_i] - r_f = \alpha + \beta_{mkt}(\mathbb{E}[r_m] - r_f) + \beta_{smb} \cdot \text{SMB} + \beta_{hml} \cdot \text{HML} + \varepsilon \tag{1.3}$$

where SMB (*Small Minus Big*) measures the excess return of small-cap stocks relative to large-cap stocks, while HML (*High Minus Low*) captures the return differential between value and growth stocks [7] [8].

From a modeling perspective, the transition from the CAPM to the Fama–French three-factor model can be interpreted as moving from a simple linear regression to a multiple linear regression framework, incorporating additional explanatory variables to capture dimensions of systematic risk that a single-factor model cannot fully represent.

The main objective of the project is not simply to compare statistical goodness-of-fit metrics, but to understand how different economic factors contribute to the generation of returns in financial markets and to evaluate how effectively models of different complexity levels represent that underlying process.

1.3 Methodology and Implementation

1.3.1 Data Collection and Preprocessing

The analysis uses daily adjusted closing prices obtained from Yahoo Finance for both Apple Inc. (`AAPL`) and the S&P 500 index (`^GSPC`). The S&P 500 is used as a proxy for the market portfolio within the CAPM framework.

The selected period spans from January 2020 to December 2024, providing a multi-year sample that includes different market conditions and volatility regimes. Adjusted prices are used in order to account for corporate actions such as stock splits and dividend adjustments when available.

After downloading the data, both time series are merged into a single dataset and aligned by trading date. Missing observations are removed prior to return calculation.

The price series are initially inspected to verify data consistency and to obtain a preliminary view of the market dynamics during the analyzed period.

Daily simple returns are then computed from the adjusted closing prices. These returns constitute the main variables used throughout the regression analysis.

Before estimating the model, the relationship between asset and market returns is visually explored through a scatter plot. As shown in Figure 1.3, the data already suggest a positive linear relationship between Apple returns and market returns, supporting the use of a linear regression framework for the CAPM estimation.

1.3.2 CAPM Implementation

Following the theoretical formulation introduced in Equation 1.2, the CAPM is implemented through a simple linear regression in which the daily returns of Apple Inc. are modeled as a function of the daily returns of the market portfolio.

For simplicity, the implementation presented here uses raw daily returns without explicitly incorporating a risk-free rate. Under this approximation, the model estimates the linear sensitivity of the selected asset relative to market movements.

The estimated regression coefficients are summarized below.

Figure 1.2: Adjusted closing prices for Apple Inc. and the S&P 500 index between 2020 and 2024. The figure is mainly used as a preliminary inspection of the downloaded financial time series before return calculation.

Figure 1.3: Scatter plot of daily Apple returns against S&P 500 daily returns. The figure provides an exploratory visualization of the positive relationship between the asset and market returns prior to the CAPM estimation.

Table 1.1: Estimated CAPM regression coefficients for Apple Inc. using the S&P 500 as the market proxy.

Term	Estimate	Std. Error	t-statistic	p-value
(Intercept)	0.000526	0.000346	1.5226	0.128105
market	1.173257	0.025682	45.6847	0.000000

The estimated market beta is approximately $\beta = 1.173257$, indicating that Apple exhibited higher sensitivity to market fluctuations than the S&P 500 during the analyzed period. Under the CAPM interpretation, this coefficient suggests that, on average, Apple returns moved approximately 17% more aggressively than the market portfolio in response to market variations.

Additional model diagnostics are reported below.

Table 1.2: Diagnostic statistics associated with the CAPM linear regression model.

Metric	Value
R-squared	0.6247
Adjusted R-squared	0.6244
F-statistic	2087.0916
Observations	1256

The estimated beta coefficient is also statistically significant, with a large associated t-statistic and a near-zero p-value. Similarly, the overall regression presents a large F-statistic, supporting the existence of a statistically meaningful linear relationship between Apple returns and market returns during the analyzed period.

The estimated coefficient of determination of approximately $R^2 = 0.625$ indicates that the market factor alone explains a substantial fraction of the observed variability in Apple daily returns during the selected period. Nevertheless, a significant portion of the variability remains unexplained, motivating the consideration of multifactor extensions beyond the single-factor CAPM specification.

The fitted linear relationship between asset and market returns is illustrated in Figure 1.4.

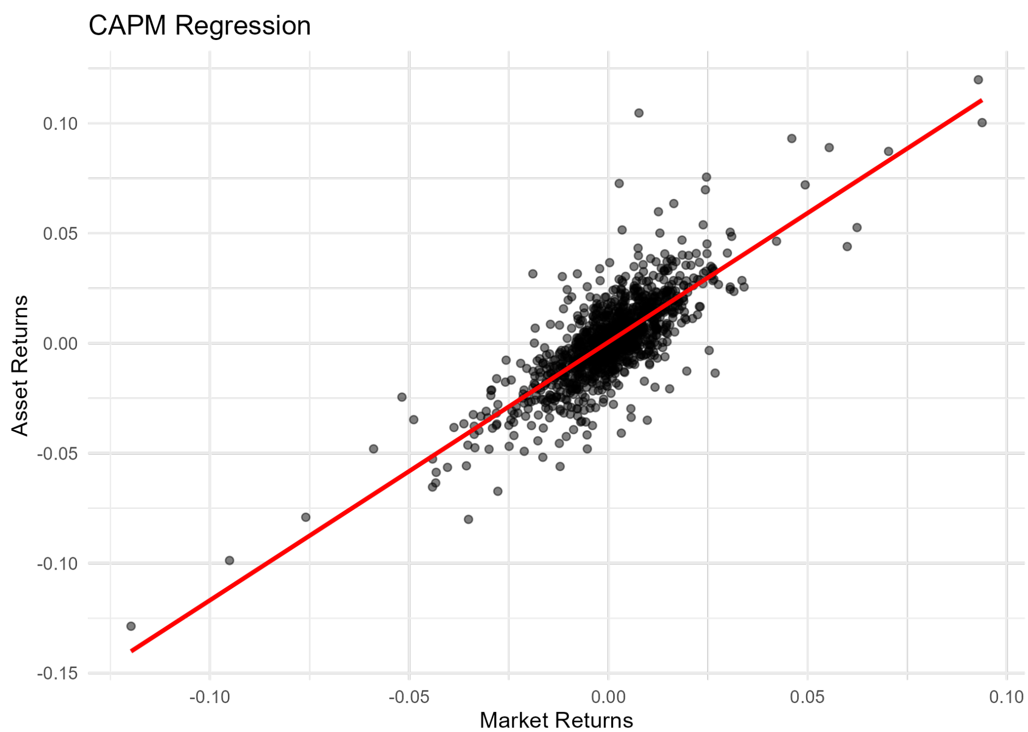

Figure 1.4: CAPM linear regression between Apple daily returns and S&P 500 daily returns. The red line represents the fitted linear relationship estimated through ordinary least squares regression.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

1.3.3 Fama–French Three-Factor Model Implementation

To extend the single-factor CAPM specification, the Fama–French three-factor model introduced in Equation 1.3 was estimated using ordinary least squares (OLS) regression. In addition to the market excess return factor, the model incorporates two additional systematic risk factors associated with firm size (`SMB`) and relative valuation (`HML`).

The implementation combines daily adjusted closing prices for Apple Inc. (`AAPL`) obtained from Yahoo Finance with the daily Fama–French factor dataset developed by Kenneth French. The final modeling dataset contains 1,256 daily observations covering the period from January 2020 to December 2024. The average daily excess return of Apple over the sample period was approximately 0.11%, with a daily standard deviation close to 2.0%.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

Figure 1.5 displays the behavior of the three explanatory factors over time. The series exhibit substantial short-term variability, particularly during periods of market stress such as the early stages of the COVID-19 pandemic in 2020. The `SMB` factor shows the highest volatility among the three factors during this period, reflecting the stronger sensitivity of small-cap firms to abrupt changes in market conditions.

The estimated regression coefficients are summarized below.

Table 1.3: Estimated Fama–French three-factor regression coefficients.

Term	Estimate	Std. Error	t-statistic	p-value
Alpha	0.000455	0.000320	1.421686	0.155366
Market Factor	1.162676	0.024001	48.442699	0.000000
Size Factor	-0.329641	0.042313	-7.790601	0.000000
Value Factor	-0.370591	0.027910	-13.278185	0.000000

The estimated market beta is approximately $\beta_{mkt} = 1.163$, indicating that Apple

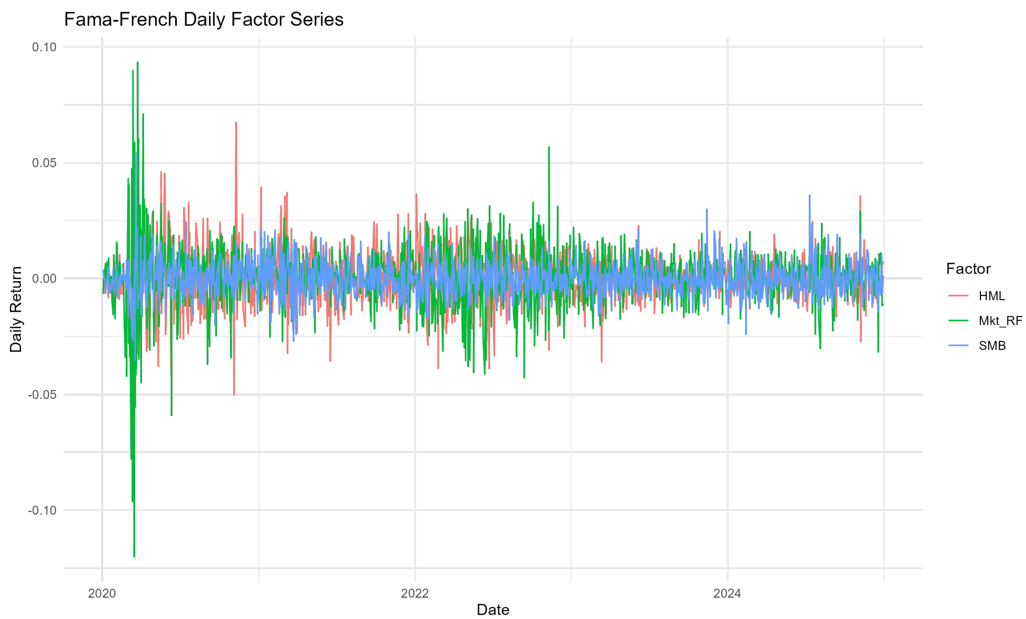

Figure 1.5: Fama–French daily factor series.

exhibits a stronger sensitivity to systematic market movements than the market portfolio itself. The coefficients associated with the size and value factors are both negative and statistically significant. The negative loading on `SMB` suggests that Apple behaves more similarly to large-cap firms than to small-cap firms, which is consistent with its market capitalization and dominant position within the technology sector. Likewise, the negative `HML` exposure indicates characteristics more closely aligned with growth-oriented firms rather than value stocks.

The estimated intercept term ($\alpha \approx 0.00046$) is positive but not statistically significant at conventional significance levels ($p \approx 0.155$), suggesting that, after accounting for the three systematic risk factors, the model does not provide strong evidence of abnormal excess returns during the analyzed period.

The overall regression fit improved relative to the single-factor CAPM specification. The coefficient of determination increased to approximately $R^2 = 0.678$, indicating that the three-factor model explains nearly 68% of the variability observed in Apple excess returns. The F-statistic associated with the regression is also highly significant, supporting the joint explanatory power of the selected factors.

Table 1.4: Diagnostic statistics associated with the Fama-French three-factor regression model.

Metric	Value
R-squared	0.677865
Adjusted R-squared	0.677093
Sigma	0.011345
F-statistic	878.191163
p-value	0.000000
Observations	1256

Figure 1.6 compares the observed excess returns against the fitted values generated by the regression model. The fitted series captures a substantial fraction of the short-term dynamics of Apple returns, although large residual fluctuations remain during periods of elevated volatility.

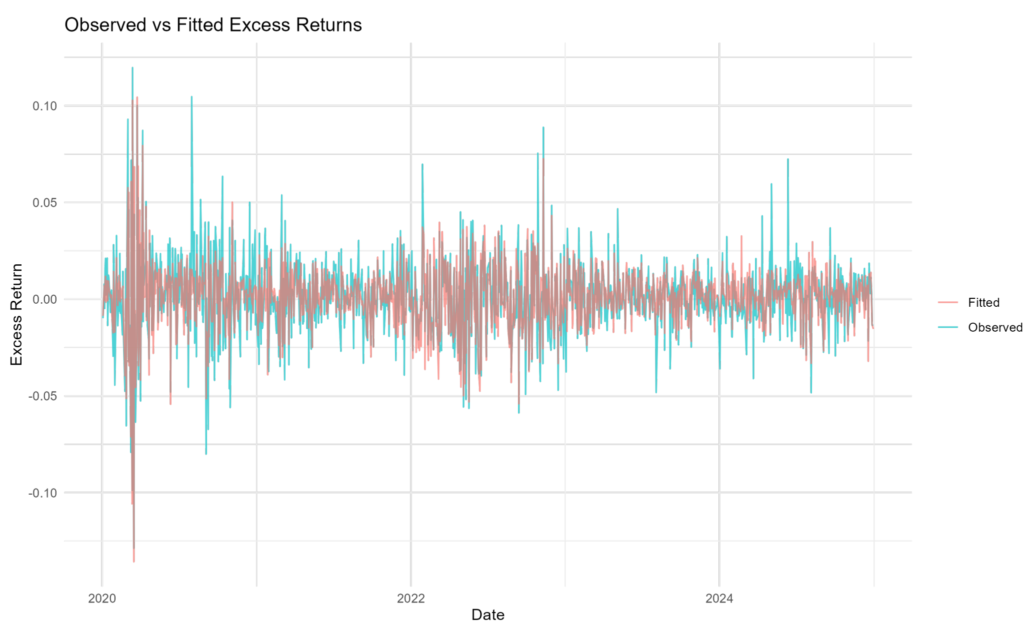

Figure 1.6: Observed and fitted excess returns under the Fama–French three-factor specification.

The residual series displayed in Figure 1.7 remains approximately centered around zero throughout the analyzed period, without obvious long-term structural drift. However, clusters of higher residual volatility are still visible during stressed market conditions, indicating that a portion of the variability remains unexplained even after incorporating the additional systematic factors.

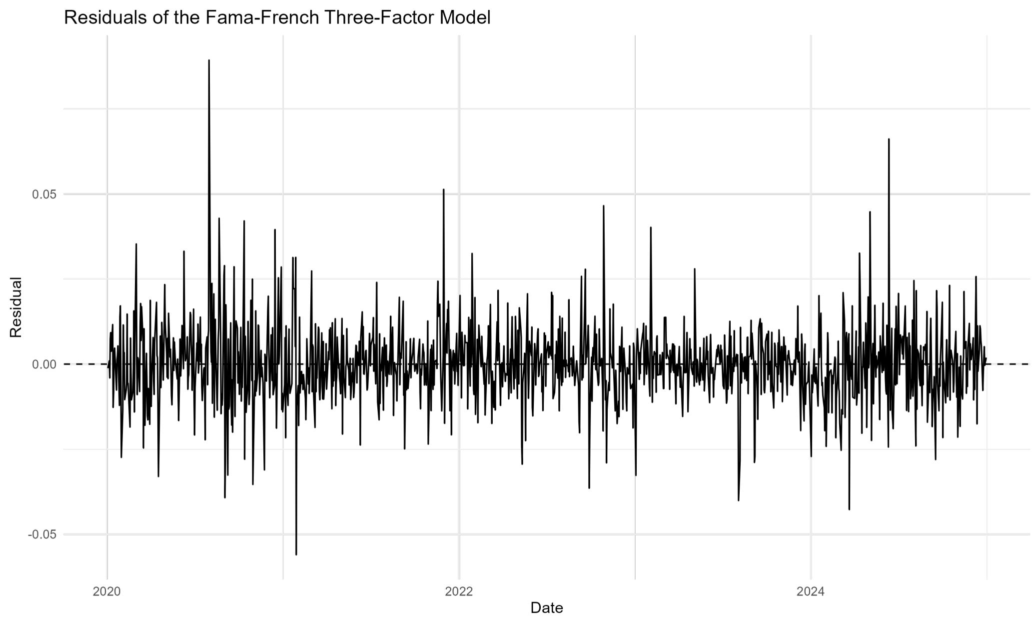

Figure 1.7: Residual series of the Fama–French three-factor model.

Finally, the residual distribution shown in Figure 1.8 exhibits a roughly symmetric shape centered near zero, although moderate tail behavior remains visible. This behavior is consistent with the presence of occasional large market movements that are not fully captured by the linear factor structure.

1.4 Results, Discussion, and Limitations

The empirical results obtained from both the CAPM and the Fama–French three-factor model provide a useful comparison between single-factor and multifactor approaches to asset pricing. Although both models capture an important fraction of the variability observed in Apple excess returns, the inclusion of additional systematic risk factors in the Fama–French specification produces a measurable improvement in explanatory performance.

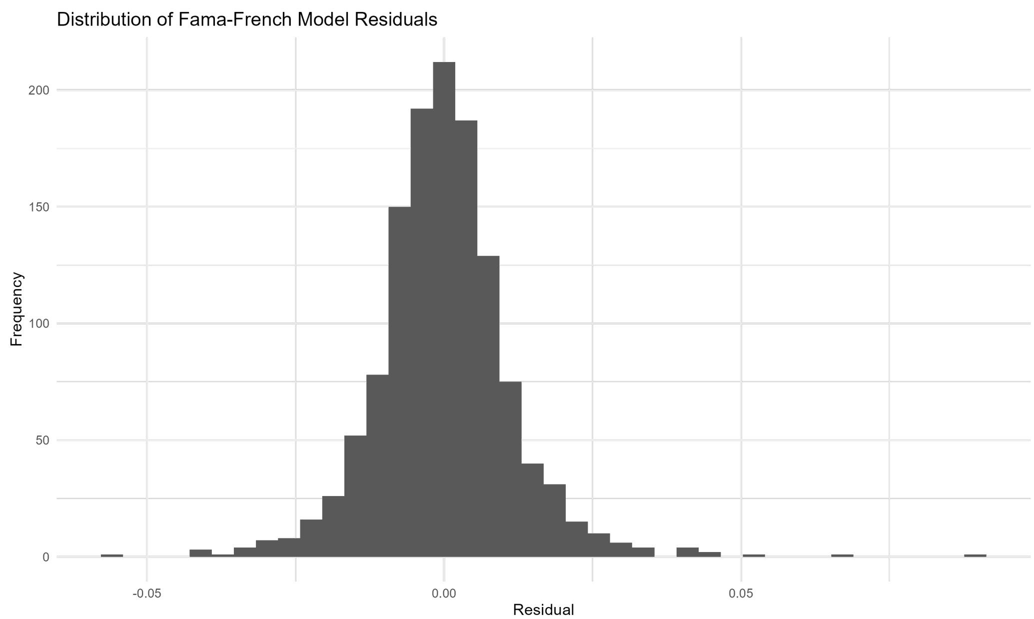

Figure 1.8: Distribution of residuals from the Fama–French three-factor model.

1.4.1 Comparative Analysis

The CAPM estimation produced a market beta close to $\beta \approx 1.17$, indicating that Apple exhibits a level of systematic risk slightly higher than that of the overall market portfolio. The market factor alone explained approximately 62.5% of the variability in Apple excess returns over the analyzed period, resulting in an estimated coefficient of determination of approximately $R^2 = 0.625$.

After incorporating the size (`SMB`) and value (`HML`) factors, the explanatory power of the regression increased to approximately $R^2 = 0.678$. Although the increase in explanatory performance is moderate, the multifactor specification captures an additional fraction of return variability that remains unexplained under the single-factor CAPM framework.

The estimated market beta remained relatively stable across both models, suggesting that market risk continues to represent the dominant systematic component driving Apple returns. However, the additional factor loadings provide further insight into the firm's structural characteristics. The negative exposure to the size factor indicates that Apple behaves more similarly to large-cap firms than to smaller companies, while the negative loading associated with the value factor reflects characteristics commonly associated with growth-oriented firms.

From a statistical perspective, both models produced highly significant overall regressions according to their respective F-statistics. Nevertheless, the Fama–French specification generated lower residual variability and a modest improvement in fitted return dynamics, particularly during periods of elevated market volatility.

The comparison between observed and fitted excess returns also suggests that the multifactor specification is better able to reproduce short-term return fluctuations than the CAPM alone. However, significant residual movements remain visible in both models, indicating that a substantial fraction of the variability in daily equity returns cannot be fully captured through linear factor structures alone.

Overall, the empirical results support the interpretation that the Fama–French three-factor model provides a richer and more flexible representation of systematic risk exposures relative to the standard CAPM specification. At the same time, the relatively modest improvement in explanatory power also highlights the dominant role played by market-wide risk in large-cap technology firms such as Apple.

1.4.2 Model Limitations

Despite their widespread use in financial economics, both the CAPM and the Fama–French three-factor model rely on simplifying assumptions that limit their ability to fully explain observed market behavior.

First, both models assume linear relationships between excess returns and systematic risk factors. In practice, financial markets frequently exhibit nonlinear dynamics, volatility clustering, structural breaks, and regime-dependent behavior that cannot be fully represented within a static linear regression framework.

Second, the analysis relies on historical daily observations and therefore assumes that the estimated factor sensitivities remain relatively stable throughout the selected time horizon. However, the relationship between individual assets and systematic risk factors may evolve over time as market conditions, macroeconomic environments, or firm-specific characteristics change.

An additional limitation arises from the selection of explanatory factors themselves. Although the Fama–French specification extends the CAPM by incorporating size and value effects, many additional sources of systematic risk have been proposed in the literature, including profitability, investment, momentum, liquidity, and volatility-related factors. Consequently, the three-factor model should not be interpreted as a complete description of equity return dynamics.

The analysis also focuses exclusively on a single large-cap technology company. While Apple represents an interesting case study due to its market relevance and liquidity, the estimated coefficients and model diagnostics cannot be directly generalized to other sectors, asset classes, or market environments without additional empirical validation.

Finally, both models are estimated under the assumption that historical relationships provide meaningful information regarding future risk-return behavior. As in most empirical financial applications, this assumption may fail under changing market conditions, particularly during periods of financial stress or structural economic transitions.

Despite these limitations, the CAPM and Fama–French frameworks continue to provide valuable benchmark models for understanding systematic risk exposures, comparing relative asset sensitivities, and constructing interpretable factor-based representations of financial returns.

1.5 Technical Appendix (Optional)

This appendix provides a compact technical background for the regression models used in this project. The main body of the chapter focuses on the financial interpretation of the CAPM and Fama-French models, while this appendix summarizes the statistical structure underlying their empirical estimation.

The CAPM specification can be interpreted as a simple linear regression model, where the excess return of an asset is explained by the excess return of the market portfolio. In contrast, the Fama-French three-factor model extends this idea to a multiple linear regression setting, where asset excess returns are explained by a set of risk factors rather than by the market factor alone.

The purpose of this appendix is not to provide a complete treatment of linear regression or statistical learning. Instead, it collects the main concepts needed to understand how the models were estimated, how the coefficients should be interpreted, and how goodness-of-fit and statistical significance measures relate to the empirical results discussed in the project. The presentation follows standard treatments of linear regression and statistical learning [9] [10].

1.5.1 Simple Linear Regression and the CAPM Model

As discussed previously in Equation 1.2, the CAPM establishes a linear relationship between the expected excess return of an asset and the expected excess return of the market portfolio. From a statistical perspective, this relationship can be interpreted as a simple linear regression model of the form

$$y_i = \theta_0 + \theta_1 x_i + \varepsilon_i \tag{1.4}$$

where θ_0 represents the intercept, θ_1 the slope coefficient, and ε_i an unobservable random error associated with observation i.

Under this interpretation, the empirical CAPM specification can be written as

$$r_i - r_f = \alpha + \beta(r_m - r_f) + \varepsilon \tag{1.5}$$

where the excess return of the asset plays the role of the response variable and the excess return of the market portfolio acts as the predictor variable. In this setting, α corresponds to the intercept term, while β measures the sensitivity of the asset return to market movements.

From the perspective of the CAPM, the intercept term α is expected to be equal to zero, implying that expected excess returns are fully explained by exposure to market risk through the coefficient β. Consequently, testing whether α differs significantly from zero provides a practical way to evaluate the empirical consistency of the model.

1.5.1.1 Ordinary Least Squares Estimation

The parameters of the regression model are estimated by minimizing the discrepancy between the observed data and the fitted regression line. In ordinary least squares (OLS), this discrepancy is quantified through the residual sum of squares (RSS), defined as

$$\text{RSS} = \sum_{i=1}^{N}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{N}(y_i - \theta_0 - \theta_1 x_i)^2 \tag{1.6}$$

where each residual represents the difference between an observed value and the value predicted by the model.

Figure 1.9 illustrates the geometric interpretation of the residuals in simple linear regression. The ordinary least squares estimator selects the parameters that minimize the squared residual distances between the observations and the fitted regression line.

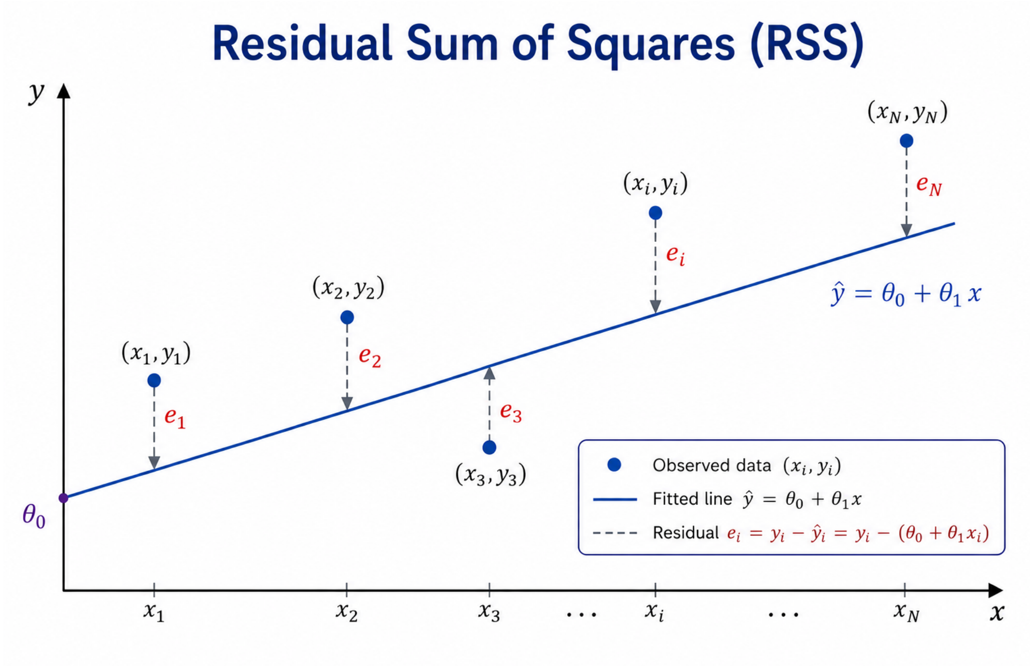

Figure 1.9: Residual Sum of Squares (RSS)

The ordinary least squares estimators are obtained by minimizing Equation 1.6 with respect to the model parameters. Since the RSS is a convex quadratic function of θ_0 and θ_1, the solution can be obtained by setting the gradient equal to zero. The convexity of the RSS follows from the positive definiteness of the associated Hessian matrix, ensuring that the stationary point corresponds to a unique global minimum.

The resulting estimators are given by

$$\hat{\theta}_1 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} \tag{1.7}$$

and

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \tag{1.8}$$

Equation 1.7 shows that the slope estimator can be interpreted as the ratio between the covariance of x and y and the variance of x. Consequently, the estimated slope increases when both variables tend to move together and decreases when their linear relationship weakens. In the context of empirical asset pricing, this interpretation leads naturally to the CAPM market beta, which can be viewed as the ordinary least squares estimator obtained by regressing the excess return of an asset against the excess return of the market portfolio. Under this interpretation, the CAPM beta measures the linear sensitivity of the asset return to market movements in the least-squares sense.

1.5.1.2 Residual Analysis and Model Assumptions

Once the regression coefficients have been estimated, the residuals provide information about the portion of the variability that is not explained by the model. For each observation, the residual is defined as

$$e_i = y_i - \hat{y}_i \tag{1.9}$$

where $\hat{y}_i$ denotes the predicted value obtained from the fitted regression model.

A distinction should be made between residuals and random errors. Residuals correspond to observable deviations between the fitted model and the data, whereas the random errors ε_i introduced in Equation 1.4 represent the unobservable stochastic component assumed by the statistical model.

Classical linear regression assumes that the random errors are independent, centered around zero, and have constant variance. Under these assumptions,

$$\varepsilon_i \sim N(0, \sigma^2) \tag{1.10}$$

where σ^2 denotes the variance of the error term.

The assumption of constant variance is commonly referred to as homoscedasticity and can be expressed as

$$\mathrm{Var}(\varepsilon_i) = \sigma^2 \tag{1.11}$$

for all observations. When the variance of the errors changes across observations, the model is said to exhibit heteroscedasticity, which may affect the reliability of confidence intervals, hypothesis tests, and statistical inference.

Although these assumptions are rarely satisfied exactly in real financial data, they provide the statistical framework underlying ordinary least squares estimation and the interpretation of the inferential quantities discussed in the following sections.

1.5.1.3 Confidence Intervals and Standard Errors

The regression coefficients estimated through ordinary least squares are subject to sampling variability. Consequently, point estimates alone do not provide information about the uncertainty associated with the estimated parameters. This uncertainty is commonly quantified through standard errors and confidence intervals.

Under the assumptions introduced previously, the variance of the random error term can be estimated from the residuals through the residual standard error (RSE), defined as

$$\mathrm{RSE} = \sqrt{\frac{\mathrm{RSS}}{N-2}} \tag{1.12}$$

where N denotes the number of observations and the subtraction of two degrees of freedom reflects the estimation of the intercept and slope coefficients.

The standard errors associated with the regression coefficients are given by

$$\mathrm{SE}(\hat{\theta}_1) = \sqrt{\frac{\sigma^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}} \tag{1.13}$$

and

$$\mathrm{SE}(\hat{\theta}_0) = \sqrt{\sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}\right]} \tag{1.14}$$

where σ^2 is typically replaced in practice by the estimator introduced in Equation 1.12.

These expressions show that the uncertainty associated with the estimated coefficients depends on both the residual variability and the dispersion of the predictor variable. In particular, the standard error of the slope coefficient decreases as the variability of x increases, leading to more stable estimates of the linear relationship.

Confidence intervals provide a range of plausible values for the regression coefficients. Under approximate normality assumptions, a 95% confidence interval for a parameter θ can be written as

$$\hat{\theta} \pm 2 \cdot \mathrm{SE}(\hat{\theta}) \tag{1.15}$$

indicating the range within which the true parameter value is expected to lie with approximately 95% confidence.

1.5.1.4 Hypothesis Testing and Statistical Significance

Beyond estimating the regression coefficients, it is often important to evaluate whether the observed linear relationship is statistically significant. In simple linear regression, this is commonly assessed through hypothesis testing on the slope coefficient.

The null and alternative hypotheses are given by

$$\begin{cases} H_0 : \theta_1 = 0 \\ H_1 : \theta_1 \neq 0 \end{cases} \tag{1.16}$$

Under the null hypothesis, the predictor variable does not contribute to explaining the variability of the response variable, implying the absence of a linear relationship between x and y.

The statistical significance of the estimated slope coefficient can be evaluated through the t-statistic,

$$t = \frac{\hat{\theta}_1}{\mathrm{SE}(\hat{\theta}_1)} \tag{1.17}$$

which measures the magnitude of the estimated coefficient relative to its standard error.

The corresponding p-value quantifies the probability of observing a statistic at least as extreme as the one obtained under the null hypothesis. Small p-values provide evidence against H_0, suggesting that the predictor variable contributes significantly to the regression model.

In the context of the CAPM, hypothesis testing is particularly relevant for evaluating whether the estimated market exposure coefficient differs significantly from zero and whether the intercept term is statistically consistent with the theoretical prediction of $\alpha = 0$.

1.5.1.5 Analysis of Variance (ANOVA)

The variability of the response variable can be decomposed into a component explained by the regression model and a residual component associated with unexplained variability. In simple linear regression, this decomposition is expressed as

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) \tag{1.18}$$

which separates the total deviation of each observation into an explained component and a residual component.

Figure 1.10 illustrates the geometric interpretation of this decomposition in the context of simple linear regression.

Summing over all observations leads to the classical ANOVA decomposition

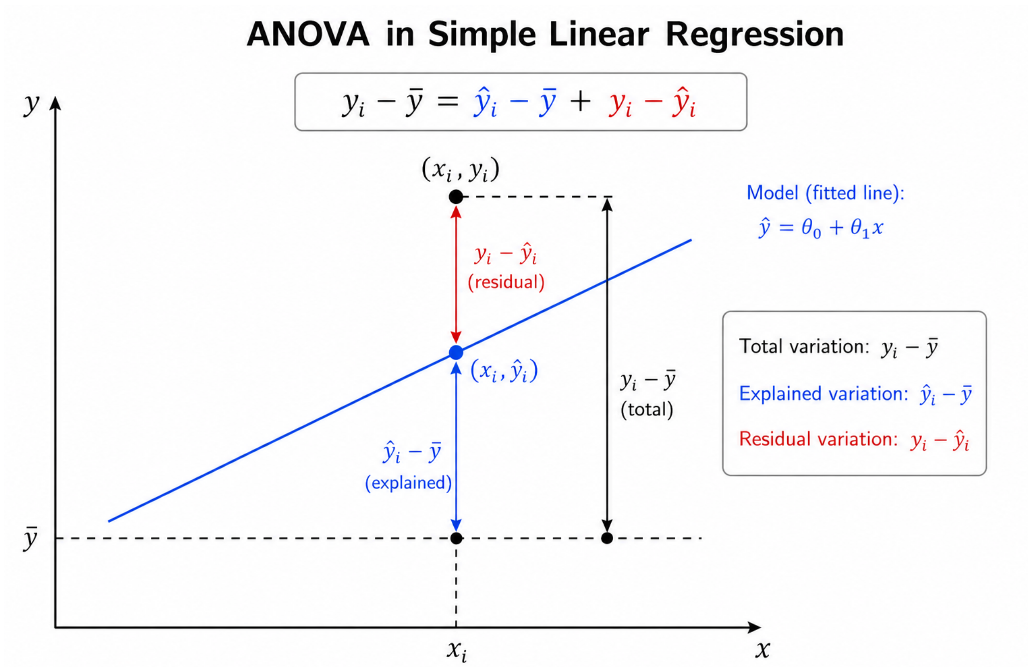

Figure 1.10: ANOVA decomposition in simple linear regression

$$\mathrm{TSS} = \mathrm{MSS} + \mathrm{RSS} \tag{1.19}$$

where

$$\mathrm{TSS} = \sum_{i=1}^{N} (y_i - \bar{y})^2 \tag{1.20}$$

represents the total variability of the response variable,

$$\mathrm{MSS} = \sum_{i=1}^{N} (\hat{y}_i - \bar{y})^2 \tag{1.21}$$

corresponds to the variability explained by the regression model, and

$$\mathrm{RSS} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{1.22}$$

measures the unexplained residual variability.

Dividing each sum of squares by its associated degrees of freedom leads to the corresponding mean squares. In simple linear regression, the mean model square and the mean residual square are given by

$$\mathrm{MMS} = \mathrm{MSS} \tag{1.23}$$

and

$$\mathrm{MSR} = \frac{\mathrm{RSS}}{N-2} \tag{1.24}$$

respectively.

These quantities allow the construction of the F-statistic,

$$F = \frac{\mathrm{MMS}}{\mathrm{MSR}} \tag{1.25}$$

which compares the variability explained by the regression model with the unexplained residual variability. Large values of the F-statistic suggest that the regression model explains a substantial fraction of the observed variability relative to the residual noise.

1.5.1.6 Coefficient of Determination (R^2)

Another commonly used measure in linear regression is the coefficient of determination, denoted by R^2. This statistic quantifies the proportion of the total variability in the response variable that is explained by the regression model.

Using the ANOVA decomposition introduced in Equation 1.19,

$$\text{TSS} = \text{MSS} + \text{RSS}$$

the coefficient of determination can be written as

$$R^2 = \frac{\text{MSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \tag{1.26}$$

Larger values of R^2 indicate that the model explains a greater fraction of the observed variability in the data. In particular, $R^2 = 1$ corresponds to a perfect fit, whereas $R^2 = 0$ indicates that the regression model does not explain the variability of the response variable better than the sample mean alone.

In the special case of simple linear regression, the coefficient of determination is directly related to the Pearson correlation coefficient through

$$R^2 = r^2 \tag{1.27}$$

where r denotes the sample correlation between the predictor variable x and the response variable y.

Although R^2 provides a useful measure of goodness of fit, a large value does not necessarily imply that the model is statistically valid or economically meaningful. For this reason, R^2 is typically interpreted together with residual diagnostics, hypothesis tests, and domain-specific considerations.

1.5.2 Multiple Linear Regression and the Fama-French Model

While the CAPM can be formulated as a simple linear regression model with a single explanatory factor, the Fama–French three-factor model extends this framework by incorporating additional sources of systematic risk. From a statistical perspective, this leads naturally to a multiple linear regression model in which the response variable depends simultaneously on several predictors.

For a model with M explanatory variables, the linear regression model can be written as

$$y_i = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_M x_{iM} + \varepsilon_i \tag{1.28}$$

where θ_0 represents the intercept, $\theta_1, \dots, \theta_M$ denote the regression coefficients associated with the predictor variables, and ε_i corresponds to the random error associated with observation i.

Using matrix notation, the model can be expressed compactly as

$$\mathbf{y} = \mathbf{X}\theta + \varepsilon \tag{1.29}$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1M} \\ 1 & x_{21} & \cdots & x_{2M} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{NM} \end{bmatrix}$$

and

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_M \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}$$

In this representation, the rows of $\mathbf{X}$ correspond to observations, while the columns represent the explanatory variables included in the model. The matrix formulation provides a compact and computationally efficient framework for parameter estimation and statistical inference.

Under this interpretation, the Fama–French three-factor model can be viewed as a particular case of multiple linear regression in which the excess return of an asset is explained simultaneously by the market factor, the SMB factor, and the HML factor.

1.5.2.1 Ordinary Least Squares Estimation

As in the simple linear regression setting, the parameters of the multiple linear regression model are estimated by minimizing the residual sum of squares. Using the matrix formulation introduced in Equation 1.29, the residual vector can be written as

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\theta \tag{1.30}$$

and the corresponding residual sum of squares is given by

$$\text{RSS} = \mathbf{e}^T\mathbf{e} = (\mathbf{y} - \mathbf{X}\theta)^T(\mathbf{y} - \mathbf{X}\theta) \tag{1.31}$$

The ordinary least squares estimator is obtained by minimizing Equation 1.31 with respect to the parameter vector θ. As in the simple linear regression case, the optimization problem is convex and admits a unique global minimum provided that the columns of the design matrix $\mathbf{X}$ are linearly independent.

The resulting estimator is given by

$$\hat{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{1.32}$$

which generalizes the ordinary least squares estimator introduced previously for simple linear regression.

The matrix formulation highlights an important aspect of multiple linear regression: each coefficient measures the linear contribution of its associated predictor while accounting for the remaining variables included in the model. In the context of the Fama–French model, this means that the estimated coefficients associated with the market, SMB, and HML factors represent their partial contributions to the excess return of the asset after controlling for the other factors included in the regression.

1.5.2.2 Statistical Interpretation of the Coefficients

In multiple linear regression, each regression coefficient measures the expected variation in the response variable associated with a one-unit change in the corresponding predictor, while holding the remaining predictors fixed. This interpretation differs from simple linear regression, where the effect of the predictor is evaluated without conditioning on additional explanatory variables.

Under the ordinary least squares framework, the uncertainty associated with the estimated parameter vector can be characterized through its covariance matrix, given by

$$\mathrm{Cov}(\hat{\theta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} \tag{1.33}$$

where σ^2 denotes the variance of the random error term.

As in the simple linear regression setting, the variance of the residuals can be estimated through the residual standard error,

$$\mathrm{RSE} = \sqrt{\frac{\mathrm{RSS}}{N - M - 1}} \tag{1.34}$$

where N represents the number of observations and M the number of explanatory variables included in the model.

The diagonal elements of Equation 1.33 determine the standard errors associated with each regression coefficient and provide a measure of the uncertainty of the parameter estimates. Larger standard errors indicate greater uncertainty in the estimated contribution of the corresponding predictor.

In the context of the Fama–French three-factor model, the estimated coefficients associated with the market, SMB, and HML factors measure the sensitivity of the asset return to each systematic risk factor after controlling for the remaining factors included in the regression. Consequently, the interpretation of each coefficient should always be understood conditionally on the presence of the other explanatory variables.

The inclusion of multiple predictors may also introduce statistical challenges such as multicollinearity, which occurs when explanatory variables exhibit strong linear dependence. In such cases, coefficient estimates may become unstable and difficult to interpret, even when the overall predictive performance of the model remains satisfactory.

1.5.2.3 Global Hypothesis Testing and Model Significance

In multiple linear regression, it is often important to evaluate whether the explanatory variables collectively contribute to the model. This can be assessed through a global hypothesis test of the form

$$\begin{cases} H_0 : \theta_1 = \theta_2 = \cdots = \theta_M = 0 \\ H_1 : \text{at least one } \theta_j \neq 0 \end{cases} \tag{1.35}$$

Under the null hypothesis, none of the explanatory variables contribute to explaining the variability of the response variable, implying that the model does not perform better than using the sample mean alone.

The global significance of the regression model can be evaluated through the F-statistic,

$$F = \frac{(\text{TSS} - \text{RSS})/M}{\text{RSS}/(N - M - 1)} \tag{1.36}$$

which compares the variability explained by the model with the unexplained residual variability after adjusting for the corresponding degrees of freedom.

Large values of the F-statistic suggest that the explanatory variables collectively explain a substantial fraction of the observed variability in the response variable. The corresponding p-value provides a quantitative measure of the evidence against the null hypothesis.

In the context of the Fama–French three-factor model, the global F-test evaluates whether the market, SMB, and HML factors jointly contribute to explaining the excess return of the asset. Consequently, the test provides a global assessment of the explanatory power of the multifactor specification relative to a model without risk factors.

1.5.2.4 Limitations of Multiple Linear Regression

Although multiple linear regression provides a flexible and interpretable framework for modeling relationships between variables, its performance depends on several assumptions and modeling choices. In practice, violations of these assumptions may affect the reliability of parameter estimates, confidence intervals, and hypothesis tests.

One important limitation is multicollinearity, which arises when explanatory variables exhibit strong linear dependence. In such cases, the regression coefficients may become unstable and highly sensitive to small changes in the data, making their interpretation more difficult.

Multiple linear regression models may also be affected by omitted variable bias when relevant explanatory factors are excluded from the specification. In empirical asset pricing, this issue motivated the development of multifactor models such as the Fama–French framework, which extends the CAPM by incorporating additional sources of systematic risk beyond the market factor alone.

Another practical limitation is that linear regression assumes a linear relationship between predictors and the response variable. While this assumption often provides a useful approximation, real financial data may exhibit nonlinear effects, structural changes, heteroscedasticity, or temporal dependence that are not fully captured by the model.

Despite these limitations, linear regression models remain fundamental tools in empirical finance due to their interpretability, computational efficiency, and strong connection with statistical inference and risk factor modeling.

Chapter 2

Unsupervised Learning for Asset Allocation

This project explores the use of unsupervised learning techniques for portfolio construction and risk diversification. The analysis combines Principal Component Analysis (PCA), hierarchical clustering, and Hierarchical Risk Parity (HRP) to study how latent market structure and correlation patterns can be used to build more stable asset allocation strategies.

Rather than focusing on return forecasting, the project examines how statistical learning methods can help identify common sources of risk, reduce redundancy across correlated assets, and improve portfolio robustness under realistic market conditions. The underlying idea is that portfolios are often considered diversified based on asset labels, sectors, or asset classes, even when a small number of common risk factors continue to drive most of their behaviour. By studying how assets actually move together, the project aims to construct allocations that better reflect the underlying structure of market risk.

2.1 Problem Definition

Constructing a diversified investment portfolio is one of the central problems in quantitative finance. In practice, investors must allocate capital across a large universe of financial instruments while balancing expected returns, risk exposure, diversification, and portfolio stability. Although the classical mean–

variance framework introduced by Harry Markowitz remains foundational in modern portfolio theory, its practical implementation presents important limitations when applied to real financial markets.

Traditional portfolio optimization methods rely heavily on the estimation and inversion of covariance matrices. In large and highly correlated markets, these matrices often become numerically unstable and highly sensitive to small estimation errors. As a result, optimized portfolios may exhibit unintuitive allocations, excessive concentration, poor robustness, and unstable out-of-sample performance. This issue becomes more severe as the number of assets increases and correlation structures evolve through time.

At the same time, financial markets naturally exhibit latent and hierarchical structures that are not fully captured by traditional optimization approaches. Assets rarely behave as isolated entities. Instead, they tend to move collectively according to sector dynamics, macroeconomic conditions, liquidity regimes, or broader market factors. Under stressed market conditions, assets that appear diversified on paper may suddenly behave in remarkably similar ways, reducing diversification precisely when it becomes most important.

The motivation behind this project is to explore whether unsupervised learning techniques can provide a more robust and structurally stable framework for asset allocation. Rather than treating portfolio construction purely as a constrained optimization problem, the project approaches it as a problem of identifying and organizing the underlying geometry of market risk.

To address this, the analysis combines two complementary perspectives. First, Principal Component Analysis (PCA) is used to study the dominant sources of variation in the market and to represent the covariance structure in terms of orthogonal risk components. This provides a compact representation of the market structure while reducing redundancy across highly correlated assets. Second, hierarchical clustering techniques are used to identify groups of statistically similar assets and to construct portfolios using Hierarchical Risk Parity (HRP), a methodology designed to improve allocation stability without requiring direct inversion of the covariance matrix.

The objective is not simply to maximize historical returns, but to investigate whether combining spectral methods and hierarchical structures can produce portfolios that are more diversified, interpretable, and robust under changing market conditions. The project therefore sits at the intersection of quantitative finance, machine learning, and statistical risk modeling, with a strong emphasis on practical portfolio construction rather than purely theoretical optimization.

2.2 Modeling Approach

The modeling approach adopted in this project treats portfolio construction as a problem of understanding the structure of market risk. The analysis focuses on how assets move together, how risk concentrates across the market, and how correlation patterns affect diversification under changing market conditions.

Financial markets are rarely composed of independent assets. Securities that belong to different sectors or asset classes often remain exposed to the same underlying drivers, particularly during periods of stress when correlations tend to increase. This means that portfolios that appear diversified on paper may still contain significant hidden concentrations of risk.

The analysis begins with Principal Component Analysis (PCA), following the general perspective of statistical risk decomposition discussed in [11] and the idea of principal portfolios introduced in [12]. PCA is used to study the covariance structure of the market and identify the dominant directions of variation across asset returns. In practice, a relatively small number of components often explains a large proportion of total market variance, suggesting that many assets continue to react to common underlying factors even when they appear superficially diversified.

The covariance matrix is decomposed according to

$$\Sigma = Q\Lambda Q^\top \tag{2.1}$$

where Q contains the eigenvectors and Λ the associated eigenvalues. The eigenvectors define orthogonal directions in the transformed risk space, while the eigenvalues measure the variance explained by each component. This decomposition provides a compact representation of the market structure and helps distinguish between broad market behaviour and more localized sources of variation.

The PCA transformation can also be interpreted as a rotation of the original return space into an orthogonal basis of uncorrelated components,

$$\tilde{\mathbf{R}} = \Gamma^{\top}\mathbf{R} \tag{2.2}$$

where $\mathbf{R}$ denotes the original return vector and $\tilde{\mathbf{R}}$ its representation in the transformed PCA space. Portfolio returns preserve their underlying structure under this transformation while becoming easier to interpret in terms of independent sources of market variation.

PCA is used as a diagnostic tool for studying market structure before constructing the final allocation. The portfolio construction stage is based on Hierarchical Risk Parity (HRP), introduced in [13] and further developed in [14]. Unlike traditional mean–variance optimization frameworks, HRP does not require direct inversion of the covariance matrix, avoiding part of the numerical instability associated with highly correlated asset universes and estimation error.

The clustering stage identifies groups of statistically similar assets using a correlation-based distance measure,

$$d_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})} \tag{2.3}$$

where ρ_{ij} denotes the correlation between assets i and j. Assets with similar behaviour are grouped together into a hierarchical structure that reflects the organization of the market more naturally than a flat covariance matrix alone.

Portfolio weights are then allocated recursively across the hierarchy, reducing concentration in highly correlated regions of the market. In practice, this tends to produce more stable allocations than traditional mean–variance optimization procedures, where small changes in covariance estimates can generate disproportionately large changes in portfolio weights.

An equally weighted $1/N$ portfolio is used as a benchmark throughout the analysis. Although simple, this benchmark provides an important reference point when evaluating whether additional model complexity leads to meaningful improvements in diversification and portfolio robustness out-of-sample.

The resulting framework combines spectral analysis, clustering techniques, and hierarchical risk allocation to study diversification from a structural perspective, with emphasis on robustness and stability under changing market conditions.

2.3 Methodology and Implementation

The empirical analysis was conducted using a diversified universe of U.S. equities representing multiple sectors of the economy, including technology, financials, energy, healthcare, consumer staples, communication services, and retail. The selected assets were chosen to provide exposure to different market dynamics while still allowing meaningful correlation structures and latent factor relationships to emerge throughout the analysis.

The asset universe used in the study is summarized in Table 2.1.

Table 2.1: Selected asset universe used throughout the portfolio construction and clustering experiments.

Ticker	Sector
AAPL	Technology
MSFT	Technology
GOOGL	Communication Services
AMZN	Consumer Discretionary
META	Communication Services
JPM	Financials
BAC	Financials
XOM	Energy
CVX	Energy
JNJ	Healthcare
PFE	Healthcare
PG	Consumer Staples
KO	Consumer Staples
WMT	Retail
HD	Consumer Discretionary

Daily adjusted closing prices were collected and transformed into continuously compounded daily returns. The analysis focused primarily on the covariance and correlation structure of returns rather than on absolute price levels. Figure 2.1 presents the evolution of adjusted prices across the selected assets during the evaluation period.

Although the selected securities belong to different sectors, the return series ex-

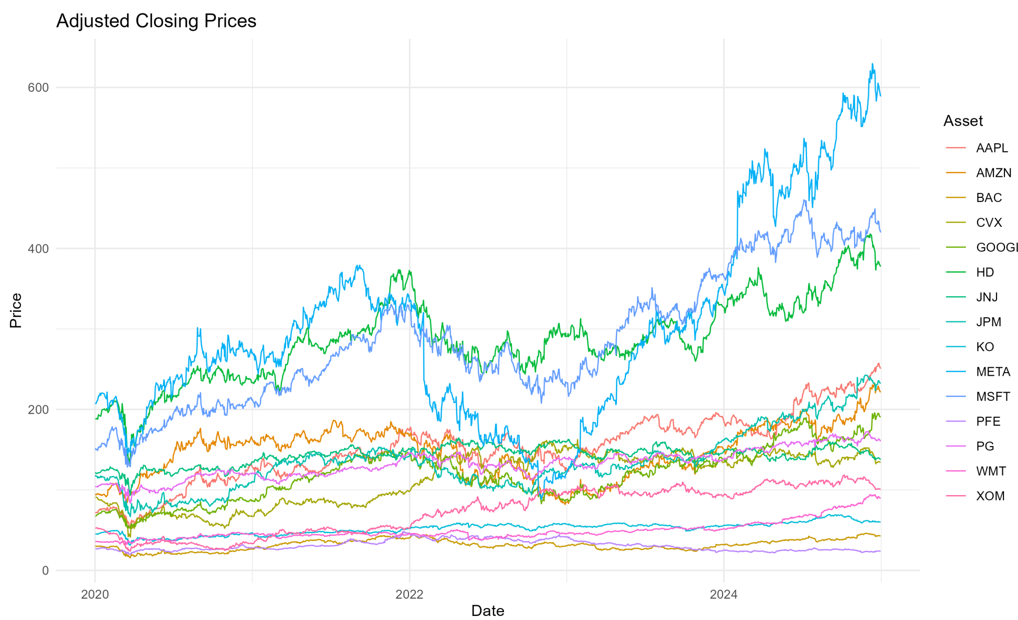

Figure 2.1: Adjusted closing prices of the selected assets over the analysis period, illustrating heterogeneous price dynamics and sector-specific trends across the asset universe.

hibit periods of synchronized behaviour, particularly during episodes of elevated market stress. This becomes more evident when examining the daily return dynamics shown in Figure 2.2.

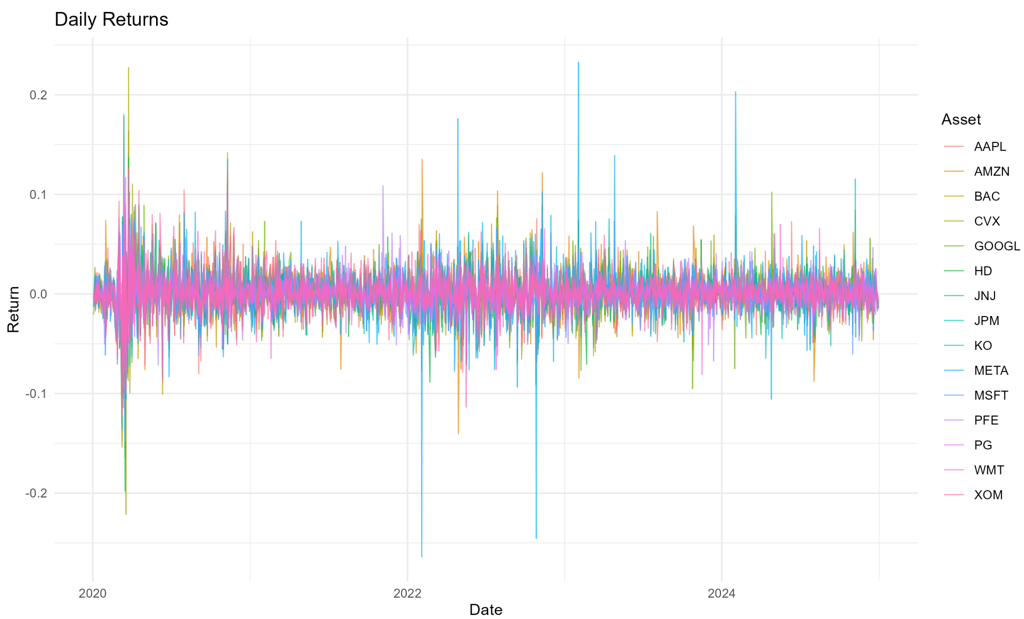

Figure 2.2: Daily returns of the selected assets over the analysis period, highlighting volatility clustering and heterogeneous risk dynamics across the asset universe.

The return series display the volatility clustering commonly observed in financial markets, together with periods of sharply increasing cross-asset dependence. The volatility spikes and correlation increases visible in the data are precisely the conditions under which naïve diversification can become less effective, providing the empirical motivation for the PCA and HRP framework adopted in this study.

The covariance matrix shown in Figure 2.3 was estimated directly from the daily return series and used throughout the PCA and HRP stages of the analysis.

The corresponding correlation matrix is presented in Figure 4.5. Several economically meaningful relationships already become visible at this stage, particularly among technology, financial, energy, healthcare, and consumer staples stocks.

Principal Component Analysis (PCA) was then applied to study the latent struc-

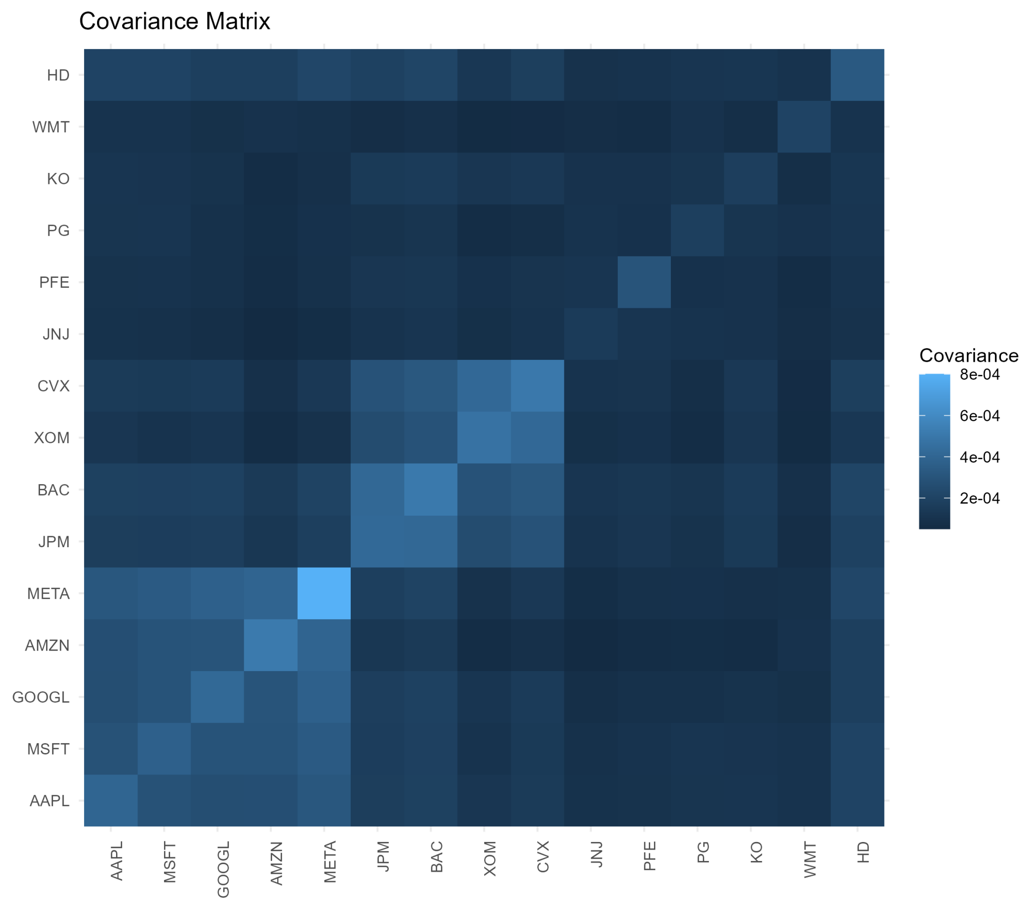

Figure 2.3: Covariance matrix of asset returns, showing the magnitude of joint variability across the selected assets used in portfolio risk estimation and allocation.

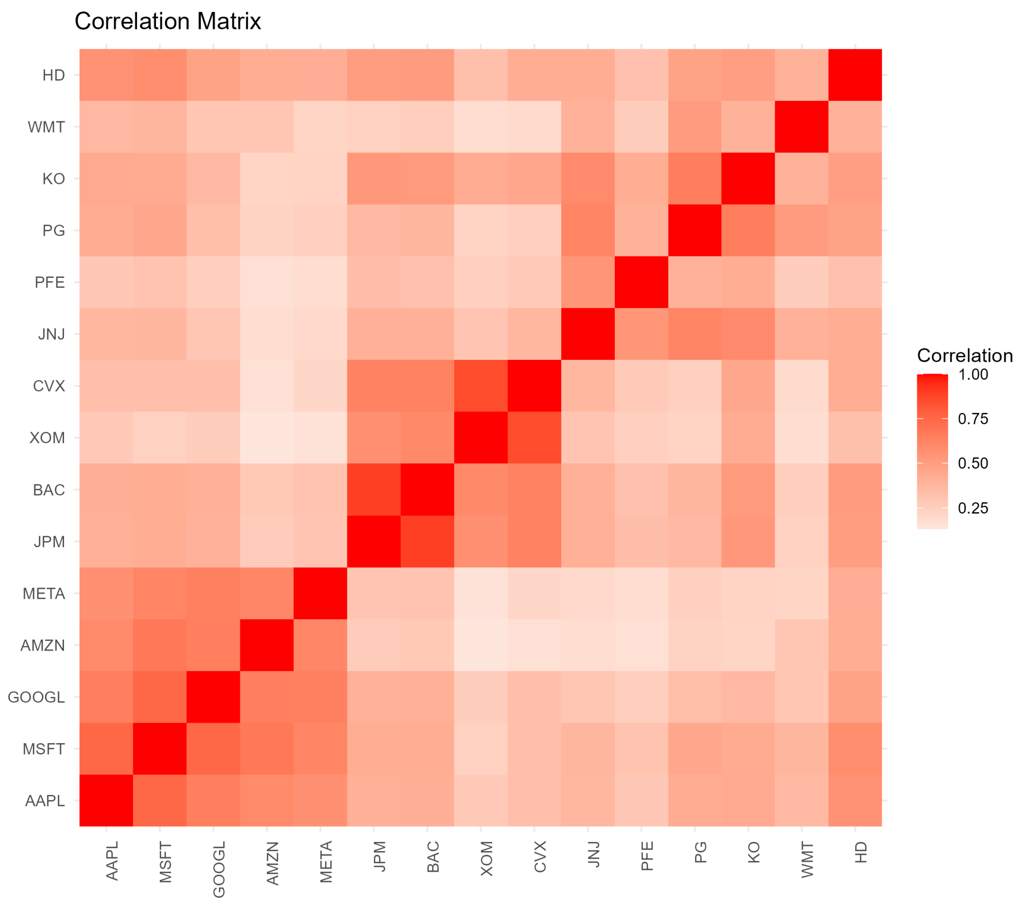

Figure 2.4: Correlation matrix of asset returns, showing the dependence structure across the selected assets prior to hierarchical clustering reordering.

ture of market risk. The decomposition was performed on the covariance matrix of returns, allowing the market structure to be represented in terms of orthogonal components ranked by explained variance.

The explained variance associated with each principal component is summarized in Table 2.2.

Table 2.2: Explained and cumulative variance ratios obtained from Principal Component Analysis (PCA) applied to asset returns.

Component	Explained Variance	Cumulative Variance
PC1	0.4638	0.4638
PC2	0.1793	0.6431
PC3	0.0740	0.7170
PC4	0.0512	0.7682
PC5	0.0466	0.8148
PC6	0.0352	0.8500
PC7	0.0297	0.8798
PC8	0.0262	0.9060
PC9	0.0230	0.9289
PC10	0.0181	0.9470
PC11	0.0144	0.9614
PC12	0.0123	0.9737
PC13	0.0102	0.9839
PC14	0.0084	0.9923
PC15	0.0077	1.0000

The first principal component explains approximately 46% of total variance, while the first two components together account for roughly 64%. The first five components explain more than 80% of total variance, indicating that a relatively small number of latent factors captures much of the variability in the asset universe. Figure 2.5 and Figure 2.6 illustrate this concentration of variance visually.

The first principal component captures a broad market-related factor affecting most assets simultaneously, consistent with its dominant contribution to total variance. Subsequent components explain progressively smaller fractions of variance and reveal more localized sectoral structures that are less visible in the orig-

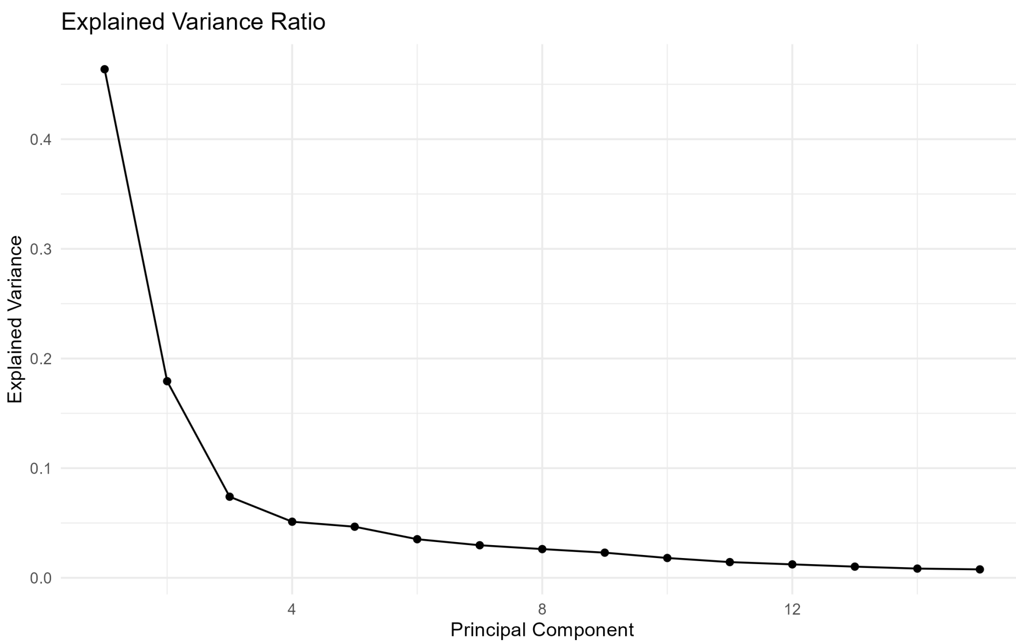

Figure 2.5: Explained variance ratio for each principal component obtained from PCA, showing that the first components capture most of the variability in the asset return space.

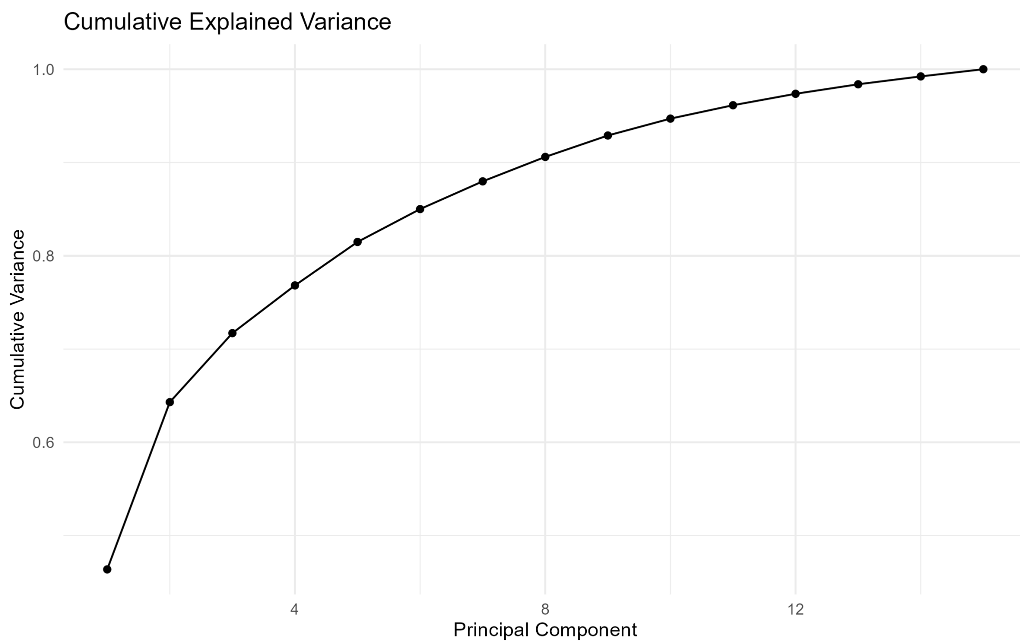

Figure 2.6: Cumulative explained variance obtained from Principal Component Analysis (PCA), showing how the first principal components capture most of the variability in the asset return space.

inal return space.

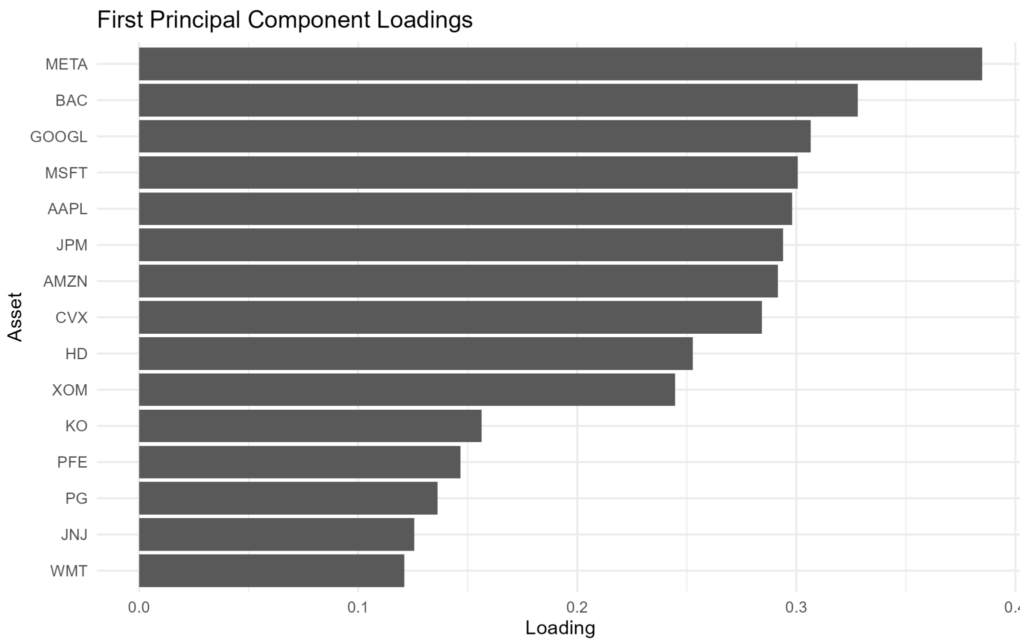

Figure 2.7: Loadings of the first principal component obtained from PCA, illustrating the assets that contribute most strongly to the dominant common risk factor in the portfolio universe.

Following the PCA analysis, hierarchical clustering was applied using the correlation-based distance metric described previously. The clustering procedure organizes assets according to statistical similarity, producing a hierarchical representation of the market structure.

The resulting dendrogram is presented in Figure 2.9.

The hierarchical organization becomes even more evident after reordering the correlation matrix according to the clustering structure, as shown in Figure 2.10.

Several economically intuitive clusters emerge naturally from the data. Technology firms tend to cluster together, energy companies form a separate branch, and defensive sectors exhibit their own hierarchical relationships.

Portfolio construction was then performed using the Hierarchical Risk Parity (HRP) allocation framework. For comparison purposes, an equally weighted portfolio was used as a benchmark throughout the analysis.

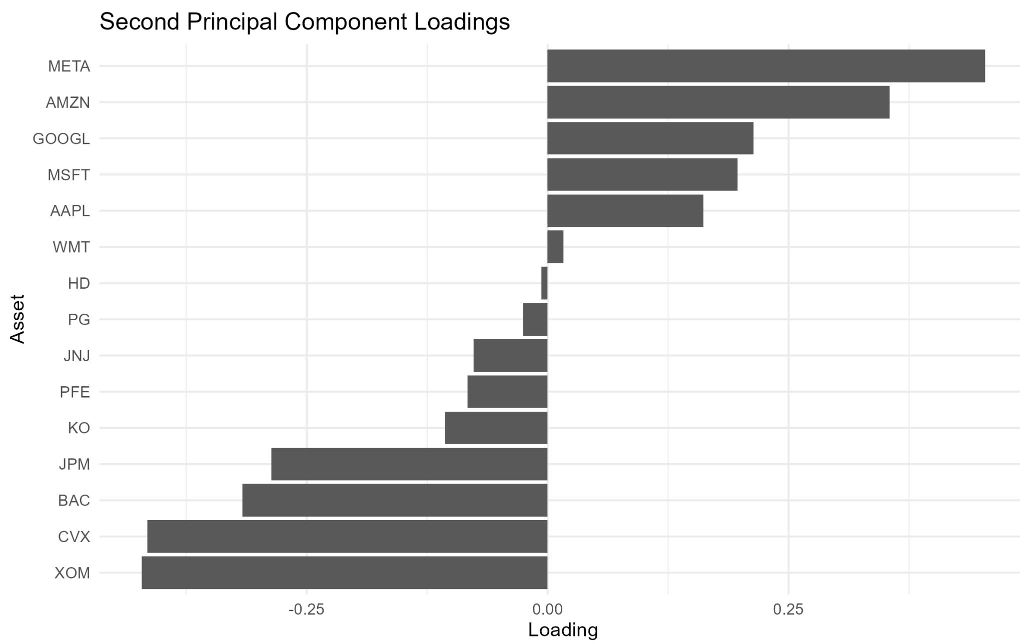

Figure 2.8: Loadings of the second principal component obtained from PCA, highlighting contrasting sectoral exposures and latent diversification patterns within the asset universe.

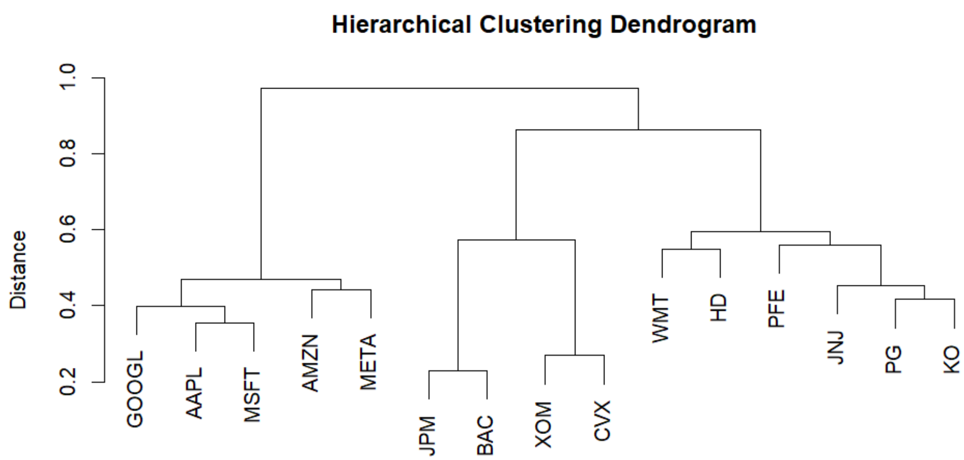

Figure 2.9: Hierarchical clustering dendrogram of the asset universe based on return correlations, revealing sectoral similarity structures used in the Hierarchical Risk Parity (HRP) allocation process.

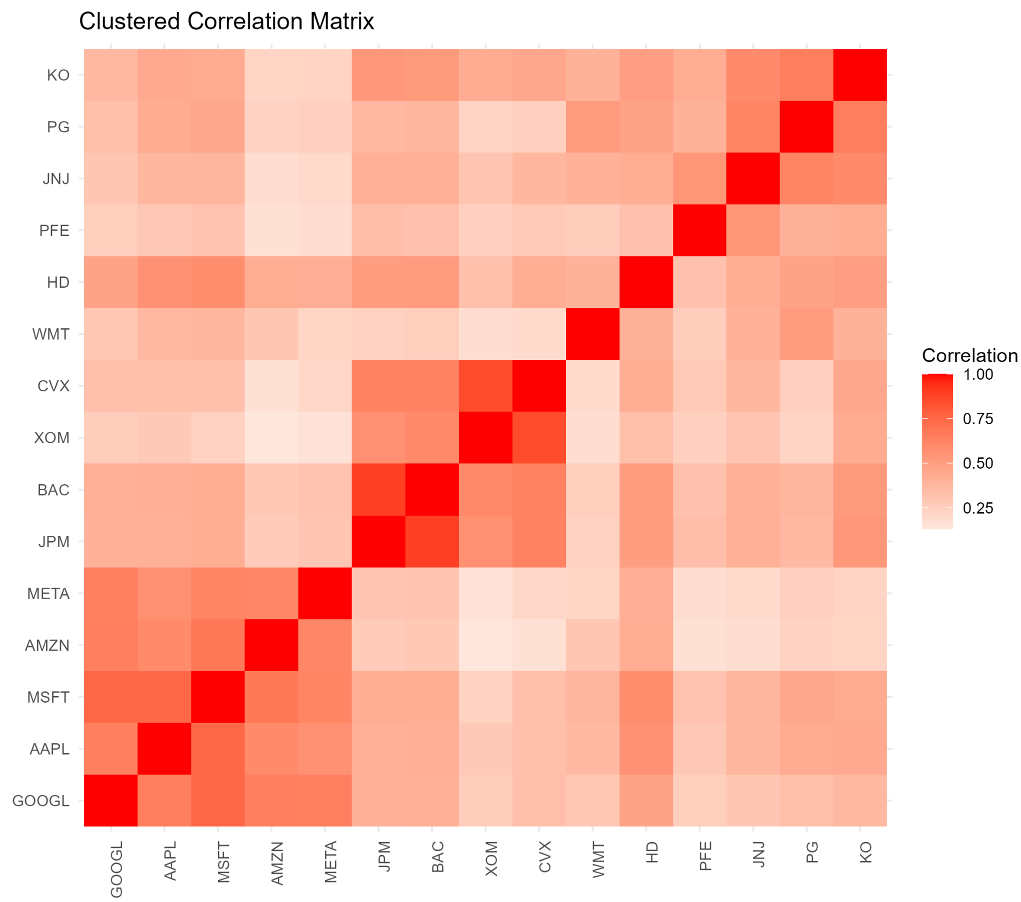

Figure 2.10: Clustered correlation matrix of asset returns, highlighting sectoral structure and similarity patterns used in the hierarchical portfolio construction process.

The equal-weight allocation is shown in Figure 2.11.

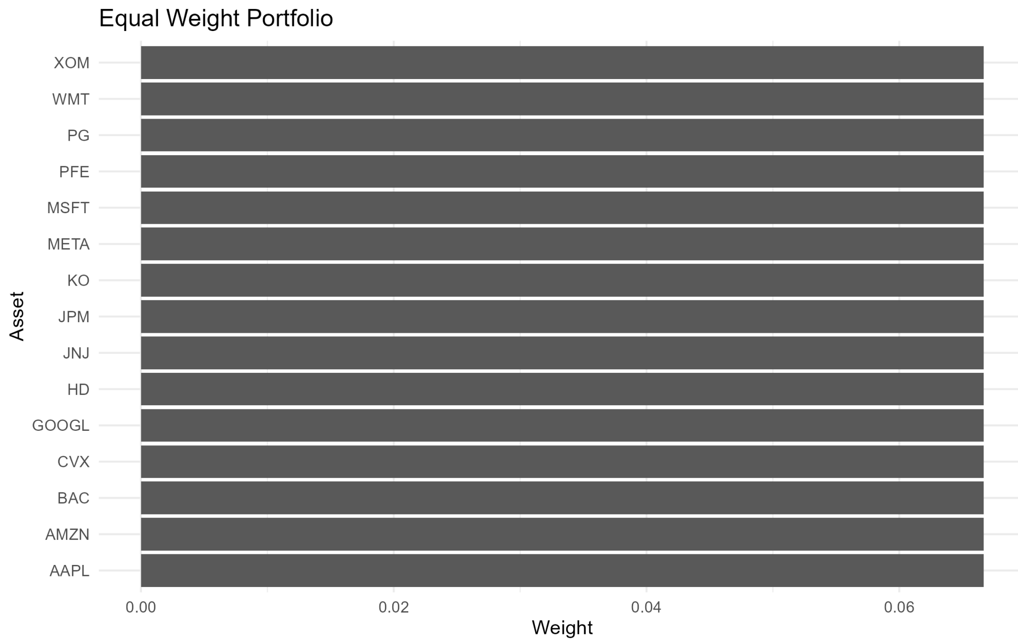

Figure 2.11: Equal-weight portfolio allocation assigning identical weights to all assets in the investment universe.

The HRP allocation produced a substantially different weighting structure. The largest portfolio weights were assigned to JNJ, KO, PG, and WMT, while exposure to several technology, financial, and energy assets was reduced relative to the equal-weight benchmark.

Table 2.3: Final portfolio weights obtained using the Hierarchical Risk Parity (HRP) allocation method.

Asset	Weight
JNJ	0.1285
KO	0.1171
PG	0.1158
WMT	0.1127
GOOGL	0.0705
HD	0.0689

Asset	Weight
PFE	0.0657
JPM	0.0471
MSFT	0.0457
AMZN	0.0439
AAPL	0.0424
BAC	0.0390
XOM	0.0386
CVX	0.0360
META	0.0282

The resulting allocation is visualized in Figure 2.12.

A direct comparison between the equal-weight benchmark and the HRP allocation is shown in Figure 2.13 and Table 2.4.

The differences between the two allocations are visible both at the individual asset level and across sectors. The larger weights assigned to JNJ, KO, PG, and WMT are broadly consistent with the lower volatility and correlation characteristics often associated with defensive sectors.

Table 2.4: Comparison between equal-weight portfolio allocation and Hierarchical Risk Parity (HRP) portfolio weights.

Asset	Equal Weight	HRP
AAPL	0.0667	0.0424
MSFT	0.0667	0.0457
GOOGL	0.0667	0.0705
AMZN	0.0667	0.0439
META	0.0667	0.0282
JPM	0.0667	0.0471
BAC	0.0667	0.0390
XOM	0.0667	0.0386
CVX	0.0667	0.0360
JNJ	0.0667	0.1285
PFE	0.0667	0.0657
PG	0.0667	0.1158

Asset	Equal Weight	HRP
KO	0.0667	0.1171
WMT	0.0667	0.1127
HD	0.0667	0.0689

Risk contributions were also evaluated in order to study how portfolio risk was distributed across assets under the HRP framework.

Table 2.5: Asset-level risk contributions under the Hierarchical Risk Parity (HRP) allocation framework.

Asset	Risk Contribution
KO	0.001158
JNJ	0.001111
PG	0.001090
GOOGL	0.001007
WMT	0.000947
HD	0.000925
JPM	0.000676
MSFT	0.000668
PFE	0.000630
BAC	0.000620
AAPL	0.000615
AMZN	0.000562
CVX	0.000489
XOM	0.000450
META	0.000443

The distribution of risk contributions is illustrated in Figure 2.14.

Portfolio performance was evaluated using cumulative returns, rolling annualized volatility, Sharpe ratio, and maximum drawdown. The cumulative return dynamics of both portfolio strategies are shown in Figure 2.15.

The temporal evolution of portfolio risk is shown in Figure 2.16.

Drawdown dynamics were analyzed separately in order to study the behaviour of both portfolios during periods of market stress.

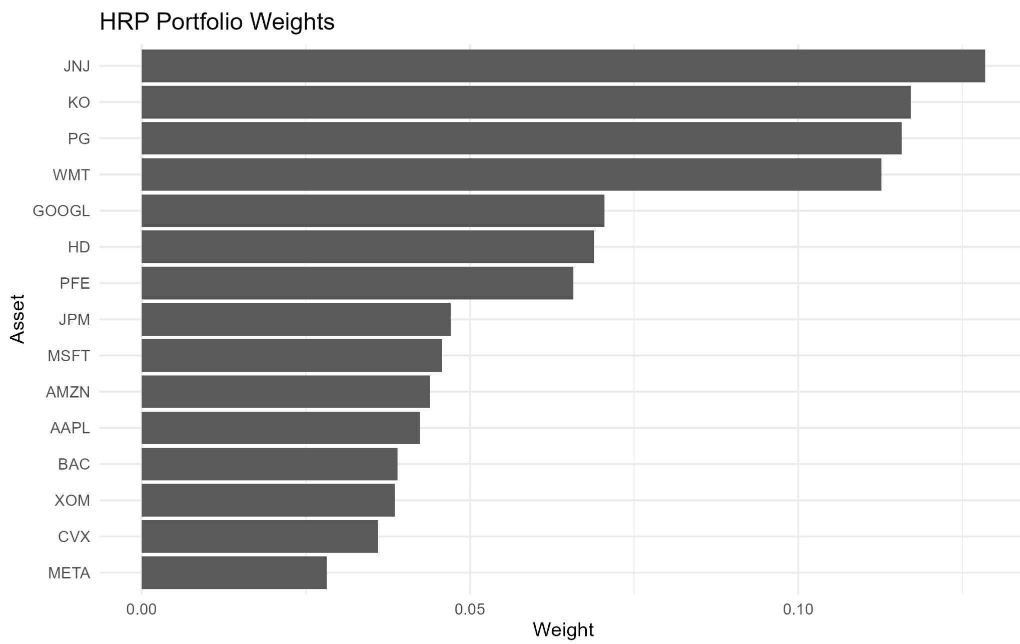

Figure 2.12: Portfolio weights obtained using the Hierarchical Risk Parity (HRP) allocation method, reflecting the hierarchical correlation structure and risk diversification across assets.

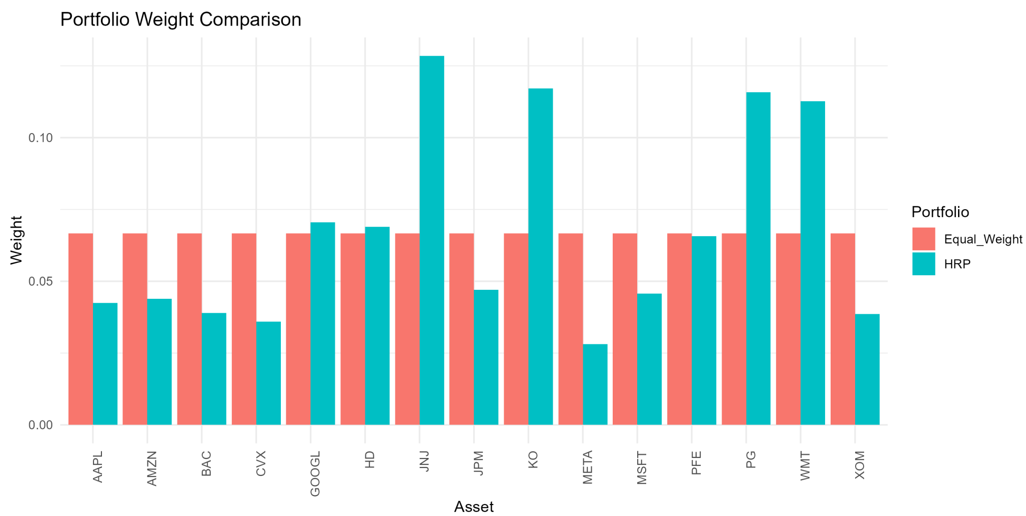

Figure 2.13: Comparison between equal-weight and Hierarchical Risk Parity (HRP) portfolio allocations, highlighting how the HRP framework redistributes exposure according to the hierarchical correlation and risk structure of the asset universe.

The final performance summary is presented in Table 2.6.

Table 2.6: Performance comparison between the Hierarchical Risk Parity (HRP) portfolio and the equal-weight benchmark portfolio.

Portfolio	Annualized Return	Annualized Volatility	Sharpe Ratio	Maximum Drawdown
HRP Portfolio	0.1754	0.2021	0.8680	-0.3139
Equal Weight Portfolio	0.1514	0.1808	0.8373	-0.2831

The HRP portfolio achieved a modest improvement in annualized return and Sharpe ratio relative to the equal-weight benchmark. However, this improvement was accompanied by a somewhat deeper maximum drawdown during the evaluation period, illustrating that diversification benefits do not necessarily translate into lower losses under all market conditions.

The resulting workflow combines PCA, hierarchical clustering, and risk-based

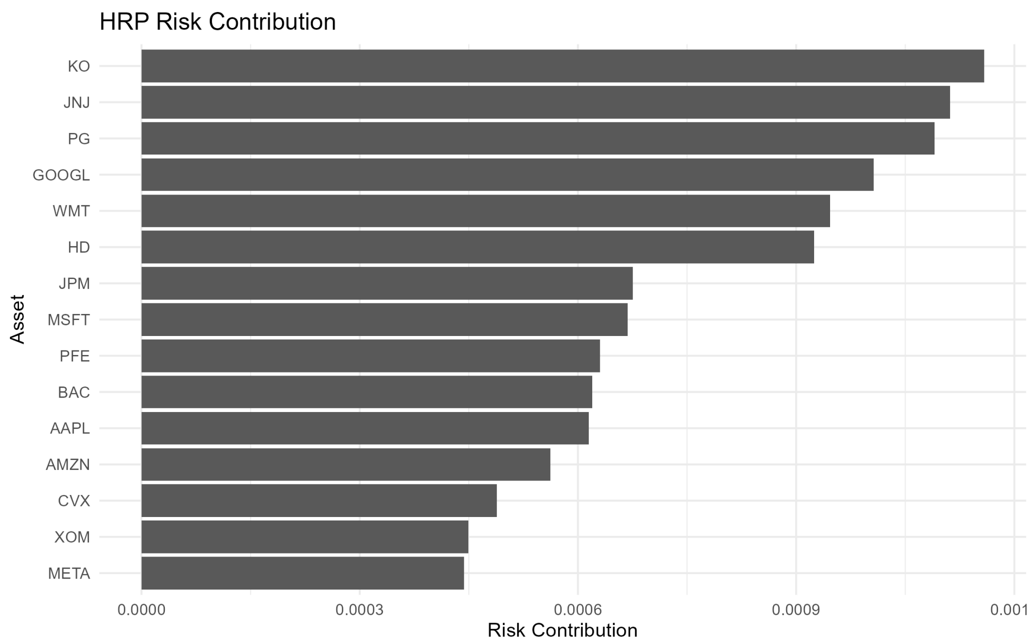

Figure 2.14: Risk contribution of each asset under the Hierarchical Risk Parity (HRP) allocation, illustrating how portfolio risk is distributed across the investment universe.

Figure 2.15: Cumulative returns of the compared portfolio strategies over the evaluation period, illustrating differences in long-term growth dynamics and drawdown behavior.

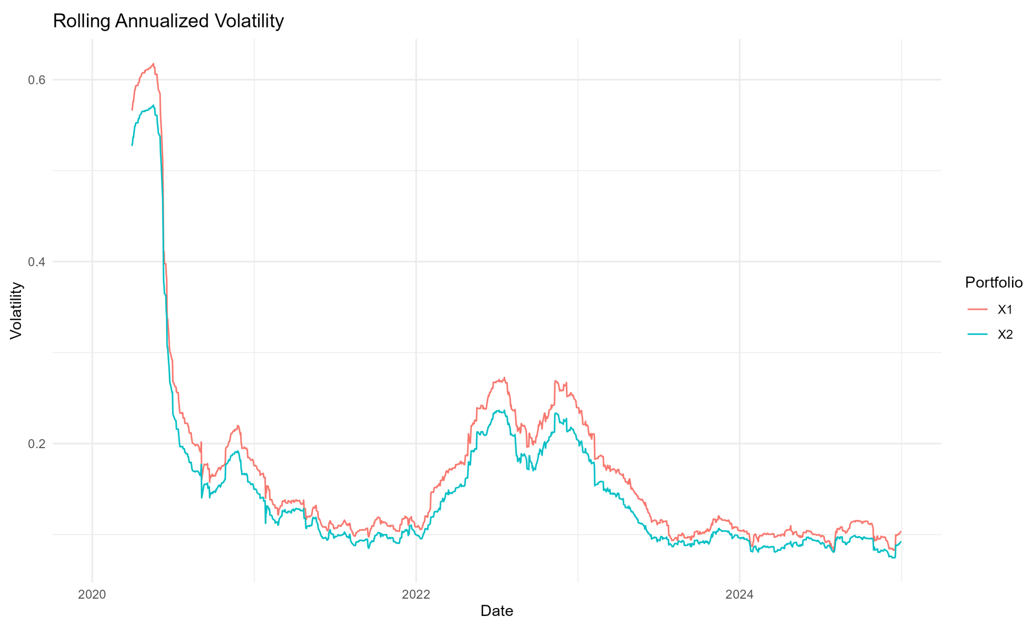

Figure 2.16: Rolling annualized volatility of the portfolio strategies over time, illustrating the dynamic evolution of portfolio risk under different allocation frameworks.

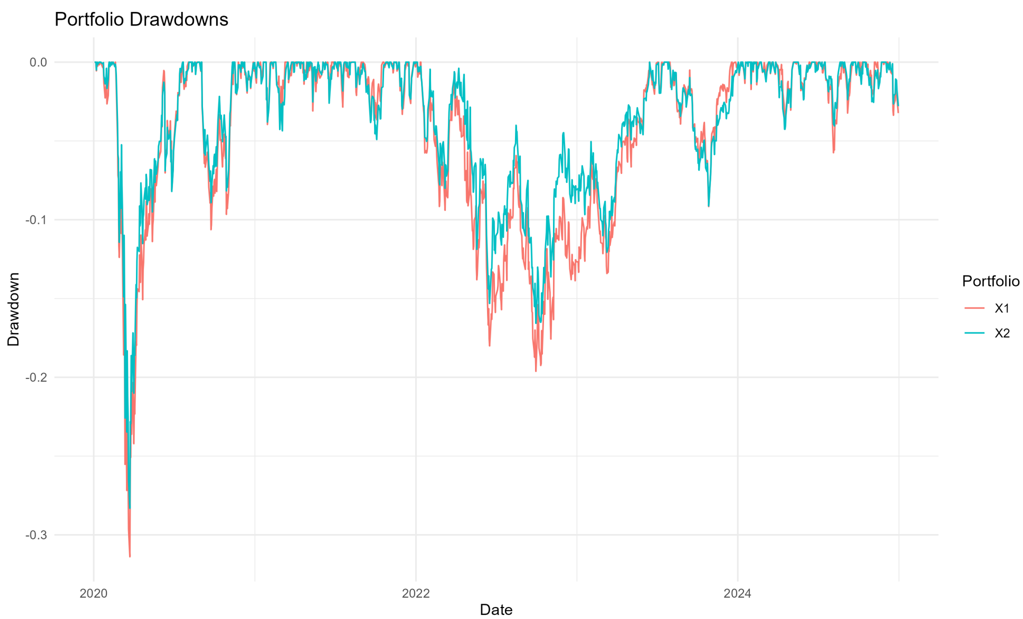

Figure 2.17: Portfolio drawdowns over the evaluation period, comparing the magnitude and persistence of temporary losses across the portfolio strategies.

portfolio allocation to study diversification from a structural rather than purely optimization-based perspective. Rather than focusing exclusively on return maximization, the analysis emphasizes robustness, interpretability, and stability under realistic market conditions.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

2.4 Results, Discussion, and Limitations

The results of the analysis suggest that the selected asset universe exhibits a considerably more structured organization of risk than might be inferred from asset labels or sector classifications alone. Although the portfolio contains companies operating across multiple industries, both the PCA decomposition and the hierarchical clustering analysis indicate that a relatively small number of latent factors account for a large proportion of observed market variability.

The PCA results provide the clearest evidence of this concentration. The first principal component alone explains approximately 46% of total variance, while the first two components account for roughly 64%. More than 80% of total variance is captured by the first five components. These findings suggest that much of the apparent diversity within the asset universe is driven by a limited set of common risk sources. From a portfolio construction perspective, this highlights an important distinction between holding many assets and achieving genuine diversification. Increasing the number of securities does not necessarily increase the number of independent sources of risk.

The principal component loadings further support this interpretation. The first component exhibits broad exposure across most assets and behaves consistently with a dominant market factor affecting the portfolio universe as a whole. Subsequent components reveal more localized patterns associated with sector-specific behaviour and relative differences between groups of assets. Together, these results illustrate how PCA can provide a compact representation of market structure while helping identify hidden concentrations of risk that may not be immediately visible in the original return space.

A complementary perspective emerges from the hierarchical clustering analysis. The dendrogram and clustered correlation matrix reveal economically intuitive groupings that arise directly from the data. Technology companies tend to cluster together, energy firms form a separate branch, and defensive sectors exhibit their own hierarchical relationships. Rather than imposing predefined classifications, the clustering procedure allows the statistical organization of the market to emerge from observed return behaviour. This provides a useful framework for understanding how diversification is distributed across the portfolio and where concentrations of risk may exist.

The portfolio construction stage extends these observations into an allocation framework. Compared with the equal-weight benchmark, the Hierarchical Risk Parity portfolio produces a noticeably different allocation structure. The largest weights were assigned to JNJ, KO, PG, and WMT, while exposure to several technology, financial, and energy assets was reduced. Importantly, this outcome should not be interpreted as a prediction that these assets will outperform in the future. Rather, it reflects their role within the correlation structure of the portfolio and the objective of distributing risk more evenly across the hierarchy identified during the clustering stage.

This distinction is central to the interpretation of the results. The primary contribution of HRP is not necessarily superior return generation, but a different approach to diversification. Traditional portfolio construction methods often allocate capital directly, whereas HRP attempts to allocate risk. As a consequence, two portfolios with similar levels of capital diversification may exhibit substantially different risk concentrations. The risk contribution analysis suggests that the HRP framework achieves a more balanced distribution of portfolio risk across assets, consistent with its underlying design objectives.

Performance comparisons between the HRP portfolio and the equal-weight benchmark reveal both strengths and trade-offs. The HRP portfolio achieved a modest improvement in annualized return and Sharpe ratio relative to the benchmark. At the same time, it experienced a somewhat deeper maximum drawdown during the evaluation period. These results illustrate an important practical point: improvements in one dimension of portfolio performance do not necessarily translate into improvements across all dimensions. Diversification, risk balancing, and return generation remain competing objectives, and no allocation methodology can fully eliminate these trade-offs.

The results should therefore be interpreted as evidence that unsupervised learning techniques can provide useful insights into market structure and portfolio construction rather than as proof of the superiority of a particular allocation strategy. The analysis demonstrates that PCA and hierarchical clustering are capable of uncovering meaningful relationships within the asset universe and that these relationships can be incorporated into a systematic allocation process. Whether this ultimately leads to superior long-term investment performance remains dependent on market conditions, investment horizons, transaction costs, and implementation choices.

Several limitations should also be acknowledged. First, the analysis was conducted using a relatively small universe of fifteen assets. While sufficient for illustrating the methodology, larger investment universes may exhibit more complex correlation structures and different clustering behaviour. Second, the study relies on a single historical sample period. Financial markets are inherently dynamic, and correlation structures evolve through time, particularly during periods of economic stress. Consequently, both PCA decompositions and clustering results may vary across market regimes.

A further limitation is that the analysis assumes frictionless portfolio implementation. Transaction costs, taxes, liquidity constraints, and rebalancing considerations were not incorporated into the evaluation. In practice, these factors can materially influence realized portfolio performance and may reduce some of the theoretical benefits of more sophisticated allocation methodologies.

Finally, although HRP was designed to improve robustness relative to covariance-sensitive optimization frameworks, it does not eliminate market risk or guarantee protection during severe market downturns. Common risk factors can still dominate portfolio behaviour when correlations rise across the market, reducing the effectiveness of diversification precisely when it is needed most.

Taken together, the results suggest that unsupervised learning techniques provide a valuable lens through which to study diversification and market structure. The main contribution of PCA and HRP in this context is not the pursuit of maximum historical return, but the ability to make sources of risk more explicit and to organize portfolio construction around the underlying structure of the market. From this perspective, the project illustrates how statistical learning methods can complement traditional portfolio theory by offering a more interpretable and structurally informed approach to asset allocation.

2.5 Technical Appendix (Optional)

This appendix provides supplementary mathematical and methodological details related to the unsupervised learning techniques used throughout the project. Additional background on PCA and clustering can be found in [9] and [15].

The objective is not to present a comprehensive treatment of the underlying methods, but rather to summarize the concepts most relevant to the portfolio construction process. The material presented here supports the interpretation of the results and clarifies some of the modeling choices discussed in the main text.

The appendix is organized into two sections. The first summarizes the principal concepts behind Principal Component Analysis (PCA), which was used to study the structure of asset return variability and identify dominant sources of common variation. The second discusses Hierarchical Clustering (HC), which was used to group assets according to their similarity patterns and support diversification decisions.

The main body of the project is intended to remain fully understandable without consulting this appendix. Readers interested primarily in the practical application and results may safely skip the material that follows.

2.5.1 Principal Component Analysis (PCA)

2.5.1.1 Centering of Returns

Let

$$X = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{np} \end{bmatrix} \tag{2.4}$$

denote the matrix of asset returns, where each column corresponds to an asset and each row represents an observation period.

Prior to applying PCA, the return series were centered by subtracting the sample mean of each asset:

$$x_{ij}^{(c)} = r_{ij} - \bar{r}_j \tag{2.5}$$

where $\bar{r}_j$ denotes the average return of asset j.

Centering removes differences in mean return levels and ensures that the subsequent analysis focuses exclusively on the covariance structure of the data. The centered return matrix was then used to construct the covariance matrix and perform the eigendecomposition underlying PCA.

2.5.1.2 Covariance Matrix

The covariance matrix summarizes the dependence structure among asset returns and is given by

$$\Sigma = \frac{1}{n-1} X_c^\top X_c \tag{2.6}$$

where X_c denotes the centered return matrix and n is the number of observations.

Each element σ_{ij} measures the extent to which the returns of assets i and j vary together. Positive values indicate that returns tend to move in the same direction, whereas negative values indicate opposite movements.

Because PCA is based on the covariance structure of the data, the matrix Σ serves as the primary object of analysis. The principal components are obtained by identifying the directions that explain the largest share of the variability captured by this matrix.

2.5.1.3 Eigenvalue Decomposition

PCA is based on the eigendecomposition of the covariance matrix,

$$\Sigma = V \Lambda V^{\top} \tag{2.7}$$

where V is the matrix of eigenvectors and Λ is a diagonal matrix containing the corresponding eigenvalues.

Each eigenvector defines a principal component, while the associated eigenvalue measures the amount of variance explained by that component.

The principal components are mutually orthogonal and provide a new coordinate system in which the dominant patterns of variation become easier to identify. Components associated with larger eigenvalues explain a greater share of total variance and therefore contain more information about the structure of asset returns.

In this project, the eigendecomposition of the covariance matrix provided the basis for evaluating how much of the variability in the investment universe could be explained by a reduced number of components.

2.5.1.4 Explained Variance

The eigenvalues obtained from the covariance matrix quantify the amount of variance explained by each principal component. Let λ_i denote the eigenvalue associated with the i-th principal component. The proportion of variance explained by that component is given by

$$\mathrm{PVE}_i = \frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j} \tag{2.8}$$

where p denotes the total number of assets.

The cumulative proportion of explained variance can be expressed as

$$\mathrm{CPVE}_k = \sum_{i=1}^{k} \mathrm{PVE}_i \tag{2.9}$$

representing the fraction of total variance retained when only the first k principal components are considered.

The explained variance analysis indicated a substantial concentration of variability in a small number of components. The first principal component accounted for approximately 46% of total variance, while the first two components explained roughly 64%. The first five components captured more than 80% of the variability observed across the investment universe.

These results suggest that a relatively small number of latent factors drive a large fraction of asset return behavior. Consequently, the effective dimensionality of the dataset is considerably smaller than the number of assets included in the analysis.

2.5.1.5 Principal Component Loadings

While explained variance quantifies the importance of each principal component, the component loadings describe how individual assets contribute to those components.

Let

$$v_k = \begin{bmatrix} v_{1k} \\ v_{2k} \\ \vdots \\ v_{pk} \end{bmatrix} \tag{2.10}$$

denote the eigenvector associated with the k-th principal component. The elements of this vector are commonly referred to as loadings.

Assets with large absolute loadings contribute more strongly to the corresponding component, whereas assets with loadings closer to zero have a smaller influence on that source of variation.

The loadings associated with the first two principal components were examined to identify the assets contributing most strongly to the dominant patterns of return variability. Because principal components are linear combinations of the original assets, the loading structure provides insight into how different securities participate in the latent factors revealed by the PCA decomposition.

Together with the explained variance analysis, the loadings help characterize the underlying structure of the investment universe and identify groups of assets that exhibit similar exposures to common sources of variation.

2.5.1.6 Interpretation for Asset Allocation

The PCA results indicate that a substantial portion of total variance can be explained by a relatively small number of principal components. From a portfolio construction perspective, this suggests that many assets are influenced by common underlying factors rather than behaving independently.

As a consequence, the number of securities held in a portfolio may overstate the true degree of diversification achieved. Assets exposed to the same dominant components can contribute similar sources of risk even when they belong to different sectors or industries.

For this reason, PCA can be viewed as a diagnostic tool for assessing the structure of risk within an investment universe. Rather than focusing on individual assets, the analysis emphasizes the latent factors that drive collective movements in returns.

In this project, PCA was used primarily as an exploratory technique to investigate the dimensionality of the dataset and to identify common sources of variation prior to the clustering analysis described in the following section.

2.5.2 Hierarchical Clustering (HC)

2.5.2.1 Correlation-Based Distance

The clustering analysis was based on the correlation structure of asset returns. Let ρ_{ij} denote the Pearson correlation coefficient between assets i and j. The corresponding distance measure is defined as

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}} \tag{2.11}$$

This transformation converts correlations into distances while preserving the properties required by a distance metric.

Several useful relationships follow directly from this definition:

- If $\rho_{ij} = 1$, then $d_{ij} = 0$.
- If $\rho_{ij} = 0$, then $d_{ij} = \sqrt{1/2}$.
- If $\rho_{ij} = -1$, then $d_{ij} = 1$.

Assets exhibiting similar return dynamics therefore appear close to one another in the resulting distance matrix, whereas assets with dissimilar behavior appear farther apart.

The correlation-based distance matrix served as the primary input to the hierarchical clustering procedure implemented in this project.

2.5.2.2 Agglomerative Clustering

Hierarchical clustering was performed using an agglomerative approach.

The procedure begins by assigning each asset to its own cluster. At each iteration, the two clusters exhibiting the smallest distance are merged into a new cluster. This process continues until all assets have been combined into a single hierarchical structure.

If the investment universe contains p assets, the algorithm initially creates p individual clusters. Each merge reduces the number of clusters by one until the hierarchy is complete.

The resulting sequence of merges reveals how assets are related at different levels of similarity. Assets that merge early in the process tend to exhibit highly similar return patterns, whereas assets that merge only at later stages are generally less related.

2.5.2.3 Ward's Linkage Method

After computing the distance matrix, the clustering algorithm must determine how distances between clusters are evaluated following each merge. This choice is known as the linkage criterion.

In this project, Ward's linkage method was employed through the Ward.D2 algorithm.

Ward's method seeks to minimize the increase in within-cluster variance produced by each merge. At every iteration, the algorithm selects the pair of clusters whose combination results in the smallest increase in overall cluster heterogeneity.

This approach tends to generate compact and relatively balanced clusters, making it particularly useful for identifying coherent groups of assets exhibiting similar return behavior.

2.5.2.4 Dendrogram Interpretation

The hierarchical structure produced by the clustering algorithm can be visualized through a dendrogram.

A dendrogram is a tree-like representation that displays the sequence of cluster merges performed during the agglomerative procedure. Each terminal node corresponds to an individual asset, while internal nodes represent cluster combinations formed throughout the algorithm.

The vertical axis represents the distance at which clusters are merged. Assets connected at lower heights tend to exhibit stronger similarity, whereas assets that merge only at higher levels are generally less related.

The dendrogram presented in the main body of the project reveals the hierarchical organization of the investment universe and highlights groups of assets exhibiting similar return behavior.

2.5.2.5 Clustered Correlation Matrix

The ordering produced by hierarchical clustering can be used to reorganize the correlation matrix according to the cluster structure identified in the data.

Rather than displaying assets in an arbitrary order, the reordered matrix places statistically similar assets next to one another. As a result, groups of highly correlated assets appear as visible blocks along the diagonal of the matrix.

This representation provides a convenient way to inspect the dependency structure of the investment universe and to identify groups of assets sharing similar return dynamics.

In this project, the clustered correlation matrix complemented the dendrogram by providing an alternative visualization of the relationships among assets.

2.5.3 Diversification Through Clustering

The clustering results indicate that similarities among assets are not distributed uniformly throughout the investment universe. Instead, groups of assets exhibiting comparable return behavior emerge naturally from the data.

From a portfolio construction perspective, these clusters can reveal hidden concentrations of risk. A portfolio may contain a large number of securities while remaining exposed to a relatively small set of common drivers if many holdings belong to the same cluster.

Hierarchical clustering provides a complementary perspective on diversification by focusing on relationships among assets rather than on individual securities. Assets belonging to different clusters are more likely to contribute distinct sources of risk and therefore may improve portfolio diversification.

Together, PCA and hierarchical clustering provide complementary views of the structure of asset returns. While PCA identifies dominant sources of variation, clustering reveals how assets are organized according to similarity patterns within the investment universe.

2.5.3.1 Methodological Limitations

PCA captures linear relationships among asset returns and may not fully represent nonlinear dependency structures present in financial markets.

Similarly, the clustering structure obtained depends on both the distance metric and the linkage criterion employed. Alternative specifications may produce different cluster configurations and lead to different interpretations of the investment universe.

Despite these limitations, PCA and hierarchical clustering remain useful exploratory tools for understanding the structure of asset returns and supporting diversification analysis.

Chapter 3

Option Pricing with Path Integrals

Most option pricing models focus on the value of a contract at maturity. Path integral methods approach the same problem from a different perspective: instead of analysing a single outcome, they evaluate the contribution of all possible price trajectories that could lead to it. This viewpoint is particularly appealing for derivatives whose value depends not only on the final price of the underlying asset, but also on the path followed to reach it.

While the Black–Scholes framework remains one of the most influential models in quantitative finance, many derivatives require numerical techniques because closed-form solutions are either unavailable or difficult to obtain. Path integral methods provide an alternative formulation that naturally expresses option values as weighted averages over entire trajectories, creating a direct connection between option pricing and simulation-based methods.

This project investigates whether a path integral formulation can serve as a practical computational framework for option pricing. The analysis focuses on European call options, where analytical Black–Scholes prices are available and can be used as a benchmark. A path integral representation of the pricing problem is derived, implemented numerically using the Metropolis–Hastings algorithm, and evaluated against both the analytical Black–Scholes solution and conventional Monte Carlo estimates. Historical market data is incorporated to illustrate how the methodology can be applied in a realistic setting.

3.1 Problem Definition

For a standard European option, the Black–Scholes model provides an elegant closed-form solution under a set of simplifying assumptions. The situation becomes more challenging when dealing with contracts that involve early exercise features, path dependence, or more complex market dynamics. In these cases, analytical solutions may no longer exist, and numerical methods become essential.

Most numerical approaches are built around either solving a pricing equation or approximating the evolution of the underlying asset through discrete structures such as lattices. Path integral methods offer a different perspective. Rather than focusing on a single terminal outcome, they formulate the valuation problem as an aggregation over all possible trajectories that the underlying asset could follow between the present and maturity.

This trajectory-based viewpoint is particularly attractive for derivatives whose value depends on the history of the underlying asset rather than on its final price alone. It also provides a natural bridge between option pricing and simulation-based techniques, allowing the valuation problem to be reformulated in terms of sampling and aggregating large ensembles of possible price paths.

The central question explored in this project is whether this alternative formulation can provide a practical foundation for option pricing while remaining consistent with established financial theory. The goal is not to outperform Black–Scholes where an analytical solution already exists, but to examine whether path integrals can serve as a flexible framework for pricing problems where closed-form solutions are no longer available.

3.2 Modeling Approach

The starting point of the model is the same stochastic process used in the Black–Scholes framework. The underlying asset is assumed to follow a geometric Brownian motion, allowing the option pricing problem to be expressed in terms of the probability distribution of future price trajectories.

Rather than working directly with the Black–Scholes partial differential equation, the pricing problem is reformulated using a path integral representation. In this formulation, the value of an option is obtained by aggregating the contribution of all admissible trajectories connecting the current asset price to a future state at maturity. Each trajectory is assigned a statistical weight determined by the underlying stochastic process and contributes to the expected discounted payoff.

To make the problem computationally tractable, the continuous time horizon is discretized into a finite number of intervals. This transforms the original path integral into a multidimensional integration problem whose dimensionality grows with the number of time steps used to represent the trajectory. The resulting formulation provides a numerical approximation of the Black–Scholes propagator while preserving the probabilistic interpretation of the original model.

Evaluating these multidimensional integrals directly quickly becomes impractical as the number of dimensions increases. For this reason, the numerical estimation is performed using Markov Chain Monte Carlo techniques. Among the available sampling methods, the Metropolis–Hastings algorithm was selected because it allows representative trajectories to be generated efficiently from the target distribution without requiring exhaustive exploration of the integration domain.

The analysis focuses on European call options, where analytical Black–Scholes prices are available and can be used as a benchmark. This provides a controlled setting in which the numerical estimates obtained from the path integral formulation can be compared against a well-established analytical solution before considering more complex derivatives.

The overall modeling strategy combines three components: a geometric Brownian motion describing the evolution of the underlying asset, a path integral formulation that expresses option values as weighted averages over entire price trajectories, and a Markov Chain Monte Carlo algorithm used to estimate those values numerically.

3.3 Methodology and Implementation

The objective of this project is not to compete with established option pricing methods, but to explore how ideas originating from the path integral formulation can be translated into a practical numerical workflow. To keep the analysis transparent, the implementation is organized around two experiments. The first uses a controlled setting in which the analytical solution is known. The second applies the same workflow to observed market data.

3.3.1 Controlled Experiment

The analysis begins with a European call option under the Black-Scholes assumptions. The parameters used throughout the baseline experiment are summarized in Table 3.1.

Table 3.1: Model parameters used in the baseline pricing experiment.

Parameter	Value
Initial Asset Price	100.0000
Strike Price	100.0000
Time to Maturity	1.0000
Risk-Free Rate	0.0500
Volatility	0.2000

Under these assumptions, the Black-Scholes formula provides an exact analytical price. This creates a useful benchmark against which numerical estimators can be evaluated.

Although the path integral formulation is naturally expressed in terms of entire asset price trajectories, a European call option depends only on the terminal asset price. For this reason, the implementation does not attempt to sample complete paths through Markov Chain Monte Carlo. Instead, the problem is simplified by sampling from the terminal log-price distribution implied by the risk-neutral geometric Brownian motion model. This reduction preserves the essential pricing problem while keeping the numerical implementation manageable.

Simulated price paths are nevertheless generated to illustrate the stochastic dynamics underlying the model and to provide a visual representation of the path-based perspective motivating the analysis.

Three pricing approaches are then compared. The Black-Scholes formula serves as the analytical benchmark, standard Monte Carlo simulation provides a conventional numerical estimator, and a Metropolis-Hastings sampler is used to generate samples from the terminal risk-neutral distribution.

The resulting estimates are reported in Table 3.2 and visualized in Figure 3.2.

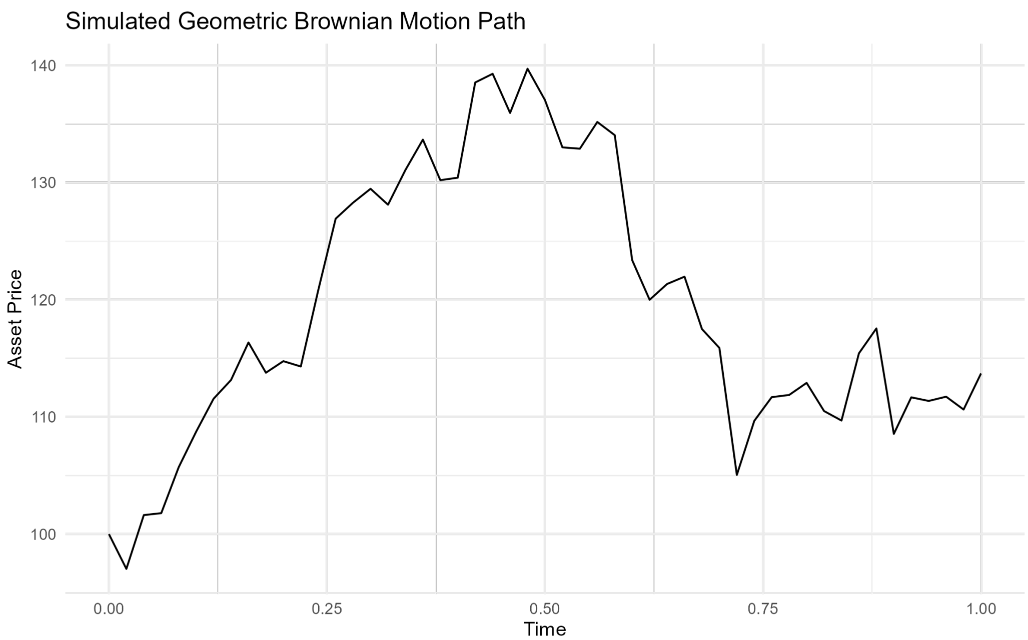

Figure 3.1: A subset of simulated asset-price trajectories under the risk-neutral geometric Brownian motion model. Although the final implementation estimates option values from the terminal distribution, the simulated paths provide a visual representation of the path-based framework motivating the analysis.

Table 3.2: Comparison of option prices obtained using Black-Scholes, Monte Carlo simulation, and Metropolis-Hastings sampling.

Method	Option Price	Absolute Error
Black-Scholes	10.4506	0.0000
Monte Carlo	10.3896	0.0610
Metropolis-Hastings	10.5800	0.1295

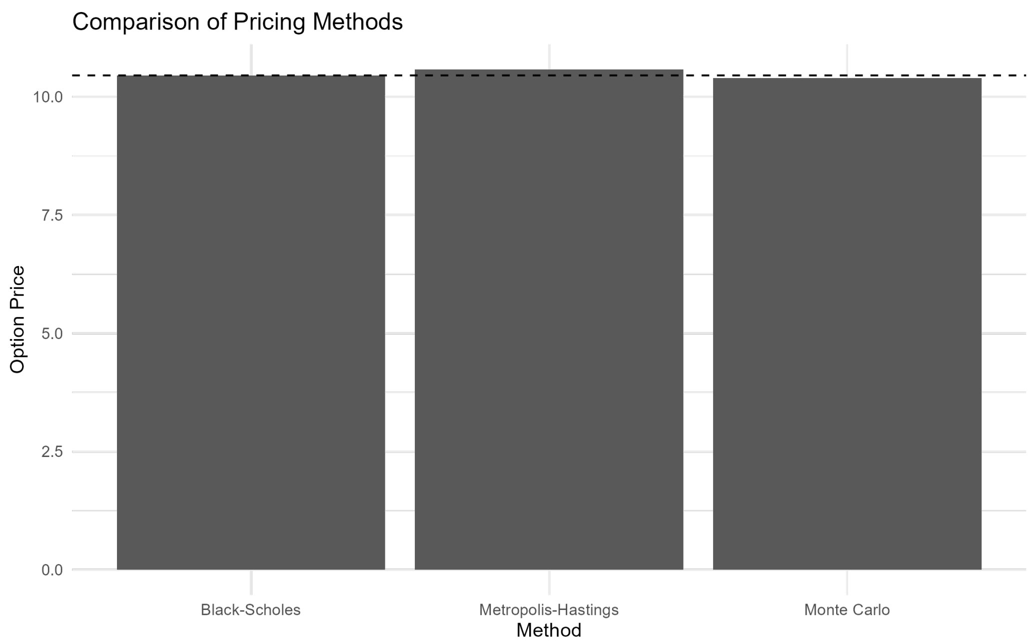

Figure 3.2: Comparison of option prices obtained using analytical and simulation-based methods.

All three methods produce estimates close to the analytical benchmark. In this simple setting, standard Monte Carlo achieves the smallest numerical error. This result is not surprising. For a European call option, direct Monte Carlo sampling is already an efficient estimator, and the additional complexity introduced by Markov Chain Monte Carlo does not provide a clear advantage.

To better understand the behavior of the numerical estimators, the experiment is repeated for different sample sizes. The results are shown in Table 3.3 and Figure 3.3.

Table 3.3: Convergence behavior of Monte Carlo and Metropolis-Hastings estimators as the number of samples increases.

Samples	Monte Carlo	Metropolis-Hastings	Black-Scholes
100	9.2822	8.3345	10.4506
500	11.4302	9.4567	10.4506
1000	10.0559	11.6238	10.4506
5000	10.3909	9.8939	10.4506
10000	10.4008	10.8390	10.4506

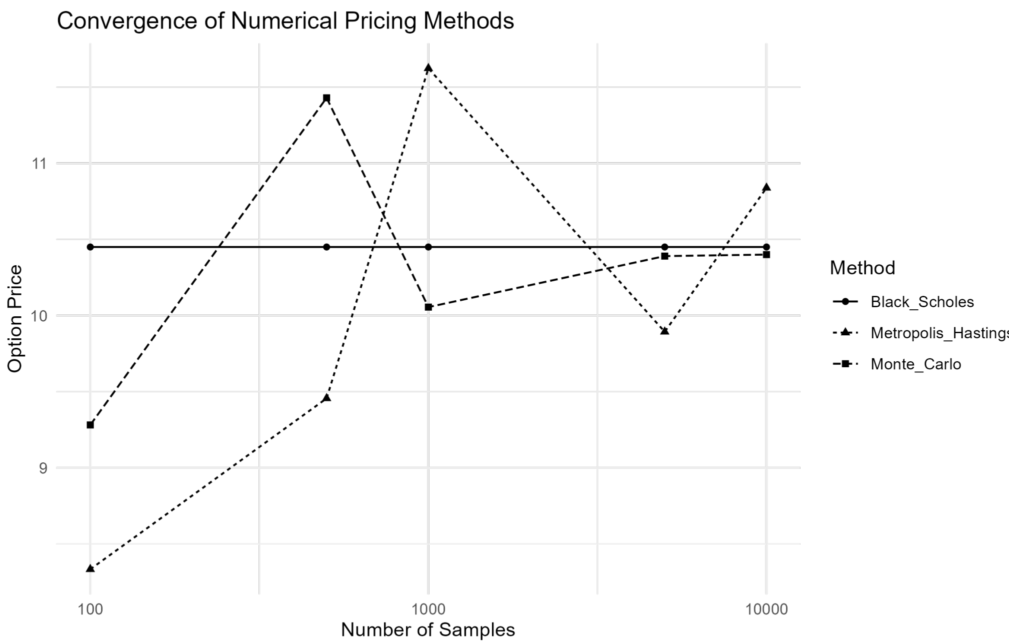

Figure 3.3: Convergence of Monte Carlo and Metropolis-Hastings estimates toward the Black-Scholes benchmark as the sample size increases.

Both estimators move toward the analytical solution as the sample size increases. The Monte Carlo estimator exhibits a more stable convergence pattern, while the Metropolis-Hastings estimates show larger fluctuations at smaller sample sizes. This behavior is consistent with the autocorrelation introduced by Markov chain sampling and illustrates one of the practical trade-offs associated with MCMC-based methods.

The purpose of this comparison is not to demonstrate that Metropolis-Hastings outperforms Monte Carlo in a problem where the analytical solution is already available. Rather, it provides a controlled environment for validating the implementation and understanding the numerical behavior of the estimator before considering more complex derivatives.

3.3.2 Market Data Example

After validating the workflow in a controlled setting, the same methodology is applied using observed market data. Historical prices for AAPL are downloaded from Yahoo Finance and used to estimate a spot price and historical volatility.

Figure 3.4: Historical AAPL price series used in the market-data example.

The resulting inputs are summarized in Table 3.4.

Table 3.4: Market-derived inputs obtained from historical AAPL data.

Parameter	Value
Ticker	AAPL

Parameter	Value
Spot Price	190.3751
Strike Price	190.3751
Time to Maturity	1
Risk-Free Rate	0.05
Historical Volatility	0.1992

The objective of this second experiment is not to construct a tradable option pricing model or a production-grade calibration. Instead, it demonstrates that the same workflow can be applied using observed financial data rather than synthetic inputs.

Pricing results obtained from the three methods are reported in Table 3.5 and visualized in Figure 3.6.

Table 3.5: Option pricing results using market-derived inputs from AAPL historical data.

Method	Option Price	Absolute Error
Black-Scholes	19.8397	0.0000
Monte Carlo	19.6705	0.1693
Metropolis-Hastings	19.5573	0.2824

The overall conclusions are consistent with those obtained in the controlled experiment. The numerical estimates remain close to the Black-Scholes benchmark, while standard Monte Carlo continues to provide the most accurate approximation among the simulation-based methods considered. At the same time, the exercise demonstrates that the implementation can be applied without modification once the required market inputs have been estimated.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

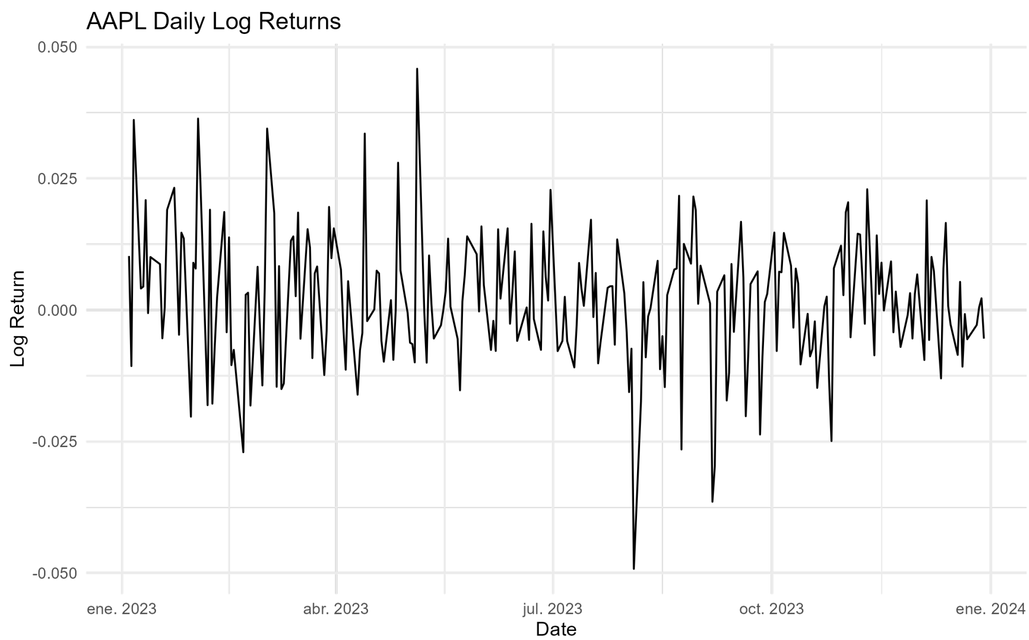

Figure 3.5: Historical log returns used to estimate the volatility parameter employed in the pricing exercise.

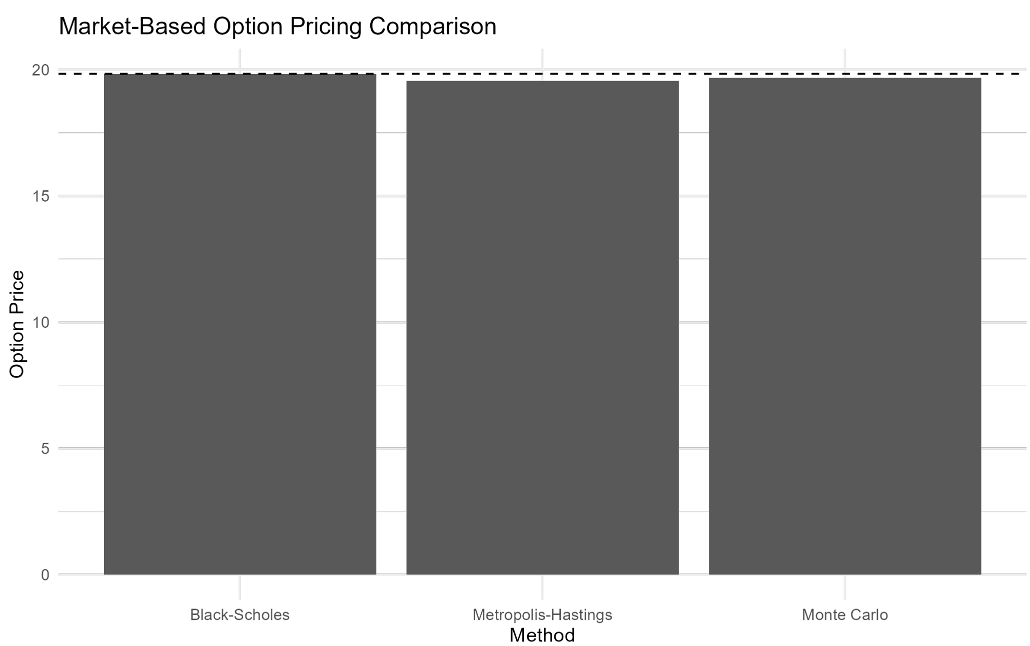

Figure 3.6: Comparison of pricing estimates obtained using market-derived inputs from AAPL historical data.

3.4 Results, Discussion, and Limitations

The results show that all three pricing approaches considered in this project produce estimates that are reasonably close to one another. In both the controlled experiment and the market-data example, the numerical estimators converge toward values consistent with the Black-Scholes benchmark, indicating that the implementation captures the essential characteristics of the pricing problem.

For the baseline European call option, standard Monte Carlo simulation produced the most accurate numerical approximation among the simulation-based methods. The Metropolis-Hastings estimator also generated reasonable price estimates, but exhibited larger deviations from the analytical solution and greater variability during the convergence analysis.

This outcome is not particularly surprising. The problem considered here is intentionally simple. Under the Black-Scholes assumptions, the terminal distribution of the asset price is known, making direct Monte Carlo simulation both straightforward and efficient. In this setting, introducing a Markov Chain Monte Carlo procedure adds computational complexity without providing a corresponding improvement in accuracy.

From a practical perspective, this observation is one of the most important findings of the project. The results suggest that Metropolis-Hastings should not be viewed as a replacement for standard Monte Carlo methods in simple European option pricing problems. When direct sampling is available, conventional Monte Carlo remains the more natural and efficient choice.

The convergence analysis reinforces this conclusion. As the number of samples increases, both estimators move toward the analytical benchmark. However, the Monte Carlo estimator exhibits a smoother convergence pattern, while the Metropolis-Hastings estimates fluctuate more noticeably at smaller sample sizes. This behavior is consistent with the autocorrelation inherent to Markov chain sampling and highlights one of the practical trade-offs associated with MCMC-based methods. Although Markov chains can explore complex probability distributions, successive samples are not independent, which can reduce sampling efficiency.

At the same time, the fact that Metropolis-Hastings does not outperform Monte Carlo in this experiment does not imply that MCMC methods are uninteresting for option pricing. The primary motivation for studying these techniques arises in situations where direct sampling becomes difficult or inefficient. Many derivatives encountered in practice depend not only on the terminal asset price but on the entire trajectory followed by the underlying asset. In such cases, the dimensionality of the problem increases substantially, and sampling-based approaches inspired by path integrals may become more attractive.

The market-data example provides an additional perspective on the methodology. Using historical AAPL prices, the same workflow produced pricing estimates that remained close to the analytical benchmark. While this exercise should not be interpreted as a production-grade option valuation framework, it demonstrates that the implementation can be applied directly to observed market data once the required inputs have been estimated. The transition from synthetic examples to real financial data required no modifications to the underlying pricing algorithms, which is an encouraging indication of the flexibility of the approach.

Several limitations should be acknowledged. First, the project focuses exclusively on European call options under the Black-Scholes assumptions. These assumptions include constant volatility, a constant risk-free interest rate, frictionless markets, and geometric Brownian motion dynamics. While these simplifications make the problem analytically tractable, they do not fully capture the behavior of real financial markets.

Second, the implementation does not perform a full path-integral simulation. The theoretical discussion motivating the project is expressed in terms of asset price trajectories, but the numerical implementation exploits the fact that European option payoffs depend only on the terminal asset price. As a result, Metropolis-Hastings is applied to the terminal log-price distribution rather than to complete asset paths. This simplification is appropriate for the problem studied here, but it leaves open the question of how the methodology would behave when applied to genuinely path-dependent derivatives.

Finally, the market-data example relies on historical volatility estimated from past returns. In practice, option pricing often incorporates additional sources of information such as implied volatility surfaces, stochastic volatility models, local volatility models, or alternative asset dynamics. Extending the framework to accommodate these features would require a more sophisticated calibration procedure than the one considered in this project.

Despite these limitations, the project successfully demonstrates that concepts originating from the path integral formulation can be translated into a practical computational workflow. For simple European options, the results confirm that standard Monte Carlo simulation remains the most efficient numerical approach among those considered. However, the implementation also provides a foundation for exploring more complex derivatives, where path-dependent payoffs and higher-dimensional probability distributions may create conditions under which MCMC-based techniques become more compelling.

3.5 Technical Appendix (Optional)

The main body of this project focuses on the practical implementation and evaluation of path-integral-inspired option pricing methods. The purpose of this appendix is to provide additional mathematical and computational details supporting the numerical results presented throughout the chapter.

The material is organized around four topics. Appendix A reviews the Black-Scholes framework underlying the pricing model. Appendix B introduces the Monte Carlo estimator used to approximate risk-neutral expectations. Appendix C summarizes the Metropolis-Hastings algorithm employed in the numerical experiments. Finally, Appendix D discusses the path integral perspective motivating the project and its connection to more general path-dependent pricing problems.

The appendix is intended as a technical reference rather than a prerequisite. Readers interested primarily in the implementation and results can safely skip this section without affecting the overall understanding of the project.

3.5.1 Appendix A. Black-Scholes Framework

The numerical experiments presented in this project are built upon the Black-Scholes framework developed by Black and Scholes and later extended by Merton [16] [17]. This appendix summarizes the mathematical assumptions underlying the model and connects them to the numerical implementation.

The starting point is the assumption that the underlying asset follows a geometric Brownian motion. The asset price dynamics are described by

$$dS_t = \mu S_t, dt + \sigma S_t, dW_t \tag{3.1}$$

where S_t denotes the asset price at time t, μ is the drift parameter, σ is the volatility, and W_t represents a standard Wiener process.

Equation 3.1 states that the evolution of the asset price contains both a deterministic component and a random component. The deterministic term controls the average growth of the process, while the stochastic term introduces uncertainty into future prices and generates the dispersion observed in the simulated trajectories presented in the main body of the project.

Applying Itô's lemma to the logarithm of the asset price yields a stochastic process with a closed-form solution. The terminal asset price can then be written as

$$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}, Z\right] \tag{3.2}$$

where $Z \sim N(0,1)$ is a standard normal random variable.

Equation 3.2 implies that the logarithm of the terminal asset price is normally distributed,

$$\log(S_T) \sim N\left(\log(S_0) + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right) \tag{3.3}$$

which means that the terminal asset price itself follows a lognormal distribution.

This result plays a central role throughout the project. The Monte Carlo simulations generate terminal prices using Equation 3.2, while the Metropolis-Hastings implementation samples from the distribution described by Equation 3.3. As discussed in the main body of the project, the payoff of a European call option depends only on the terminal asset price, making this distribution the key object of interest.

Under the risk-neutral valuation framework, the expected growth rate of the asset is replaced by the risk-free interest rate r. The value of a European call option can then be expressed as

$$C = e^{-rT}\mathbb{E}\left[\max(S_T - K, 0)\right] \tag{3.4}$$

where C denotes the option value and K is the strike price.

Evaluating Equation 3.4 under the lognormal distribution of S_T leads to the Black-Scholes formula

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \tag{3.5}$$

with

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{3.6}$$

and

$$d_2 = d_1 - \sigma\sqrt{T} \tag{3.7}$$

Here, $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

For the purposes of this project, the availability of an analytical solution is particularly valuable. It provides a reference against which the Monte Carlo and Metropolis-Hastings estimators can be evaluated, allowing numerical errors and convergence behavior to be studied in a controlled setting. The Black-Scholes model therefore serves not only as a pricing framework, but also as a benchmark for validating the numerical methods explored throughout the chapter.

3.5.2 Appendix B. Monte Carlo Option Pricing

The Black-Scholes pricing formula provides a closed-form solution for European call options. However, many derivatives do not admit analytical solutions, making numerical methods necessary. One of the most widely used approaches in quantitative finance is Monte Carlo simulation [18].

The starting point is the risk-neutral valuation formula introduced in Equation 3.4,

$$C = e^{-rT}\mathbb{E}\left[\max(S_T - K, 0)\right] \tag{3.8}$$

This expression states that the option value is equal to the discounted expected payoff under the risk-neutral probability measure. In most practical applications, the expectation cannot be evaluated analytically and must be approximated numerically.

Monte Carlo simulation replaces the theoretical expectation with an empirical average computed from a large number of simulated outcomes. If $S_T^{(1)}, S_T^{(2)}, \dots, S_T^{(N)}$ denote N independent realizations of the terminal asset price, the option value can be approximated by

$$\hat{C}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \max\left(S_T^{(i)} - K, 0\right). \tag{3.9}$$

Equation 3.9 is the estimator implemented in the Python notebook. Each simulation generates a possible terminal asset price according to the geometric Brownian motion model described in Appendix A, computes the corresponding payoff, and averages the results across all simulated scenarios.

An important property of Monte Carlo methods is that the estimator converges to the true expected value as the number of simulations increases. Under standard regularity conditions,

$$\hat{C}_N \to C \qquad \text{as} \qquad N \to \infty. \tag{3.10}$$

This result follows from the Law of Large Numbers and provides the theoretical justification for the simulation approach.

The accuracy of the estimator is commonly measured through its standard error. For Monte Carlo methods, the estimation error decreases proportionally to

$$\mathcal{O}\left(N^{-1/2}\right), \tag{3.11}$$

which implies that reducing the error by a factor of two requires approximately four times as many simulations.

This characteristic is one of the principal strengths and weaknesses of Monte Carlo methods. On the one hand, the approach is conceptually simple and can be applied to a wide range of pricing problems. On the other hand, convergence can be relatively slow, particularly when high precision is required.

For the European call option considered in this project, Monte Carlo simulation performs remarkably well. The results presented in the main body of the chapter show that the numerical estimates converge rapidly toward the Black-Scholes benchmark as the number of simulations increases. This behavior is expected because the terminal distribution of the underlying asset is known and can be sampled directly.

The role of Monte Carlo simulation in this project is therefore twofold. First, it provides a practical numerical approximation to the Black-Scholes pricing problem. Second, it serves as a baseline against which the Metropolis-Hastings implementation can be evaluated. Since direct sampling is available in this setting, Monte Carlo represents a natural benchmark for assessing the advantages and limitations of alternative sampling-based approaches.

3.5.3 Appendix C. Metropolis-Hastings Sampling

The Monte Carlo estimator discussed in Appendix B relies on direct sampling from the target distribution. In the Black-Scholes setting considered in this project, this approach is straightforward because the terminal asset price distribution is known analytically. However, many problems in computational finance involve probability distributions that are difficult to sample from directly.

Markov Chain Monte Carlo (MCMC) methods address this challenge by constructing a stochastic process whose long-run behavior reproduces a desired target distribution. Rather than generating independent samples, MCMC algorithms produce a sequence of correlated samples that gradually explore the probability space.

The Metropolis-Hastings algorithm is one of the most widely used MCMC methods. The algorithm begins from an initial state x and proposes a new candidate state x' from a proposal distribution $q(x'|x)$. The proposed state is then accepted with probability

$$\alpha(x, x') = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right), \tag{3.12}$$

where $\pi(\cdot)$ denotes the target probability distribution.

If the proposed state is accepted, the Markov chain moves to x'. Otherwise, the chain remains at its current position. Repeating this procedure produces a sequence of samples whose stationary distribution converges to the target distribution.

In the present project, the objective is not to sample complete asset-price trajectories. Instead, the algorithm is applied to the terminal log-price distribution derived in Equation 3.3. This simplification is possible because the payoff of a European call option depends only on the terminal asset price.

The target distribution therefore corresponds to

$$\log(S_T) \sim N\left(\log(S_0) + \left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right), \tag{3.13}$$

where the drift parameter has been replaced by the risk-free interest rate in accordance with the risk-neutral valuation framework.

Once samples from the terminal distribution have been generated, option values can be estimated using the same discounted payoff expression introduced in Appendix B. The difference lies not in the pricing equation itself, but in the mechanism used to obtain the samples.

An important consequence of this approach is that successive observations generated by the Markov chain are not independent. This property introduces autocorrelation into the sample and can reduce statistical efficiency relative to direct Monte Carlo simulation.

This observation helps explain the behavior reported in the main body of the project. The Metropolis-Hastings estimator produced option values consistent with the Black-Scholes benchmark, but exhibited larger fluctuations than standard Monte Carlo at smaller sample sizes. Since direct sampling from the terminal distribution is available in this problem, the additional flexibility of MCMC provides little practical advantage.

The purpose of including Metropolis-Hastings in this project is therefore not to outperform conventional Monte Carlo simulation. Rather, it serves as an introduction to sampling techniques that become increasingly relevant when direct sampling is difficult or when the pricing problem depends on higher-dimensional probability distributions. These situations arise naturally in many path-dependent derivatives and motivate the broader path integral perspective discussed in the next appendix.

3.5.4 Appendix D. Path Integral Perspective

The title of this project refers to the path integral formulation of option pricing. Although the numerical implementation developed throughout the chapter does not perform a full path integral simulation, the underlying motivation originates from a broader perspective on stochastic processes and probability distributions.

The path integral formalism was originally developed in quantum mechanics by Feynman as an alternative formulation of physical dynamics [19]. Rather than describing the evolution of a system through a single trajectory, the theory considers all possible trajectories connecting an initial and a final state. Observable quantities are then obtained by aggregating the contributions of these trajectories.

A similar idea appears naturally in stochastic finance. The future value of a derivative depends on the possible evolutions of the underlying asset, each associated with a particular probability. From this viewpoint, option pricing can be interpreted as an average over a space of possible asset-price paths [20].

In a schematic form, the value of a derivative can be expressed as

$$C(S_0, t_0) = \int \mathcal{D}[S(t)], P[S(t)], \Phi[S(t)] \tag{3.14}$$

where $\mathcal{D}[S(t)]$ denotes integration over all possible trajectories, $P[S(t)]$ represents the probability associated with a given path, and $\Phi[S(t)]$ denotes the payoff functional.

The key distinction between this expression and the formulations presented in the previous appendices is that the object being integrated is no longer a single terminal value, but an entire trajectory.

For the European call option considered in this project, the payoff depends only on the terminal asset price,

$$\Phi[S(t)] = \max(S_T - K, 0). \tag{3.15}$$

As a consequence, the pricing problem can be reduced to the terminal distribution of S_T. This observation explains why the implementation developed in the chapter focuses on sampling terminal prices rather than complete paths. For this particular derivative, the additional information contained in the intermediate trajectory is unnecessary for computing the payoff.

This simplification is one of the reasons why standard Monte Carlo simulation performs so well in the experiments presented earlier. Direct sampling from the terminal distribution is available, making more sophisticated sampling strategies difficult to justify on efficiency grounds alone.

The situation changes when the payoff depends on the entire history of the asset rather than its terminal value. Examples include Asian options, lookback options, barrier options, and many other path-dependent derivatives. In these cases, the trajectory itself becomes part of the pricing problem, and the path integral viewpoint provides a natural conceptual framework for describing the associated probability space [21] [22].

From this perspective, the implementation developed in this project can be viewed as a simplified introduction to a broader class of numerical techniques. The Monte Carlo and Metropolis-Hastings algorithms operate on the terminal distribution because the European call option permits this reduction. More complex derivatives would generally require sampling higher-dimensional objects, including entire asset-price trajectories.

The objective of this chapter was not to build a complete path integral pricing engine, but rather to investigate whether ideas originating from the path integral framework could be translated into a practical computational workflow. The results suggest that, for simple European options, direct Monte Carlo simulation remains the most efficient approach among those considered. At the same time, the project establishes a foundation for exploring more sophisticated derivatives in which the trajectory itself becomes the central object of interest.

Chapter 4

Credit Card Fraud Detection

Every credit card transaction generates a simple decision: approve it or block it. Most of the time the answer is obvious. The difficulty lies in the small fraction of transactions that appear legitimate but are actually fraudulent. In large payment systems, these events are rare, costly, and constantly changing, creating a learning problem that differs substantially from many conventional classification tasks.

This project uses a publicly available European credit card transaction dataset containing highly anonymized transaction records. The analysis combines unsupervised learning techniques to explore the structure of fraudulent activity with a supervised benchmarking framework designed to evaluate the predictive performance of several classification algorithms.

4.1 Problem Definition

The central difficulty in credit card fraud detection is severe class imbalance. Fraudulent transactions represent only a tiny fraction of overall activity, meaning that a model can achieve high accuracy while failing to identify most fraud cases [23].

This imbalance creates an unavoidable trade-off. Missing a fraudulent transaction may result in direct financial losses, while incorrectly flagging a legitimate transaction may interrupt a customer purchase and generate unnecessary friction. Effective fraud detection therefore requires more than maximizing a single performance metric. The objective is to identify as many fraudulent transactions as possible while maintaining an acceptable level of false alarms [24] [25].

Most machine learning studies approach this problem through supervised classification, using historical examples of fraudulent and legitimate transactions to train predictive models. This approach has produced substantial improvements over traditional rule-based systems and remains the foundation of modern fraud analytics [26] [27].

A less explored question is whether fraudulent transactions exhibit recognizable structures before any classification model is trained. If fraud is not a single behavior but a collection of different behaviors, clustering techniques may reveal groups of transactions that share common characteristics. Such patterns could provide additional insight into how fraudulent activity is organized within the feature space and help explain why certain classification models perform better than others [28].

The project therefore investigates the problem from two complementary perspectives. First, clustering methods are used to explore whether fraudulent transactions form identifiable groups within the feature space. Second, a supervised benchmarking study compares several machine learning algorithms under a common evaluation framework. The objective is not only to determine which models achieve the strongest predictive performance, but also to better understand the structure of fraudulent behavior in a highly imbalanced transaction dataset.

4.2 Modeling Approach

The analysis combines unsupervised and supervised learning techniques to examine credit card fraud from two complementary perspectives. The first investigates whether fraudulent transactions exhibit identifiable structure within the feature space. The second evaluates the extent to which that structure can be exploited for predictive purposes.

The unsupervised stage applies clustering algorithms to the transaction data in order to explore the distribution of fraudulent and legitimate transactions. K-Means, Agglomerative Clustering, and DBSCAN are used because they represent different notions of similarity and cluster formation. K-Means partitions observations around centroids, Agglomerative Clustering builds nested groups through successive merges, and DBSCAN identifies groups based on local density while explicitly accounting for noise points. Together, these methods allow the analysis to examine whether fraudulent transactions form concentrated groups, distinct fraud profiles, or isolated anomalies.

The supervised stage is formulated as a benchmarking study. Rather than focusing on a single predictive model, the project compares a diverse set of classification algorithms representing different modeling philosophies. The benchmark includes Logistic Regression, Naive Bayes, Linear and Quadratic Discriminant Analysis, K-Nearest Neighbors, Linear Support Vector Machines, Decision Trees, Random Forests, and XGBoost. To examine the impact of class imbalance, the models are evaluated under three training strategies: the original transaction distribution, random undersampling, and the Synthetic Minority Oversampling Technique (SMOTE).

Because fraudulent transactions represent only a small fraction of the observations, model evaluation focuses on metrics that remain informative under severe class imbalance. Particular attention is given to precision, recall, F1-score, ROC-AUC, Precision-Recall AUC, and the Kolmogorov-Smirnov statistic. Stratified cross-validation is used throughout the benchmark to ensure consistent comparisons across models.

The underlying assumption of the project is that fraudulent transactions leave detectable signatures in the available features. If such signatures exist, clustering methods should reveal some degree of structure within the transaction space, while supervised learning algorithms should be able to exploit that structure for predictive purposes. Examining both perspectives provides a more complete view of fraud behavior than either approach in isolation.

4.3 Methodology and Implementation

The analysis begins with an unsupervised exploration of the feature space to determine whether fraudulent transactions exhibit identifiable structure without using the target labels. Clustering algorithms are used to examine the distribution of transactions and to assess whether fraud tends to concentrate in specific regions of the data.

The second stage introduces the transaction labels and evaluates a range of supervised learning algorithms within a common benchmarking framework. Together, these two perspectives provide a broader understanding of the problem, combining exploratory analysis with predictive modeling.

4.3.1 Unsupervised Learning Methodology

The unsupervised learning stage investigates whether fraudulent transactions exhibit identifiable structure in the feature space without using the target labels during model construction. If fraudulent transactions tend to concentrate in specific regions of the transformed feature space, clustering algorithms should be able to reveal these patterns.

The analysis begins with an exploratory examination of the dataset. Particular attention is given to the severe class imbalance that characterizes the fraud detection problem.

The class distribution is summarized below.

Table 4.1: Class distribution in the credit card transaction dataset.

Class	Transactions	Percentage (%)
Legitimate	284,315	99.827
Fraudulent	492	0.173

Figure 4.1 illustrates the extreme imbalance between legitimate and fraudulent transactions. Fraudulent transactions represent less than one percent of all observations, making fraud detection a highly imbalanced classification problem.

Transaction amounts were also examined to characterize the range and concentration of transaction values.

Table 4.2: Summary statistics for transaction amounts.

Statistic	Value
Minimum	0.00
1st Quartile	5.60
Median	22.00
Mean	88.35
3rd Quartile	77.17
Maximum	25,691.16

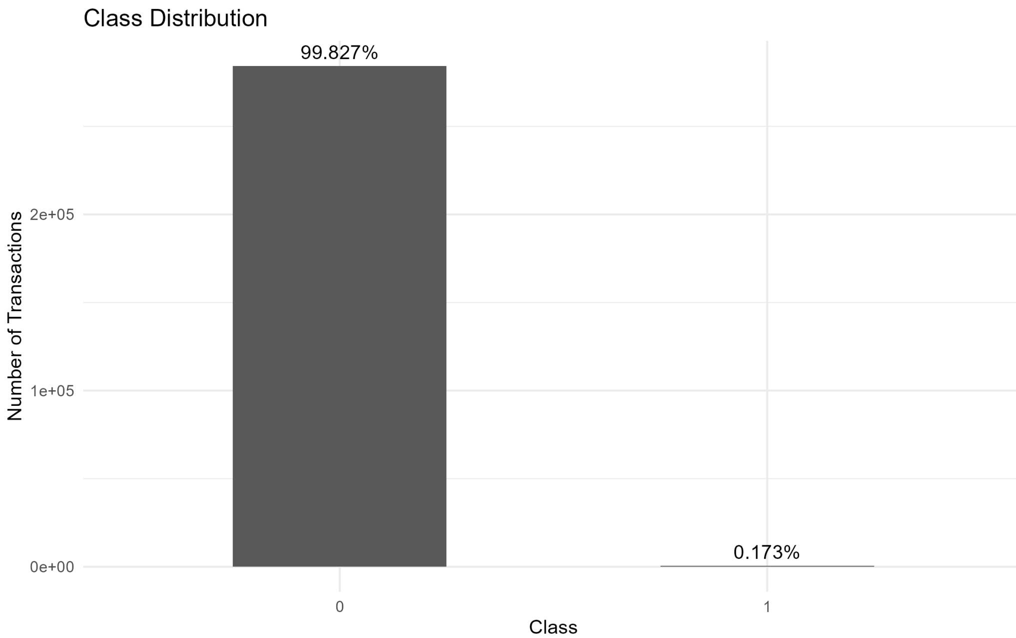

Figure 4.1: Class distribution in the credit card transaction dataset.

The distribution of transaction amounts is highly right-skewed, with most transactions concentrated at relatively small values and a small number of extreme observations. This behavior can be observed in Figure 4.2.

Transaction amounts were further compared across classes. Although fraudulent and legitimate transactions overlap substantially, Figure 4.3 suggests differences in the distribution of transaction values between the two groups.

The variables V1–V28 correspond to principal components obtained through a prior anonymization process. Selected features were visualized to examine differences between fraudulent and legitimate transactions. Several variables exhibit noticeable shifts between classes, suggesting that fraudulent transactions occupy distinct regions of the transformed feature space.

To complement the univariate analysis, a correlation matrix was computed using all available variables.

Following the exploratory analysis, the target variable was separated from the predictors and the Time variable was removed. The data was then partitioned into training and test sets using stratified sampling to preserve the original class

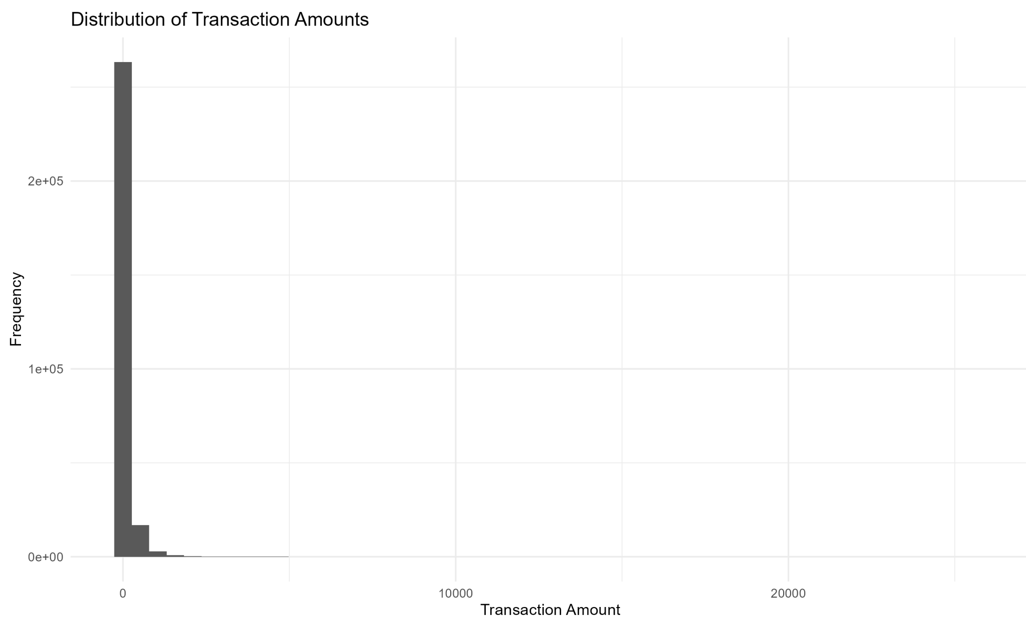

Figure 4.2: Distribution of transaction amounts.

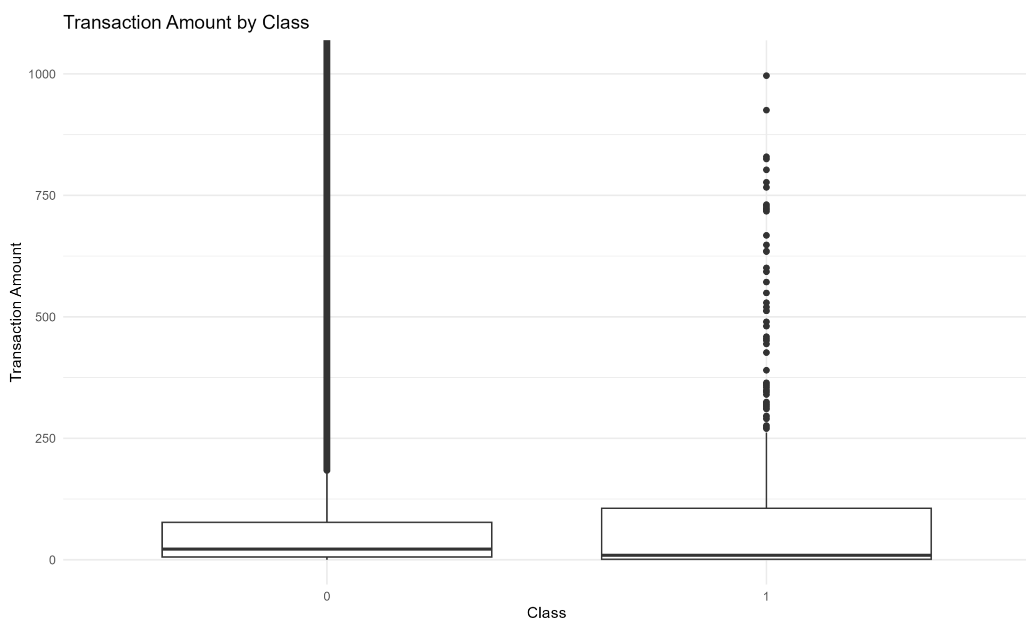

Figure 4.3: Transaction amount by class.

Figure 4.4: Selected PCA feature distributions by class.

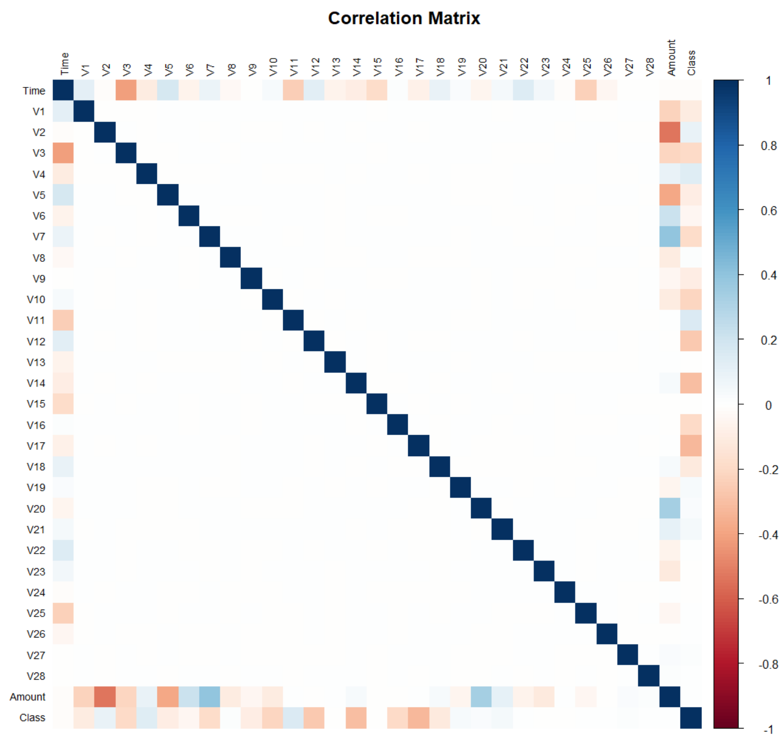

Figure 4.5: Correlation matrix of the credit card transaction dataset.

distribution. Feature scaling was applied after the train-test split, with scaling parameters estimated exclusively from the training data.

Because the training set contains more than two hundred thousand observations, a stratified sample was extracted from the training data for the clustering analysis. The sample preserves the original fraud rate while reducing computational requirements and allowing a consistent comparison across clustering algorithms.

Three clustering approaches were evaluated:

- K-Means Clustering
- Agglomerative Clustering
- DBSCAN

For K-Means, multiple values of (k) were explored. The elbow method was evaluated using the within-cluster sum of squares (inertia).

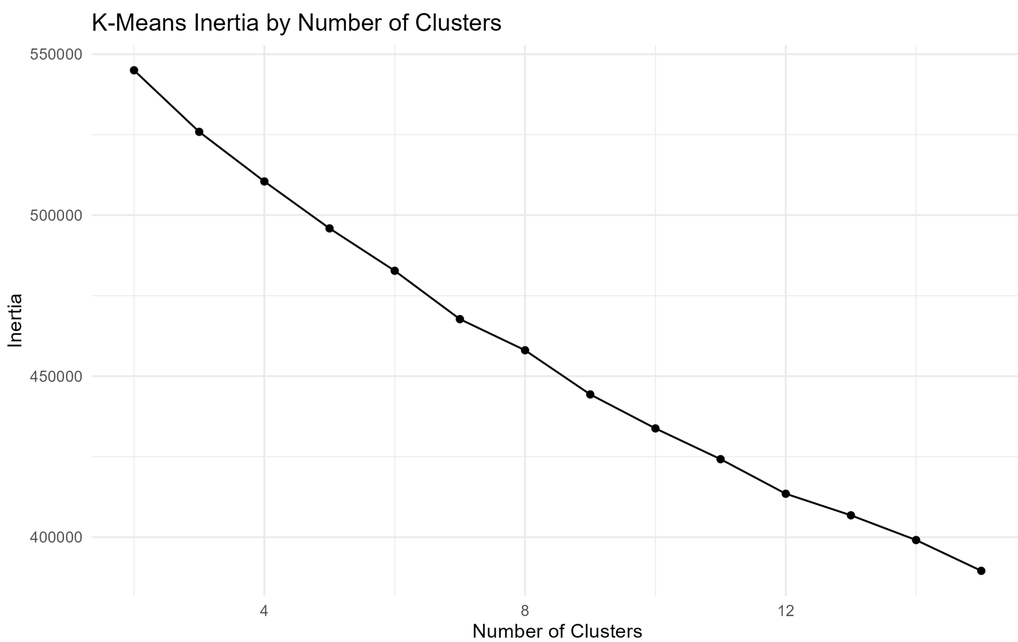

Figure 4.6: K-Means inertia as a function of the number of clusters.

The elbow plot does not reveal a clear point at which additional clusters provide diminishing returns. As a result, the number of clusters cannot be determined reliably using inertia alone. Additional validation metrics were evaluated and used as complementary diagnostic tools. Since no single criterion produced a definitive solution, the final choice was guided by interpretability and fraud concentration within the resulting clusters.

A three-cluster solution was retained because it revealed the strongest concentration of fraudulent transactions while remaining straightforward to interpret.

The composition of the resulting clusters was then examined by comparing cluster-level fraud rates against the baseline fraud rate observed in the clustering sample.

Table 4.3: Summary of the main findings obtained from the clustering algorithms.

Method	Key Finding
K-Means	One cluster exhibited a fraud rate approximately 21 times higher than the baseline fraud rate.
Agglomerative Clustering	A small cluster exhibited a fraud rate approximately 490 times higher than the baseline fraud rate.
DBSCAN	Fraudulent transactions were primarily identified as noise points rather than members of dense clusters.

The clustering results provide complementary perspectives on the structure of fraudulent transactions. K-Means identified a relatively large region characterized by elevated fraud concentration, while Agglomerative Clustering isolated a much smaller cluster with an exceptionally high fraud rate. In contrast, DBSCAN classified fraudulent transactions primarily as noise, suggesting that some fraud patterns behave as isolated anomalies rather than members of dense groups.

Taken together, these results indicate that fraudulent transactions are not uniformly distributed across the feature space. Although the exact structure varies across clustering algorithms, all three methods reveal behavior that differs substantially from the baseline distribution of legitimate transactions.

These findings provide preliminary evidence that fraudulent transactions leave detectable signatures in the available feature space and motivate the supervised learning stage of the analysis.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

4.3.2 Supervised Learning Benchmark

The supervised stage uses the transaction labels to evaluate how well different classification models can identify fraudulent transactions. The purpose is not to find a model that looks good under a single metric, but to compare modelling choices under conditions that resemble the operational reality of fraud detection: the positive class is rare, the cost of missed fraud is not the same as the cost of a false alarm, and a model that performs well on average may still be unhelpful if it fails to rank suspicious transactions near the top.

The benchmark was implemented in two complementary forms. The Python notebook runs the full comparison across the complete set of candidate algorithms and class imbalance strategies. The R / Quarto notebook focuses on the two retained configurations that are most useful for the final discussion: a Random Forest trained on the original class distribution, and an XGBoost model trained after applying SMOTE. This split keeps the implementation practical while preserving the main modelling comparison.

The training and test sets were created using stratified sampling so that the rare fraud class remained represented in both subsets. Scaling was estimated from the training data only and then applied to the test data, avoiding leakage from unseen transactions into the model preparation step. This is especially important in fraud detection, where small forms of leakage can make a model appear much stronger than it would be in deployment.

The original training distribution is shown in Table 4.4. The table also shows the SMOTE-resampled version used by the XGBoost model. The resampled dataset is balanced for training purposes only; the test set remains in the original imbalanced distribution.

Table 4.4: Class distribution by training strategy.

Strategy	Class	Transactions	Percentage (%)
Original	Legitimate	227,446	99.825
Original	Fraudulent	399	0.175
SMOTE	Legitimate	227,446	50.000
SMOTE	Fraudulent	227,446	50.000

Three imbalance strategies were considered in the broader supervised benchmark: the original class distribution, random undersampling, and SMOTE. The original distribution preserves the natural frequency of fraud, which is useful because it reflects the problem as it is encountered in practice. Random undersampling creates a balanced training set by removing legitimate transactions, which may help some models learn the minority class but also discards a large amount of information. SMOTE creates synthetic minority-class observations, allowing the model to see a denser representation of the fraud region without removing legitimate transactions.

The evaluation framework uses precision, recall, F1-score, ROC-AUC, Precision-Recall AUC, and the Kolmogorov-Smirnov statistic. In this setting, accuracy is deliberately avoided as a headline metric. With fraud representing only 0.173% of transactions, a classifier could obtain high accuracy by predicting almost everything as legitimate.

Precision measures how many flagged transactions are actually fraudulent:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.1}$$

Recall measures how many fraudulent transactions are successfully identified:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.2}$$

The F1-score combines both quantities into a single summary:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.3}$$

For model selection, Precision-Recall AUC is particularly important because it focuses on the minority class. ROC-AUC remains useful, but under severe imbalance it can sometimes look reassuring even when the model generates too many false positives for practical use. The KS statistic adds a further diagnostic view by measuring the maximum separation between the empirical score distributions of legitimate and fraudulent transactions.

The broader Python benchmark selected one leading model from each imbalance strategy. Random Forest performed best under the original class distribution, while XGBoost was the strongest model under both undersampling and SMOTE. The final test-set inspection then focused on whether those benchmark findings translated to unseen transactions.

In the final R / Quarto implementation, the comparison was narrowed to the two most informative retained configurations. Random Forest trained on the original class distribution represents the best-performing model that does not alter the training distribution. XGBoost with SMOTE represents the strongest imbalance-aware alternative. The undersampling-based XGBoost model was not retained for the final comparison because, despite strong cross-validation results on the balanced training data, its test-set behaviour produced too many false positives to be attractive as a practical fraud-screening model.

The final comparison is summarized in Table 4.5.

Table 4.5: Final model comparison on the test set.

Strategy	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC	KS statistic
Original	Random Forest	0.9625	0.8280	0.8902	0.9656	0.8782	0.9056
SMOTE	XGBoost	0.7080	0.8602	0.7767	0.9823	0.8594	0.9040

The two models make a useful trade-off visible. Random Forest is more conservative. It achieves higher precision, a higher F1-score, and the strongest Precision-Recall AUC. In practical terms, when it flags a transaction as fraudulent, it is more likely to be correct. This matters because false alarms are not free: they can block legitimate purchases, inconvenience customers, and create manual review work.

XGBoost with SMOTE is more aggressive. It achieves higher recall and a higher ROC-AUC, meaning it captures a larger share of the fraudulent transactions and ranks the two classes well across thresholds. The cost is lower precision: more legitimate transactions are pulled into the fraud queue. Depending on the operational context, this may still be acceptable. A bank or payment provider with a low-friction review process may prefer higher recall, while a retailer or platform with a strong customer-experience focus may place greater weight on precision.

Figure 4.7 compares the retained models using Precision-Recall AUC. The difference is not large, but Random Forest remains slightly ahead on the metric that is most directly aligned with rare-event detection.

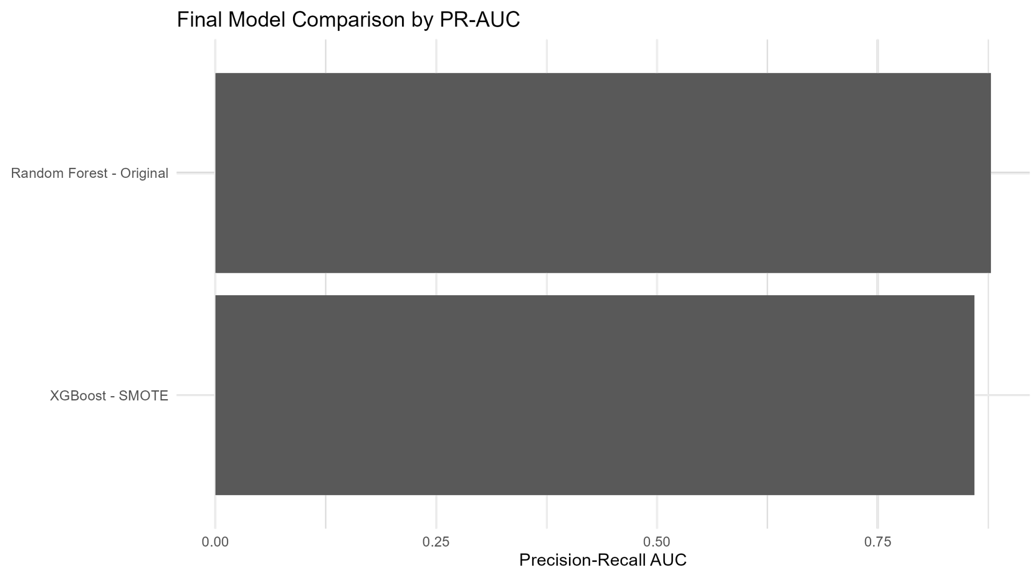

Figure 4.7: Final model comparison by Precision-Recall AUC.

The confusion matrices in Figure 4.8 show the same trade-off from an operational perspective. Random Forest produces fewer false positives, while XGBoost with SMOTE detects slightly more fraudulent cases at the default threshold. This is the kind of result where the "best" model depends less on the leaderboard and more on the business rule attached to the score.

The ROC curves in Figure 4.9 show that both models have strong ranking ability across classification thresholds. XGBoost with SMOTE has the higher ROC-AUC, which is consistent with its stronger recall. However, ROC curves should be read carefully in this problem because the number of legitimate transactions is so large that a small false positive rate can still translate into many legitimate transactions being flagged.

The Precision-Recall curves in Figure 4.10 provide a more direct view of rare-class performance. They show how precision changes as recall increases, which is often closer to the decision that a fraud team needs to make: how many alerts can be reviewed, and how many missed frauds are acceptable?

The KS statistic provides another way to assess whether the model scores separate legitimate and fraudulent transactions. The values in Table 4.6 are almost

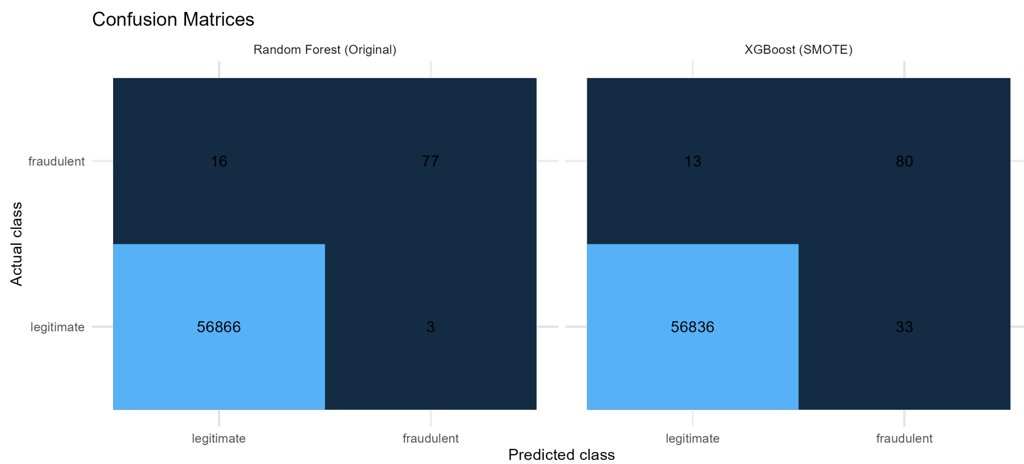

Figure 4.8: Confusion matrices for the retained supervised models.

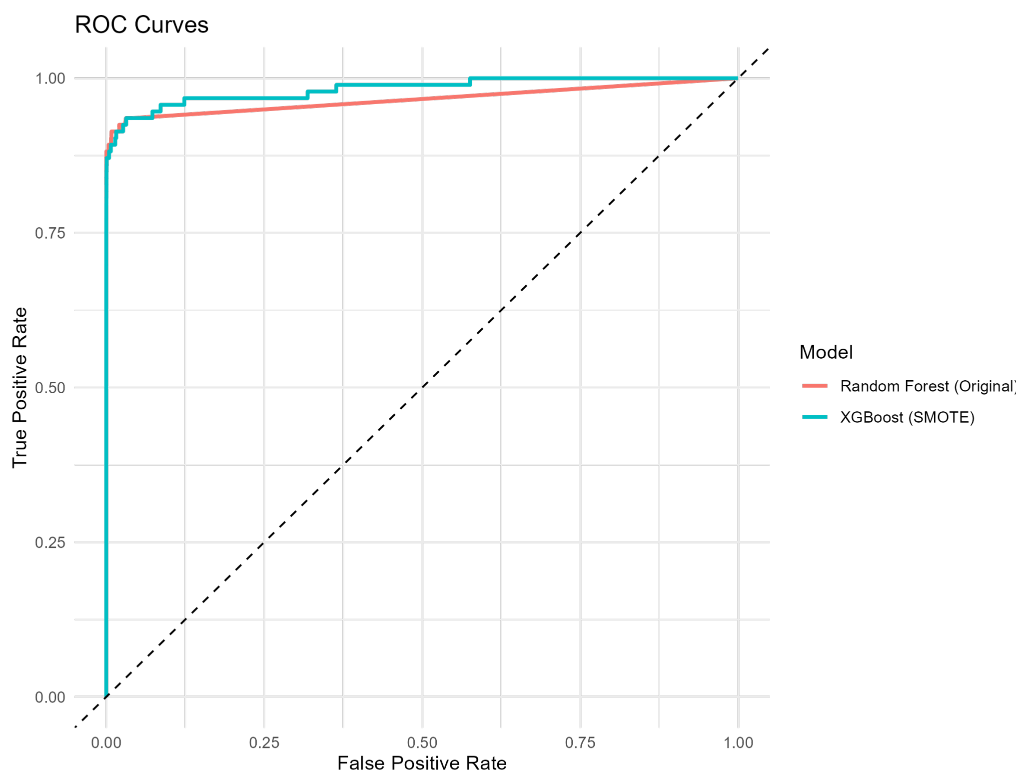

Figure 4.9: ROC curves for the retained supervised models.

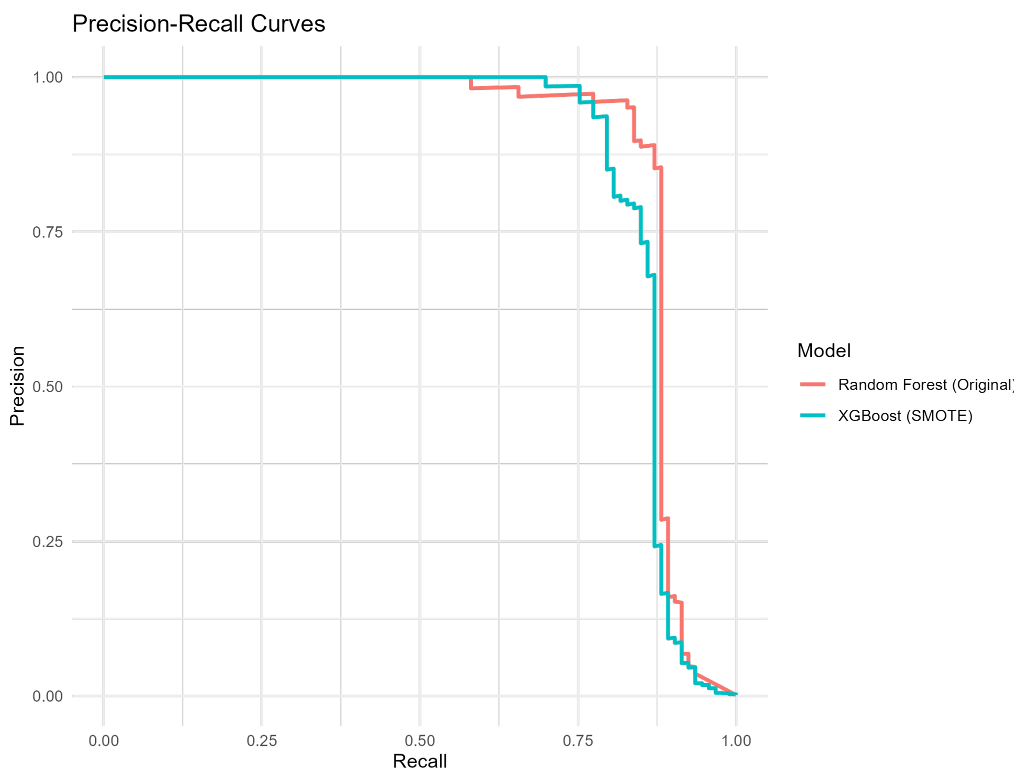

Figure 4.10: Precision-Recall curves for the retained supervised models.

identical for the two retained models.

Table 4.6: Kolmogorov-Smirnov statistic for the retained supervised models.

Model strategy	KS statistic
Random Forest - Original	0.9056
XGBoost - SMOTE	0.9040

The diagnostic plot in Figure 4.11 shows the empirical cumulative distributions of the predicted fraud probabilities for the two classes. A wider separation between the curves indicates stronger discrimination. The near-identical KS values suggest that both models assign noticeably different score distributions to legitimate and fraudulent transactions, even though their default-threshold behaviour differs.

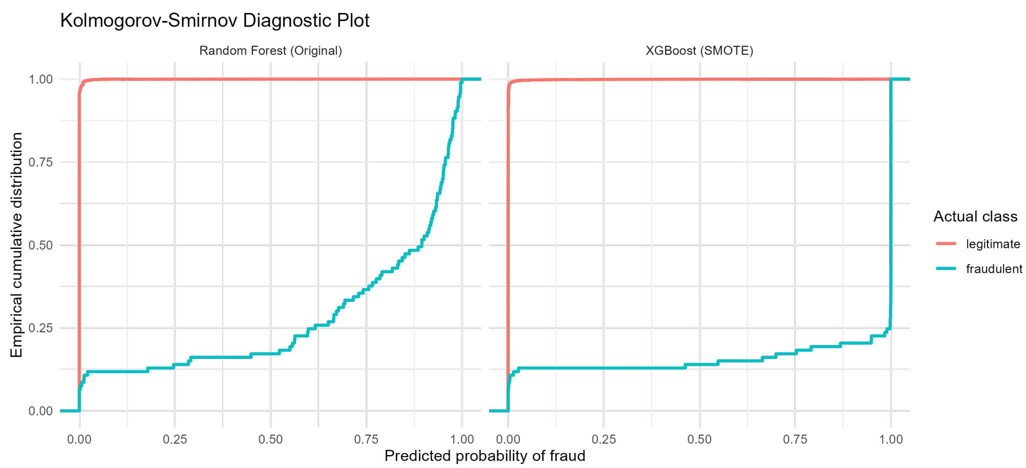

Figure 4.11: Kolmogorov-Smirnov diagnostic plot for the retained supervised models.

From a modelling point of view, the main conclusion is that tree-based ensemble methods are well suited to this dataset. They can capture nonlinear relationships among the anonymized PCA features without requiring a large amount of manual feature engineering. The original Random Forest is a strong and stable option, while SMOTE makes XGBoost more sensitive to fraudulent transactions. The stronger choice depends on the preferred operating point. For a portfolio project aimed at a practical analytics audience, this is an important distinction: the model is not simply selected because it has the largest headline score, but because its error profile fits the decision context.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

4.4 Results, Discussion, and Limitations

The benchmark highlights two consistent findings. First, tree-based ensemble methods provided the most reliable discrimination between legitimate and fraudulent transactions. Second, explicitly addressing the class imbalance produced substantial improvements across almost every classifier.

Random Forest and XGBoost achieved the strongest overall performance because both algorithms are well suited to heterogeneous tabular data containing complex, nonlinear relationships. Rather than relying on a single global decision boundary, they partition the predictor space into many local decision rules, allowing subtle interactions among the anonymized variables to be captured. This flexibility is particularly valuable in fraud detection, where fraudulent transactions rarely follow a single, easily separable pattern.

The severe class imbalance proved to be the central modelling challenge. With fraudulent transactions representing less than 0.2% of the observations, models trained on the original data naturally favored the majority class. Under these conditions, high overall accuracy can be achieved while missing a substantial fraction of fraudulent operations, making accuracy a poor indicator of practical performance.

Applying resampling strategies before training shifted the learning process toward the minority class. Random oversampling increased the exposure of fraudulent examples, while SMOTE generated additional synthetic observations that expanded the minority decision region instead of simply duplicating existing cases. Although the magnitude of the improvement varied across algorithms, the overall effect was consistent: recall increased markedly, while precision typically declined because more legitimate transactions were classified as suspicious.

This trade-off between precision and recall is unavoidable in operational fraud detection. A model optimized for precision reduces the number of false alarms but may fail to detect fraudulent transactions. A model optimized for recall identifies a larger proportion of fraud at the expense of investigating more legitimate payments. The preferred operating point ultimately depends on business costs, including the financial impact of missed fraud, investigation capacity, customer experience, and regulatory considerations.

For this reason, the Precision-Recall AUC provides a more informative summary than the traditional ROC AUC. When the positive class is extremely rare, ROC curves may appear optimistic because the false positive rate changes only slightly even after many legitimate transactions are incorrectly flagged. Precision-Recall curves focus exclusively on the minority class by evaluating how many detected alerts are actually fraudulent and how many fraudulent transactions are successfully recovered. As a result, they offer a more realistic assessment of model quality under extreme class imbalance.

Several limitations should be considered when interpreting these results. The predictors available in the public dataset were anonymized through Principal Component Analysis before release, preventing any interpretation of which original transaction characteristics contributed most to the predictions. The analysis also relies on a single historical collection of European card transactions observed over a limited time window. Fraud patterns evolve continuously as attackers adapt to existing detection systems, meaning that a model calibrated on historical behaviour may gradually lose predictive power through concept drift. In practice, production systems require continuous monitoring, periodic recalibration of decision thresholds, and regular retraining as new labelled transactions become available. Consequently, the benchmark should be viewed as a comparison of modelling strategies under controlled conditions rather than as a final fraud detection system ready for deployment.

4.5 Technical Appendix

This appendix summarizes the methodological concepts required to understand the modelling decisions presented in the benchmark. The objective is to provide enough theoretical background to interpret the results without developing the algorithms in full mathematical detail.

4.5.1 Random Forest

Random Forest is an ensemble learning algorithm that combines the predictions of many decision trees. Rather than relying on a single tree, the model aggregates a collection of independently constructed trees, producing predictions that are typically more stable and less sensitive to random fluctuations in the training data.

4.5.1.1 Decision Trees

A decision tree partitions the predictor space into a sequence of increasingly homogeneous regions. Starting from the root node, the algorithm repeatedly selects the feature and split point that best separate the observations according to a chosen impurity criterion. The final leaves contain observations with similar characteristics, and each leaf is assigned a class probability based on the proportion of fraudulent transactions it contains.

For a binary classification problem, the estimated probability of fraud at a terminal node is simply

$$\hat{p} = \frac{\text{Number of fraudulent observations}}{\text{Total observations in the node}}. \tag{4.4}$$

Although decision trees are easy to interpret and naturally capture nonlinear relationships and feature interactions, a single tree is usually unstable. Small changes in the training data may produce substantially different tree structures.

4.5.1.2 Ensemble Prediction

Random Forest addresses this instability by combining many trees built from slightly different versions of the training data while also considering random subsets of predictors at each split. The final prediction is obtained through majority voting,

$$\hat{y} = \text{mode}\left(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_B\right), \tag{4.5}$$

where $\hat{y}_b$ denotes the prediction of the b-th tree.

Class probabilities are estimated by averaging the probabilities produced by the individual trees,

$$\hat{P}(y = 1 \mid x) = \frac{1}{B} \sum_{b=1}^{B} \hat{P}_b(y = 1 \mid x). \tag{4.6}$$

Because prediction is based on many independent trees rather than a single model, Random Forest generally achieves lower variance and better generalization. These characteristics explain why it consistently performs well on structured tabular datasets such as credit card transaction records. Its main limitations are reduced interpretability compared with a single decision tree and larger computational requirements as the number of trees increases.

4.5.2 XGBoost

XGBoost is an optimized implementation of Gradient Boosting in which trees are added sequentially instead of independently. Each new tree is trained to reduce the prediction errors made by the ensemble constructed so far, allowing the model to progressively refine its representation of complex decision boundaries.

If the prediction after $t-1$ trees is $\hat{y}^{(t-1)}$, the next tree contributes a correction,

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + f_t(x), \tag{4.7}$$

where f_t represents the new decision tree.

Rather than minimizing only the training loss, XGBoost optimizes a regularized objective function,

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{t} \Omega(f_t), \tag{4.8}$$

where $l(\cdot)$ measures the prediction error and $\Omega(\cdot)$ penalizes overly complex trees.

The regularization term discourages unnecessarily deep or highly specialized trees, reducing the risk of overfitting while preserving predictive accuracy. Together with efficient optimization procedures and effective handling of nonlinear feature interactions, this design explains why XGBoost frequently ranks among the strongest performers on structured tabular datasets.

4.5.3 Handling Class Imbalance

Fraud detection datasets are naturally imbalanced because fraudulent transactions represent only a very small fraction of all operations. Standard classification algorithms are designed to minimize overall prediction error, so they tend to prioritize the majority class unless corrective measures are introduced during training.

One widely used approach is the Synthetic Minority Over-sampling Technique (SMOTE). Instead of duplicating existing minority observations, SMOTE generates new synthetic examples by interpolating between neighboring fraudulent transactions in the feature space.

For each minority observation, the algorithm selects one of its nearest minority neighbours and creates a synthetic sample along the line joining the two observations,

$$x_{\text{new}} = x_i + \lambda(x_j - x_i), \tag{4.9}$$

where x_i is an existing minority observation, x_j is one of its nearest minority neighbours, and $\lambda \in [0, 1]$ is a random interpolation factor.

By populating the minority region with plausible synthetic observations, SMOTE encourages the classifier to learn broader decision boundaries instead of focusing on a handful of isolated fraud cases. This generally improves the ability to recover fraudulent transactions while avoiding the excessive duplication that can occur with simple random oversampling.

Several extensions combine synthetic oversampling with subsequent cleaning procedures that remove ambiguous or overlapping observations near the class boundary. Although such hybrid methods can further improve performance in some applications, the benchmark presented in this chapter focuses on the standard resampling strategies needed to evaluate the effect of class imbalance on the candidate classifiers.

Chapter 5

Customer Value and Retention

Customer relationships are among the most valuable assets of a business, yet they are also among the most difficult to quantify. While transactional databases record what customers have purchased, they rarely explain how purchasing behaviour evolves over time or which customers deserve the greatest commercial attention. As a result, organisations often rely on simple indicators such as historical revenue or purchase counts, even though these metrics provide only a partial view of customer behaviour.

Database marketing has long recognised that effective customer management requires combining behavioural analysis with predictive and financial perspectives rather than relying on a single performance indicator [29]. Understanding who customers are, who is likely to leave, and which relationships are expected to generate the greatest long-term value are closely related questions that should be addressed within a unified analytical framework.

This chapter develops a complete customer analytics workflow for a non-contractual retail business using transactional data from the AdventureWorks database. Unlike subscription businesses, where customer churn is explicitly observed through contract cancellations, retail businesses rarely record when a customer relationship has ended. Instead, customer behaviour must be inferred from historical purchasing activity.

The analysis is organised around three complementary tasks. First, customers are segmented using Recency, Frequency, and Monetary (RFM) variables to identify groups with similar purchasing behaviour. Second, churn is formulated as a supervised learning problem by defining customer inactivity over a future observation window. Finally, customer lifetime value (CLV) is estimated by combining customer profitability with retention risk and explicit discount-rate assumptions.

Rather than presenting these models as independent analyses, the chapter demonstrates how descriptive analytics, predictive modelling, and economic valuation can be integrated into a single decision-support workflow. The emphasis is not on producing a universal customer score, but on developing a transparent framework in which every modelling assumption remains explicit and interpretable.

5.1 Problem Definition

Customer value is inherently difficult to assess in a non-contractual business. Unlike subscription services, where customers can be classified as active or cancelled, retail customers may purchase irregularly, return after long periods of inactivity, or buy only seasonally. Consequently, the absence of recent transactions does not necessarily indicate that a customer has churned.

This uncertainty complicates several common business decisions. Which customers should receive retention efforts? Which customers appear valuable but are gradually becoming inactive? Which recently acquired customers show early signs of future growth? Which inactive customers are unlikely to return?

Historical revenue alone cannot answer these questions because it ignores how recently customers have purchased, how frequently they buy, and how likely they are to continue purchasing in the future. Likewise, prioritising customers solely according to churn probability overlooks the fact that not every customer contributes equally to the business.

The AdventureWorks database provides a realistic environment for studying these problems because it stores detailed transactional information across customers, orders, products, costs, and sales territories. Rather than relying on pre-computed customer summaries, this project reconstructs customer behaviour directly from transactional records and derives the features required for segmentation, churn prediction, and customer value estimation.

The analysis therefore addresses three related questions.

- How can customers be grouped according to their purchasing behaviour?

- Which customers appear most likely to become inactive?
- How does expected customer value change once profitability, retention risk, and discounting are considered simultaneously?

Although each question can be investigated independently, combining them provides a substantially richer understanding of customer relationships than any single model alone.

5.2 Modeling Approach

The analytical workflow follows a layered structure in which each stage provides information for the next.

The first stage focuses on behavioural segmentation. Customer-level RFM variables are computed from the complete transaction history available for each individual. Because these variables exhibit strong positive skewness, they are transformed and standardised before applying K-Means clustering. The resulting clusters are then interpreted using the original RFM variables to preserve their business meaning.

The second stage introduces prediction through churn modelling. Since customer churn is not explicitly recorded in the AdventureWorks database, an operational target is constructed by separating the available transaction history into calibration and observation periods. Customers who make no purchases during the observation window are labelled as churned, allowing the problem to be formulated as a supervised classification task while avoiding information leakage.

The final stage estimates customer lifetime value. Customer profitability is derived from historical revenue and product costs, while segment-specific retention behaviour and discount-rate assumptions are used to construct a risk-adjusted valuation framework. A regression model is then trained to approximate this valuation using the customer-level features engineered throughout the previous stages.

Viewed as a whole, the workflow progresses naturally from behavioural description to future risk estimation and finally to economic value. Rather than producing isolated analytical models, each stage complements the previous one, resulting in a coherent framework for customer prioritisation and retention planning.

5.3 Methodology and Implementation

The methodology is organised into three complementary stages that progressively transform transactional sales records into actionable customer insights. Although each stage addresses a different analytical problem, they are connected through a common customer-level feature engineering pipeline built from the AdventureWorks transactional database.

The workflow begins by describing customer purchasing behaviour through RFM segmentation. These behavioural profiles then provide additional context for a supervised churn prediction model, where customer inactivity is defined using future purchasing behaviour observed within a separate validation period. Finally, the outputs of the previous stages are combined with profitability information to construct a risk-adjusted Customer Lifetime Value (CLV) framework that supports customer prioritisation from both behavioural and financial perspectives.

Rather than treating segmentation, churn prediction, and lifetime value estimation as independent modelling exercises, the implementation integrates them into a single analytical workflow in which each stage provides information that strengthens the interpretation of the next.

5.3.1 Customer Segmentation with RFM Analysis

Customer segmentation provides the behavioural foundation for the remainder of the chapter. Instead of analysing thousands of customers individually, the objective is to identify groups with similar purchasing patterns that can later support both churn prediction and customer value estimation.

The implementation begins by extracting transaction-level data directly from the AdventureWorks database rather than relying on pre-aggregated customer tables. Sales orders, order details, products, product categories, customers, and sales territories are combined to create a reusable transactional dataset from which customer-level features can be derived consistently throughout the project.

Customer behaviour is summarised using the classical Recency, Frequency, and Monetary (RFM) framework. Recency measures the time elapsed since the customer's most recent purchase, Frequency counts completed purchase events, and Monetary represents cumulative customer spending over the observation period. Because these variables exhibit strong positive skewness, particularly Frequency and Monetary value, they are transformed using the Box-Cox transformation and standardised before clustering to reduce the influence of extreme observations.

Customer segments are then identified using the K-Means algorithm. Rather than selecting the number of clusters arbitrarily, several candidate solutions are evaluated using complementary internal validation metrics, including the silhouette coefficient, the Calinski–Harabasz Index, and the Davies–Bouldin Index. The final segmentation balances statistical quality with business interpretability, allowing each cluster to be characterised according to the original RFM variables rather than the transformed feature space.

To complement the discrete segmentation, the implementation also introduces a continuous within-segment RFM score that ranks customers according to their relative purchasing behaviour inside each cluster. This additional layer helps distinguish customers who belong to the same behavioural segment but differ in their overall commercial importance, providing a practical prioritisation mechanism for subsequent retention activities.

The resulting customer segments constitute the descriptive layer of the analytical workflow and are subsequently incorporated as explanatory variables in both the churn prediction and customer lifetime value models.

The complete Python implementation of this workflow, including data extraction, feature engineering, clustering diagnostics, segment interpretation, and customer scoring, is available in the companion notebook linked above.

A complete Python implementation of the RFM analysis presented in this section is available in the project repository .

5.3.2 Churn Prediction in Non-Contractual Relationships Based on RFM Analysis

Once the customer segments have been established, the analysis shifts from behavioural description to predictive modelling. The objective is no longer to characterise purchasing behaviour, but to identify customers who appear at risk of becoming inactive in the near future. In a non-contractual retail environment, however, this is not straightforward because the absence of recent purchases does not necessarily imply that a customer has permanently left. Instead, churn must be defined operationally from observed purchasing behaviour.

To reproduce a realistic prediction scenario, the available transaction history is divided into calibration and observation periods. All explanatory variables are computed exclusively from the calibration window, while customer activity during the subsequent observation period is used only to validate the churn outcome. This chronological separation ensures that every predictor reflects information that would have been available at the time the prediction was made, thereby preventing information leakage.

Rather than defining churn solely as the absence of future purchases, the implementation combines several behavioural indicators derived during the calibration period. Customers are first identified as potential churners if they belong to a low-activity behavioural segment, their recency exceeds their historical average purchase latency, and their within-segment RFM score falls below the average score of their corresponding segment. These behavioural conditions identify customers whose purchasing patterns are consistent with progressive disengagement rather than temporary inactivity. A customer is ultimately labelled as churned only if no additional purchases are observed during the subsequent validation period. This two-stage definition combines historical behaviour with future purchasing activity, producing a target that remains consistent with the non-contractual nature of retail customer relationships while avoiding information leakage.

The predictive feature set extends the variables developed during the segmentation stage. In addition to the original RFM variables and customer segment assignments, the implementation derives customer-level descriptors that capture purchasing cadence, average order value, customer tenure, product diversity, profitability, purchase intervals, geographic information, and several aggregated transaction statistics. Together, these variables provide a more comprehensive representation of customer behaviour than RFM alone while preserving a clear business interpretation.

Several supervised learning algorithms are evaluated under a common modelling framework using cross-validation and hyperparameter optimisation. Because churned customers represent only a minority of the customer base, particular attention is given to model evaluation under class imbalance. Performance is therefore assessed using complementary metrics, including ROC AUC, Average Precision, Precision, Recall, and F1-score. In addition, cumulative gains analysis is used to evaluate how effectively customers can be ranked according to their estimated churn risk, which is often more valuable for retention planning than binary classification alone.

Rather than producing deterministic decisions, the selected classifier estimates the probability that each customer will become inactive during the observation period. These probabilities provide a practical ranking of retention risk that can support targeted marketing campaigns, customer prioritisation, and proactive retention strategies. They also constitute one of the principal inputs for the risk-adjusted customer lifetime value framework developed in the following section.

A complete Python implementation of the churn prediction model presented in this section is available in the project repository.

5.3.3 Risk-Adjusted Customer Lifetime Value Prediction

The final stage of the workflow extends the analysis from customer retention to customer value. While the churn model estimates the likelihood that a customer will become inactive, it does not indicate the potential economic impact associated with that outcome. In practice, customers with similar churn probabilities may contribute very differently to the business, making it necessary to consider both retention risk and expected profitability when prioritising commercial actions.

Customer Lifetime Value (CLV) provides a natural framework for integrating these dimensions. Following the general principles described by [29], customer value is interpreted as the present value of future customer profits rather than a simple summary of historical revenue. The implementation adopts this perspective while adapting it to a non-contractual retail environment in which future purchasing behaviour cannot be observed directly.

The modelling process begins by estimating customer-level profitability from historical sales and product costs extracted from the AdventureWorks database. These observed profits are then combined with the behavioural information obtained during the previous stages of the workflow. Instead of assuming identical future behaviour for every customer, the implementation incorporates segment-specific retention characteristics derived from the RFM analysis together with the predicted probability of churn estimated by the classification model. Segment-specific discount-rate assumptions are subsequently introduced to account for differences in the uncertainty associated with future customer relationships.

These components are combined to construct a risk-adjusted customer lifetime value that serves as the target variable for the predictive model. Since this valuation cannot be observed directly in the transactional database, it should be interpreted as a transparent economic framework based on explicit behavioural and financial assumptions rather than as a measured quantity.

Once the target has been constructed, the problem is formulated as a supervised regression task. The customer-level features engineered throughout the project—including behavioural variables, profitability measures, customer characteristics, and the outputs of the segmentation and churn models—are used to train several regression algorithms under a common cross-validation framework. Their predictive performance is assessed using complementary regression metrics together with residual analysis and predicted-versus-observed comparisons to evaluate both numerical accuracy and ranking capability.

The resulting model provides a scalable approximation of the underlying valuation framework. Rather than recomputing the complete economic model for every new customer, organisations can estimate expected customer value directly from observable behavioural characteristics while maintaining consistency with the assumptions adopted throughout the analytical workflow.

A complete Python implementation of the customer lifetime value model presented in this section is available in the project repository .

An interactive Streamlit application implementing the methods presented in this chapter is available .

5.4 Results, Discussion, and Limitations

The three stages of the analytical workflow provide complementary perspectives on customer behaviour. Customer segmentation offers an interpretable description of purchasing patterns, churn modelling estimates future customer inactivity, and customer lifetime value integrates these behavioural insights with expected economic contribution. Rather than evaluating each component independently, the results are interpreted as successive layers of information that progressively enrich the understanding of customer relationships.

5.4.1 Customer Segmentation

The exploratory analysis confirms the strong asymmetry typically observed in transactional retail data. As shown in Table 5.1, Frequency and Monetary value exhibit substantial positive skewness, indicating that most customers purchase infrequently while a relatively small number account for a disproportionate share of total revenue. Such distributions are common in customer analytics and motivate the use of transformations before applying distance-based clustering algorithms.

Table 5.1: Summary statistics of the RFM variables before transformation.

Variable	Mean	Standard Deviation	Skewness
Recency	191.2675	150.4236	2.4352
Frequency	1.6457	1.4571	7.3882
Monetary	5745.4041	38800.3833	12.7974

The distributions of the original RFM variables are illustrated in Figure 5.1. After applying the Box-Cox transformation and standardisation, the feature space becomes considerably more suitable for clustering while preserving the original variables for business interpretation.

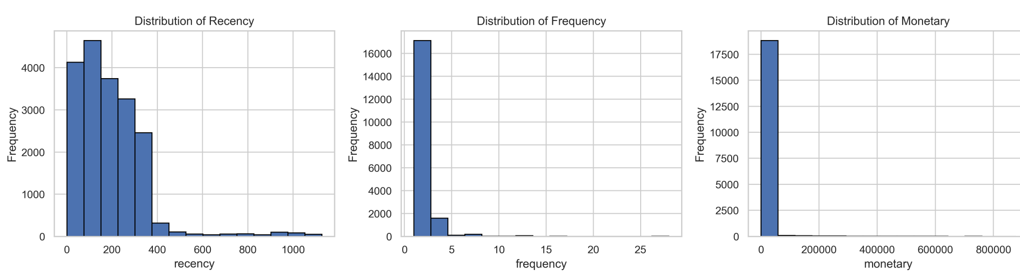

Figure 5.1: Distribution of the original RFM variables before transformation.

Several candidate values of (k) were evaluated using complementary clustering diagnostics. Rather than relying on a single criterion, the final solution balances cluster separation, compactness, and interpretability. As summarised in Table 5.2, the nine-cluster solution provides the best overall compromise across the internal validation metrics without producing unnecessarily fragmented customer groups.

Table 5.2: Clustering quality metrics used to determine the optimal number of customer segments.

k	Silhouette	Calinski–Harabasz	Davies–Bouldin
2	0.4435	15024.18	0.9683
3	0.3890	13311.31	1.0315
4	0.4035	12695.18	0.8897
5	0.4237	13171.97	0.9205
6	0.4267	13676.56	0.8533
7	0.4348	14390.70	0.7965
8	0.4556	15406.58	0.7315
9	**0.4655**	**16264.65**	**0.7046**
10	0.4622	16513.19	0.7133

The corresponding diagnostic curves are presented in Figure 5.2.

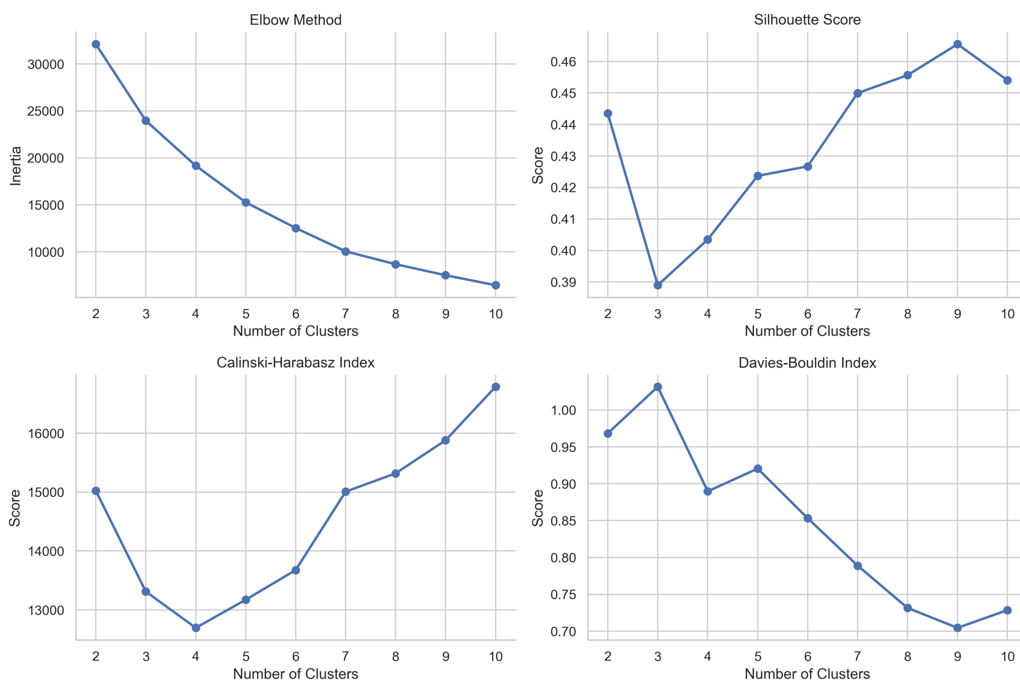

Figure 5.2: Cluster selection diagnostics for different values of (k).

The selected segmentation identifies nine customer profiles that capture meaningful differences in purchasing behaviour. Rather than representing fixed customer categories, these segments describe behavioural tendencies ranging from highly engaged customers to inactive, low-value buyers. A summary of the resulting segments is presented in Table 5.3.

Table 5.3: Summary of the final customer segments obtained from the RFM analysis.

Segment	Customers	Recency	Frequency	Monetary	Avg. Score
Champions	2767	87.1	3.05	28042.2	0.74
High Value but Cooling	2832	264.0	2.27	8903.2	0.51
Frequent Low-Spend Customers	791	67.5	2.77	776.8	0.76
Recent Low-Value Customers	1724	87.3	1.00	1397.6	0.66
Active Minimal Buyers	2463	147.6	1.20	2145.7	0.63
Declining Frequent Buyers	1075	107.1	2.35	122.3	0.57
Inactive Minimal Buyers	4987	328.4	1.00	1054.6	0.35
Inactive Low-Value Customers	2046	274.7	1.00	1352.8	0.40
Lost Customers	434	845.6	1.01	2852.1	0.18

The three-dimensional representation shown in Figure 5.3 highlights the gradual transition between behavioural groups rather than sharply separated customer categories. This reinforces an important practical point: customer segmentation should be interpreted as a descriptive representation of purchasing behaviour rather than a definitive classification of customer types.

5.4.2 Churn Prediction

Customer segmentation provides a descriptive representation of purchasing behaviour, but it does not indicate whether customers are likely to remain active. The second stage of the workflow therefore introduces a predictive component

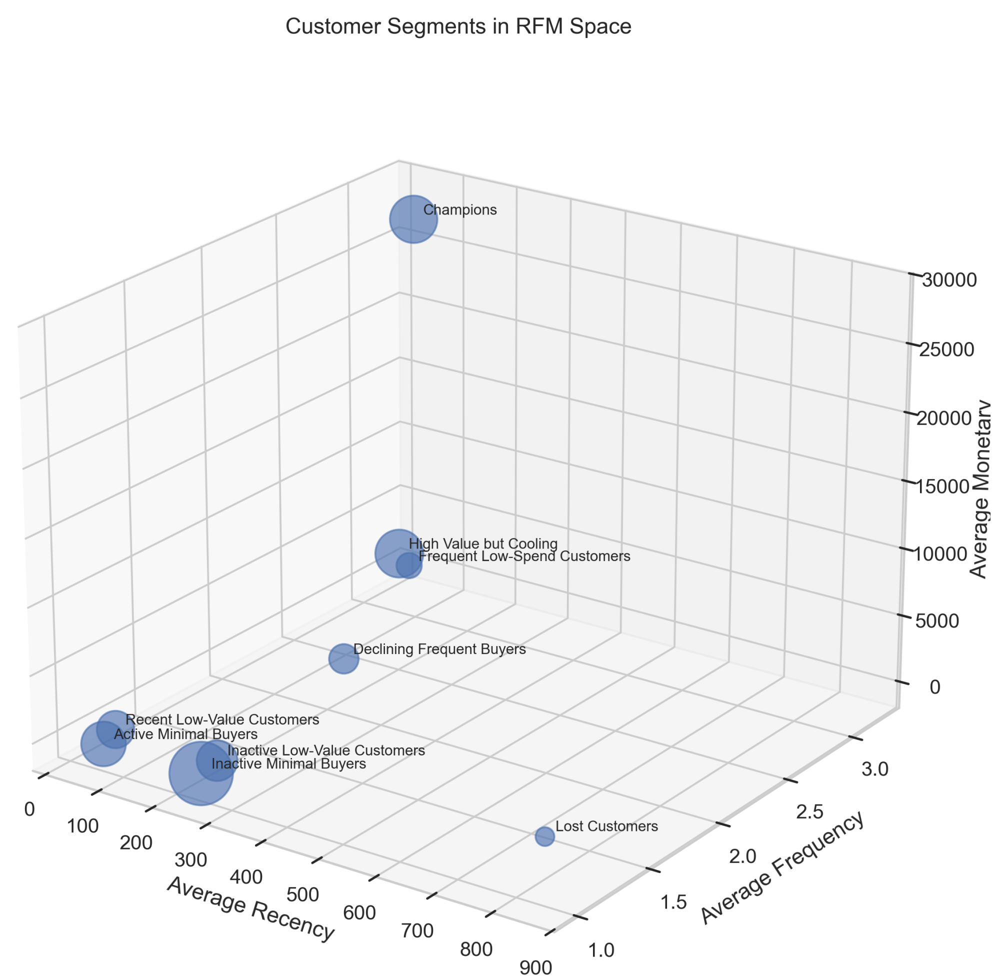

Figure 5.3: Three-dimensional representation of the customer segments in the original RFM space.

by estimating the probability that each customer will become inactive during the observation period.

Several supervised learning algorithms were evaluated using the engineered customer-level features described in the previous section. Their performance is summarised in Table 5.4. Rather than relying on a single evaluation metric, the comparison considers both discrimination and ranking performance to identify the model that provides the most reliable estimates of customer retention risk.

Table 5.4: Performance comparison of the evaluated churn prediction models. Abbreviations: RF = Random Forest; XGB = XGBoost; LR = Logistic Regression; AP = Average Precision; BA = Balanced Accuracy.

Model	Sampling	AUC	AP	Acc.	BA	Prec.	Rec.	F1
RF	SMOTE	1.000	0.971	0.997	0.999	0.800	1.000	0.889
XGB	Under	1.000	0.951	0.987	0.994	0.455	1.000	0.625
RF	Original	1.000	0.959	0.997	0.949	0.857	0.900	0.878
XGB	SMOTE	0.999	0.960	0.997	0.949	0.857	0.900	0.878
XGB	Original	0.999	0.951	0.998	0.950	0.900	0.900	0.900
RF	Under	0.999	0.939	0.965	0.982	0.233	1.000	0.377
LR	SMOTE	0.999	0.892	0.987	0.994	0.455	1.000	0.625
LR	Original	0.998	0.871	0.996	0.849	0.875	0.700	0.778
LR	Under	0.997	0.829	0.969	0.984	0.256	1.000	0.408

The best-performing specification was the Random Forest model trained with SMOTE. As shown in Table 5.5, it achieved excellent discrimination on the independent test set, with a ROC AUC of 0.9996 and an Average Precision of 0.9707. More importantly, the model maintained full recall for the churn class while keeping precision at 0.8000, which is a useful balance for retention-oriented applications where missing high-risk customers is usually more costly than reviewing a small number of false positives.

Table 5.5: Performance of the selected churn prediction model on the independent test set.

ROC AUC	Average Precision	Accuracy	Balanced Accuracy	Precision	Recall	F1
0.9996	0.9707	0.9973	0.9986	0.8000	1.0000	0.8889

The class-level results in Table 5.6 provide a more detailed view of the same behaviour. The model correctly identifies all churn cases in the test set, while most retention cases are also classified correctly. This is particularly relevant because the churn class is very small relative to the retained customer population.

Table 5.6: Precision, Recall, F1-score, and support for the selected churn prediction model.

Class	Precision	Recall	F1-score	Support
Retention	1.0000	0.9973	0.9986	1842
Churn	0.8000	1.0000	0.8889	20
Accuracy	0.9973	0.9973	0.9973	0.9973
Macro avg	0.9000	0.9986	0.9438	1862
Weighted avg	0.9979	0.9973	0.9975	1862

The Receiver Operating Characteristic curve presented in Figure 5.4 demonstrates that the model separates active and inactive customers across a broad range of decision thresholds. Since customer retention datasets are inherently imbalanced, the Precision–Recall curve shown in Figure 5.5 provides an equally important perspective by evaluating the model's ability to identify genuinely high-risk customers without generating an excessive number of false positives.

The confusion matrix shown in Figure 5.6 reinforces this interpretation. Most customers are correctly classified, while the remaining errors reflect the intrinsic uncertainty associated with non-contractual customer behaviour. In this setting, some customers naturally purchase at irregular intervals, making it difficult to distinguish perfectly between temporary inactivity and progressive disengagement.

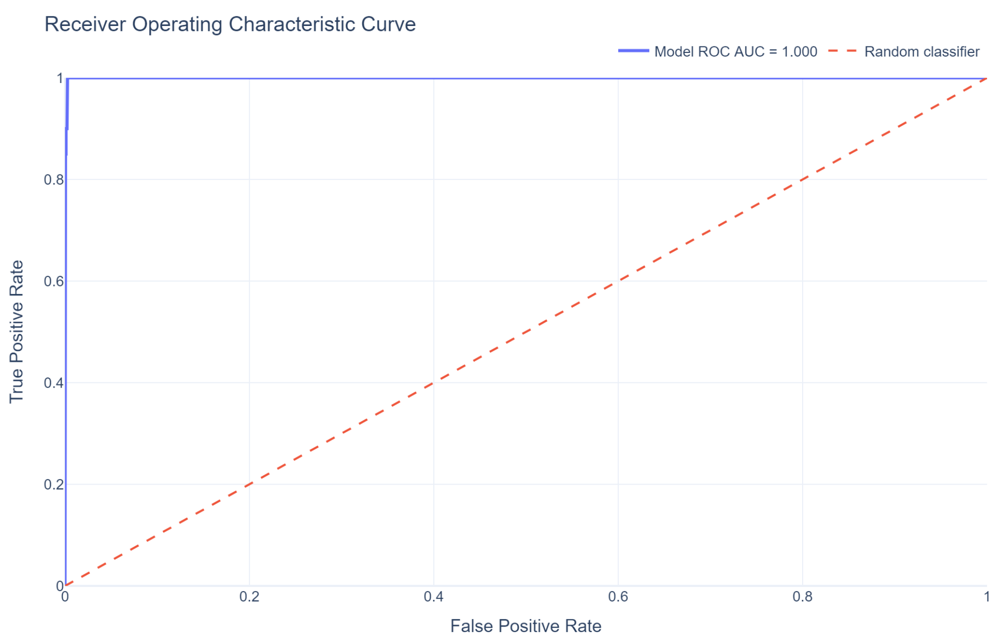

Figure 5.4: Receiver Operating Characteristic curve for the selected churn prediction model.

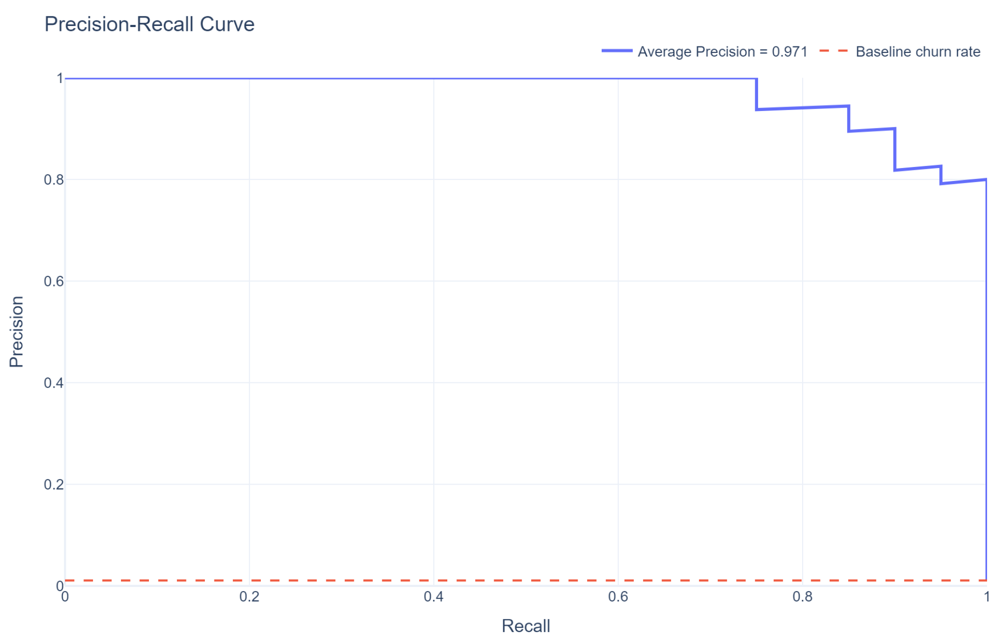

Figure 5.5: Precision–Recall curve for the selected churn prediction model.

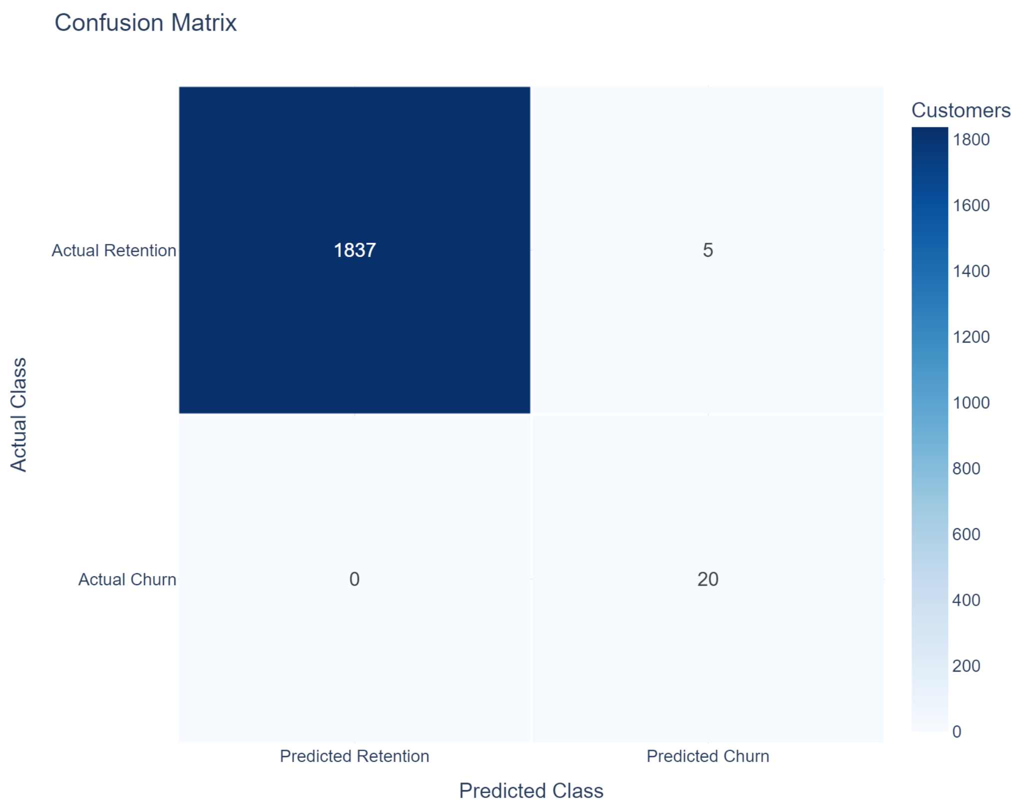

Figure 5.6: Confusion matrix for the selected churn prediction model.

From a business perspective, ranking customers by estimated churn probability is often more useful than producing binary predictions. Marketing teams typically have limited resources and cannot intervene with every customer simultaneously. The cumulative gains curve shown in Figure 5.7 demonstrates that the model concentrates a substantial proportion of future churners within the highest-risk customers, enabling retention campaigns to be prioritised more effectively.

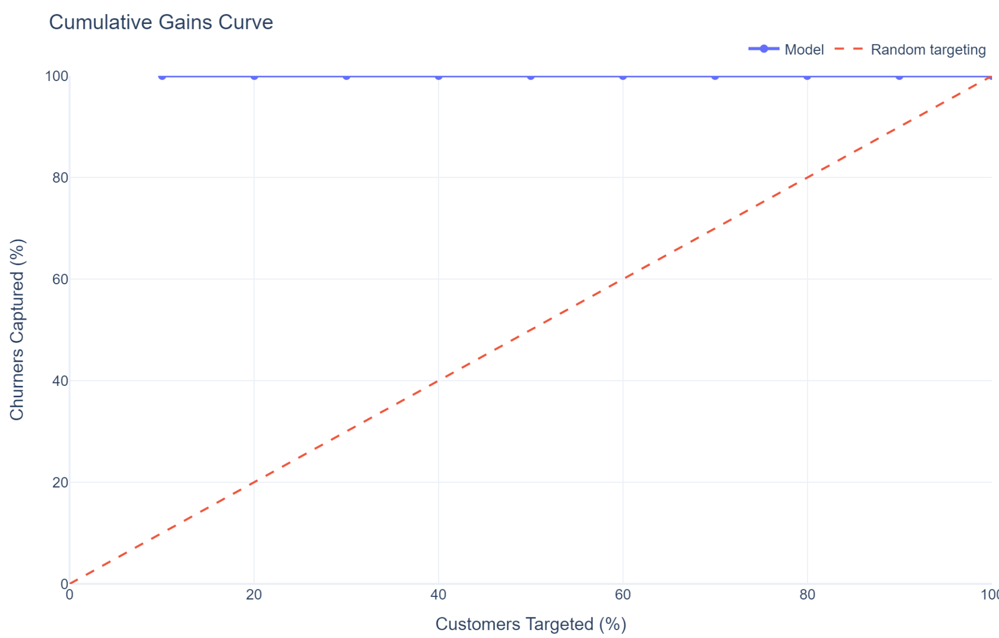

Figure 5.7: Cumulative gains curve for the selected churn prediction model.

Inspection of the feature importance rankings further supports the behavioural interpretation of the model. As shown in Figure 5.8, variables related to purchasing recency, customer tenure, purchasing cadence, and behavioural segmentation contribute substantially to the predicted probabilities. This consistency suggests that the model is learning meaningful patterns associated with customer engagement rather than relying on spurious statistical relationships.

Although the predictive performance is strong, several limitations should be acknowledged. The model depends on the operational definition of churn adopted in this project, meaning that alternative calibration or observation

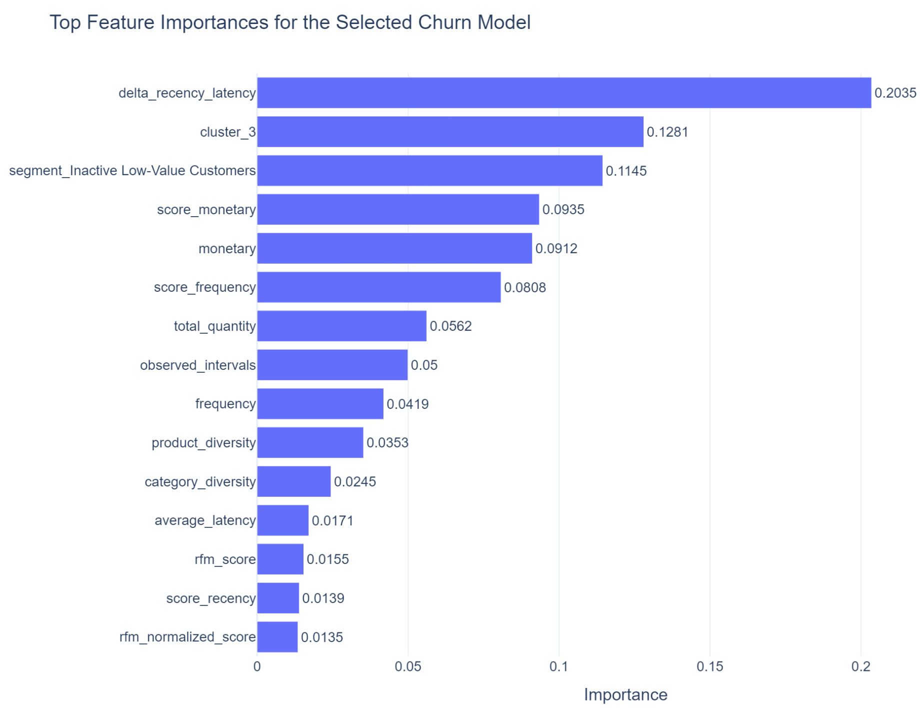

Figure 5.8: Feature importance for the selected churn prediction model.

windows would naturally produce different labels. The churn class is also small in absolute terms, so the excellent performance metrics should be interpreted with caution. The predicted probabilities quantify behavioural risk rather than certainty, and customers identified as high risk should be understood as candidates for proactive retention strategies rather than customers who have definitively left the business.

5.4.3 Risk-Adjusted Customer Lifetime Value Prediction

The final stage of the workflow integrates customer profitability with the behavioural information obtained during the segmentation and churn analyses. Rather than estimating customer value exclusively from historical purchases, the objective is to approximate a risk-adjusted valuation that combines observed profitability with expected customer retention. This produces a more realistic prioritisation framework by recognising that customers with similar historical revenue may contribute very differently to future business performance.

Several regression algorithms were evaluated using the customer-level features engineered throughout the project. Their predictive performance is summarised in Table 5.7. The Random Forest regressor achieved the best overall performance, with the lowest RMSE and MAE and the highest (R^2), indicating that it reproduced the constructed valuation framework more accurately than the other candidate models.

Table 5.7: Performance comparison of the evaluated Customer Lifetime Value prediction models.

Model	RMSE	MAE	R^2
Random Forest	1936.5286	345.9793	0.9966
XGBoost	2178.4778	473.2691	0.9957
Regression Tree	2607.8713	456.9903	0.9938
K-Nearest Neighbors Regression	4066.1658	880.0326	0.9850
Ridge Regression	8251.9469	4268.8566	0.9384
LASSO Regression	8300.8116	4313.9835	0.9377
Linear Regression	8316.9955	4332.1693	0.9374
Support Vector Regression	35181.7991	19710.3313	-0.1193

The selected model was then evaluated on the independent test set. As reported in Table 5.8, the final Random Forest model retained very strong predictive performance, explaining most of the variation in the constructed CLV target while keeping the average absolute error relatively small.

Table 5.8: Predictive performance of the selected Customer Lifetime Value model on the independent test set.

Model	RMSE	MAE	R^2
Random Forest	2028.6151	363.8659	0.9963

The relationship between predicted and observed customer lifetime values is illustrated in Figure 5.9. Most observations are concentrated around the identity line, indicating that the regression model successfully reproduces the underlying valuation framework across a broad range of customer values. The residual analysis shown in Figure 5.10 further suggests that no major systematic prediction bias is present, although variability naturally increases for the highest-value customers, which represent only a small fraction of the customer base.

Feature importance analysis provides additional insight into the behaviour of the regression model. As shown in Figure 5.11, the estimated customer value is influenced not only by historical profitability but also by behavioural characteristics derived during the previous stages of the workflow, including purchasing recency, transaction frequency, customer tenure, and the outputs of the churn prediction model. This demonstrates that the estimated lifetime value reflects a combination of historical performance and expected future behaviour rather than historical spending alone.

The segment-level results in Table 5.9 show why customer value should not be interpreted only at the individual level. Some segments contribute strongly because they contain many customers, while others contribute through high average expected value. The largest aggregate predicted value is concentrated in Declining Frequent Buyers, mainly because this segment combines a large customer base with a very high average predicted CLV. By contrast, Lost Customers and Inactive Low-Value Customers produce negative expected value under the adopted framework, reflecting weak profitability and unfavourable behavioural assumptions.

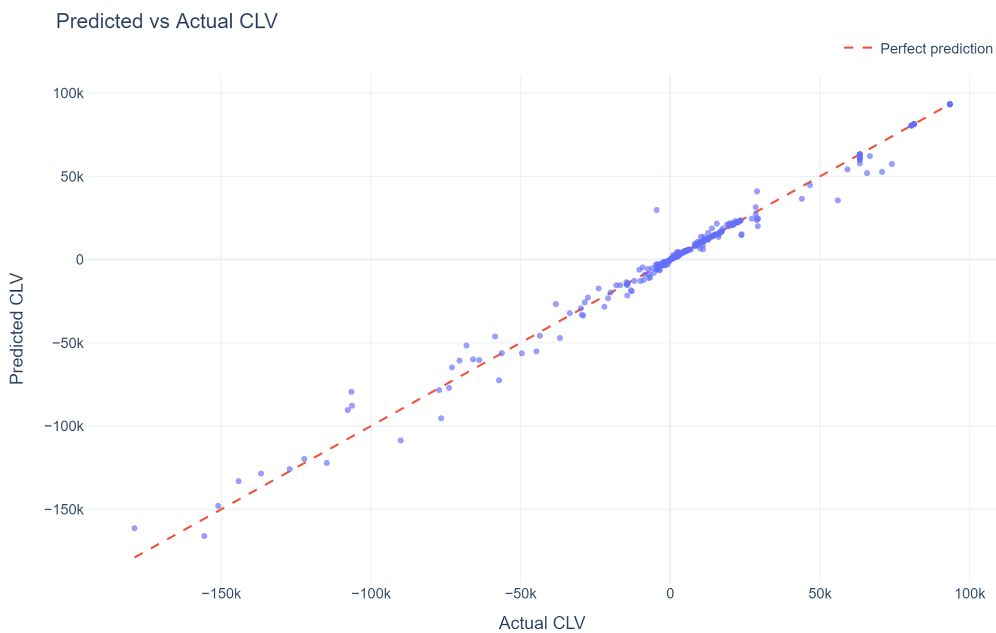

Figure 5.9: Predicted versus observed Customer Lifetime Value on the independent test set.

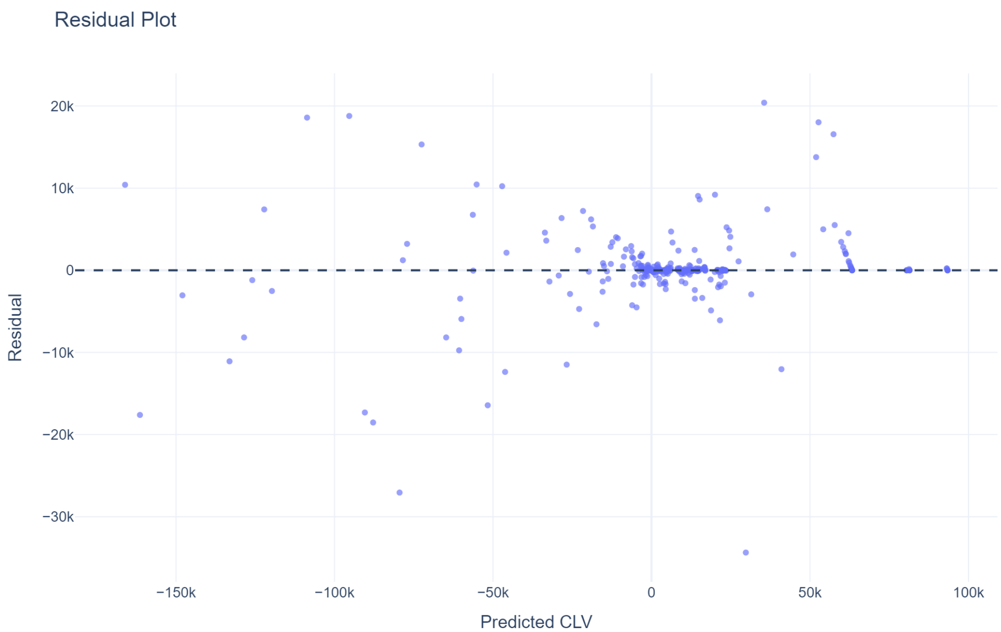

Figure 5.10: Residual analysis for the selected Customer Lifetime Value model.

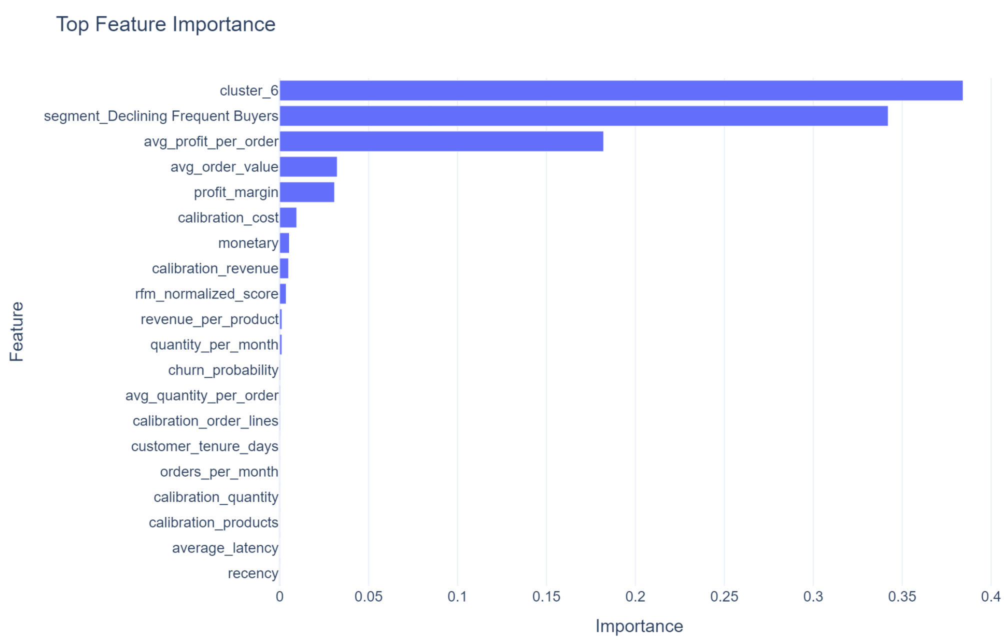

Figure 5.11: Feature importance for the selected Customer Lifetime Value model.

Table 5.9: Customer Lifetime Value summarised by behavioural segment.

Segment	Customers	Avg. Predicted CLV	Total Predicted CLV
Declining Frequent Buyers	1395	76859.60	107219140.34
High Value but Cooling	454	11367.20	5160706.87
Frequent Low-Spend Customers	928	10412.29	9662602.51
Active Minimal Buyers	1767	10098.15	17843434.86
Inactive Minimal Buyers	686	2776.18	1904456.08
Recent Low-Value Customers	609	2748.03	1673548.64
Champions	41	242.58	9945.82
Lost Customers	53	-71046.79	-3765480.03
Inactive Low-Value Customers	273	-18539.27	-5061219.47

These differences are also visible in Figure 5.12 and Figure 5.13. The first figure shows the distribution of predicted CLV within each behavioural segment, while the second aggregates predicted value across the customer portfolio. Together, they show that customer prioritisation should consider both individual-level value and the broader segment structure in which that value is concentrated.

These results should be interpreted within the assumptions adopted to construct the valuation framework. The response variable used for model training is not directly observed in the AdventureWorks database but is derived from historical profitability, behavioural segmentation, predicted retention, and explicit discount-rate assumptions. Consequently, the regression model should be understood as learning to approximate this internally consistent valuation framework rather than predicting an objectively measurable quantity. Alternative assumptions regarding customer retention or discounting would naturally produce different numerical estimates, although the overall modelling framework would remain unchanged.

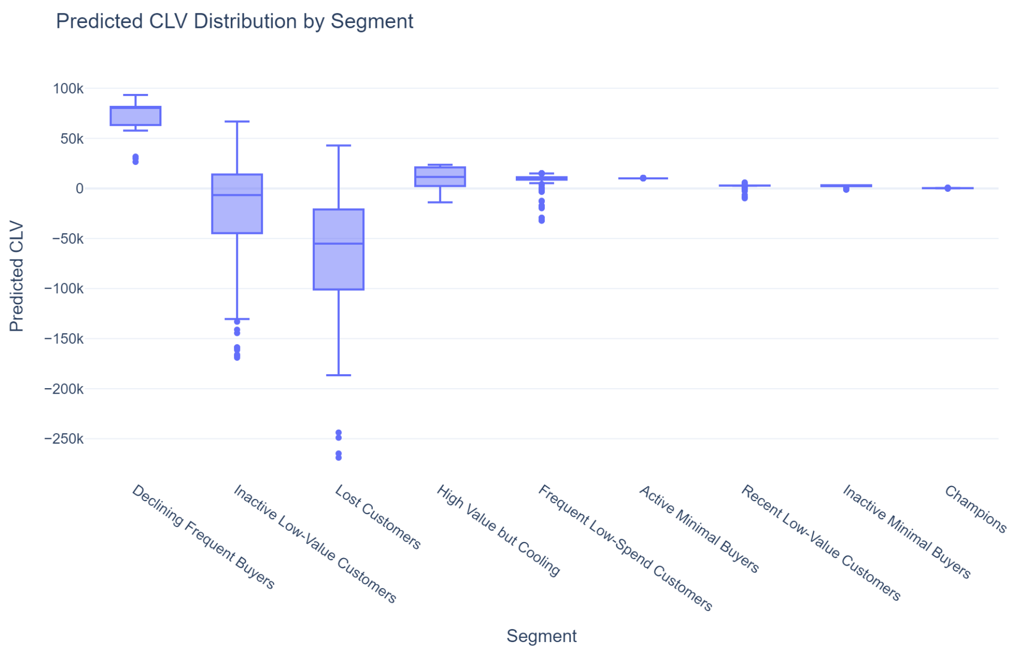

Figure 5.12: Distribution of predicted Customer Lifetime Value across customer segments.

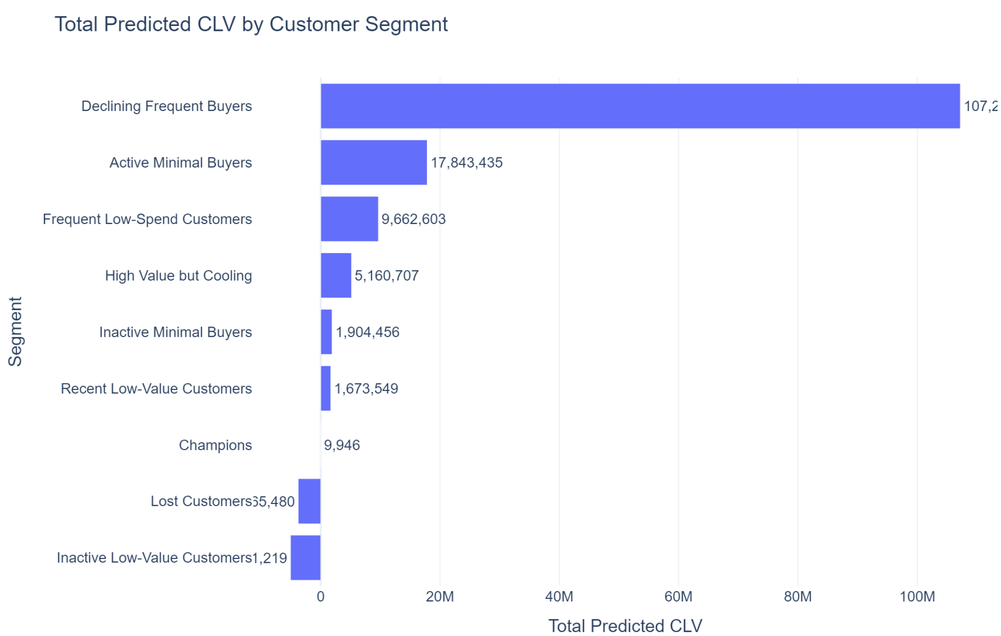

Figure 5.13: Total predicted Customer Lifetime Value aggregated by customer segment.

5.5 Technical Appendix (Optional)

The main chapter emphasises the practical application of customer analytics using transactional retail data. This appendix complements the discussion by summarising the principal mathematical concepts underlying the implementation. Rather than providing exhaustive theoretical derivations, it introduces the core formulations used throughout the workflow, including behavioural customer segmentation, the operational definition of churn, and the construction of the risk-adjusted Customer Lifetime Value (CLV) framework.

5.5.1 RFM Analysis

Recency, Frequency, and Monetary (RFM) analysis is one of the most widely used approaches for summarising customer purchasing behaviour in transactional businesses. Instead of modelling the complete purchase history, it represents each customer through three complementary behavioural variables:

- **Recency**, measuring the time elapsed since the customer's most recent purchase.
- **Frequency**, representing the total number of completed purchases during the observation period.
- **Monetary value**, describing the customer's cumulative spending.

For customer i, Recency is computed as

$$R_i = t_{\text{ref}} - t_{\text{last},i}, \tag{5.1}$$

where t_{ref} denotes the reference date used for the analysis and $t_{\text{last},i}$ is the date of the customer's most recent transaction.

Frequency is defined as

$$F_i = n_i, \tag{5.2}$$

where n_i is the number of completed purchases.

Monetary value is calculated as

$$M_i = \sum_{j=1}^{n_i} v_{ij}, \tag{5.3}$$

where v_{ij} represents the monetary value of transaction j.

Although conceptually simple, these three variables capture complementary aspects of customer behaviour and provide the behavioural representation used throughout the remainder of the workflow. Because Frequency and Monetary value typically exhibit strong positive skewness, additional preprocessing is required before applying distance-based clustering algorithms.

5.5.1.1 Box-Cox Transformation

The RFM variables exhibit markedly different distributions, with Frequency and Monetary value generally dominated by a relatively small number of high-value observations. Since K-Means relies on Euclidean distances, these extreme values can disproportionately influence the clustering process.

To reduce this effect, each RFM variable is transformed using the Box-Cox transformation,

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \lambda \neq 0, \\ \log(y), & \lambda = 0, \end{cases} \tag{5.4}$$

where the transformation parameter λ is estimated from the observed data.

After transformation, each variable is standardised to zero mean and unit variance before clustering. The transformed variables are used exclusively during model fitting; all business interpretation remains based on the original RFM variables.

5.5.1.2 K-Means Clustering

Customer segmentation is performed using the K-Means clustering algorithm, which partitions customers into K groups by minimising the within-cluster sum of squared Euclidean distances,

$$\min_{C_1,\dots,C_K} \sum_{k=1}^{K} \sum_{x_i \in C_k} |x_i - \mu_k|^2 , \tag{5.5}$$

where μ_k denotes the centroid of cluster k.

The optimisation alternates between assigning observations to their nearest centroid and updating the centroids according to the mean of the assigned observations until convergence.

Since no ground-truth customer labels exist, the optimal number of clusters is determined using complementary internal validation metrics together with business interpretability. Once the clusters have been identified, they are interpreted using the original RFM variables rather than the transformed feature space, allowing each segment to be associated with a meaningful purchasing profile.

5.5.2 Behavioural Churn Definition

Unlike subscription-based businesses, transactional retail datasets rarely contain explicit churn events. Customer inactivity must therefore be inferred from observed purchasing behaviour rather than directly measured. The implementation adopted in this project defines churn through a behavioural rule that combines customer engagement during the calibration period with observed activity during a subsequent validation period.

Let S_i denote the behavioural segment assigned to customer i, R_i the customer's recency, $\bar{L}_i$ the average time between consecutive purchases (purchase latency), and Score_i the within-segment RFM score. Furthermore, let $\overline{\text{Score}}_{S_i}$ represent the average RFM score of the customer's segment.

A customer is considered a potential churn candidate when all behavioural conditions

$$S_i \in \mathcal{L},$$

$$R_i > \bar{L}_i,$$

and

$$\text{Score}_i < \overline{\text{Score}}_{S_i}$$

are simultaneously satisfied, where $\mathcal{L}$ denotes the set of low-activity customer segments identified during the RFM analysis.

The final churn label is then determined by inspecting future purchasing behaviour during the validation period,

$$y_i = \begin{cases} 1, & \text{if the customer makes no purchases during the validation window,} \\ 0, & \text{otherwise.} \end{cases} \tag{5.6}$$

This formulation differs from the common approach of defining churn solely through future inactivity. Instead, it first identifies customers whose historical purchasing behaviour is already consistent with progressive disengagement and subsequently verifies whether this behaviour persists during the validation period. By separating behavioural identification from future validation, the resulting target remains suitable for supervised learning while reducing the risk of labelling temporarily inactive customers as permanent churners.

The chronological separation between calibration and validation periods also prevents information leakage, since every explanatory variable is computed exclusively from information that would have been available at the time the prediction is made.

5.5.2.1 Classification Metrics

Customer churn prediction is formulated as a binary classification problem. Let

- TP: true positives,
- FP: false positives,
- TN: true negatives,
- FN: false negatives.

Several complementary metrics are used throughout the chapter to evaluate predictive performance.

Precision measures the proportion of customers predicted to churn who actually become inactive,

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{5.7}$$

where higher values indicate fewer false positive predictions.

Recall, also referred to as sensitivity, measures the proportion of actual churners correctly identified by the classifier,

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{5.8}$$

making it particularly relevant when missing high-risk customers carries a significant commercial cost.

The harmonic mean of Precision and Recall is given by the F1-score,

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{5.9}$$

which provides a balanced summary of classifier performance under class imbalance.

Threshold-independent evaluation is performed using the Receiver Operating Characteristic (ROC) curve and the Precision–Recall curve. While ROC analysis measures the trade-off between the true positive rate and false positive rate across all possible classification thresholds, Precision–Recall curves provide a more informative assessment when the positive class is relatively uncommon, as is typically the case in customer retention problems.

From a business perspective, ranking quality is often more important than binary classification accuracy. For this reason, the implementation also evaluates cumulative gains, which quantify how effectively the model concentrates future churners within the highest-ranked customers. This ranking perspective is particularly useful for prioritising retention campaigns when marketing resources are limited.

5.5.3 Risk-Adjusted Customer Lifetime Value

Customer Lifetime Value (CLV) is commonly defined as the present value of the future profits expected from a customer relationship. A general formulation is given by

$$\mathrm{CLV} = \sum_{t=1}^{T} \frac{\mathbb{E}[P_t]}{(1+d)^t}, \tag{5.10}$$

where $\mathbb{E}[P_t]$ denotes the expected customer profit during period t, d is the discount rate, and T represents the planning horizon.

In practice, however, neither future profits nor customer retention are directly observable in non-contractual retail businesses. The implementation presented in this chapter therefore constructs a risk-adjusted valuation framework that combines observed customer profitability with behavioural information derived from the previous stages of the workflow.

Let $\hat{P}_i$ denote the estimated average profit generated by customer i, c_{S_i} the historical churn rate associated with the behavioural segment (S_i), and (d_{S_i}) the corresponding segment-specific discount rate. Assuming that the probability of customer retention remains approximately constant within each behavioural segment, the expected Customer Lifetime Value is estimated as

$$\widehat{\mathrm{CLV}}_i = \sum_{t=1}^{T} \frac{\hat{P}_i \left(1 - c_{S_i}\right)^{t-1}}{\left(1 + d_{S_i}\right)^t}, \tag{5.11}$$

where the term

$$\left(1 - c_{S_i}\right)^{t-1}$$

represents the probability that customer i remains active up to period t.

Unlike the classical formulation, both the expected retention and the discount rate depend on the customer's behavioural segment rather than being assumed identical across the entire customer base. This allows customers exhibiting different purchasing patterns to be valued under different behavioural assumptions while preserving a transparent and reproducible valuation framework.

The resulting quantity should not be interpreted as an objectively measurable future lifetime value. Instead, it represents a model-based valuation constructed from observed profitability together with explicit assumptions regarding customer retention and the time value of money.

Once the target variable has been constructed, Customer Lifetime Value prediction becomes a supervised regression problem,

$$\widehat{\mathrm{CLV}}_i = f(\mathbf{x}_i) + \varepsilon_i, \tag{5.12}$$

where $\mathbf{x}_i$ denotes the customer-level feature vector engineered throughout the project, $f(\cdot)$ represents the regression model, and ε_i captures the unexplained prediction error.

Rather than estimating future profits directly, the regression model learns to approximate the valuation framework defined by Equation 5.11. This considerably reduces the computational effort required to evaluate new customers while maintaining consistency with the behavioural and financial assumptions adopted throughout the analytical workflow.

Chapter 6

Understanding Customer Feedback through NLP

Customer reviews capture an aspect of product performance that is difficult to obtain from structured transactional data alone. Beyond the numerical rating, written feedback provides context about what customers value, the problems they encounter after purchase, and the language they use to describe their experience. Analysing this information at scale allows organisations to move beyond aggregate satisfaction metrics and identify recurring themes that can support product improvement, quality assurance, and customer service.

This chapter develops a sentiment classification workflow using the **Amazon Reviews 2023** dataset, focusing on the `All_Beauty` product category. Customer reviews are transformed into numerical representations using TF–IDF features and classified with a multinomial Logistic Regression model. The project also extends beyond model development by packaging the trained Scikit-Learn pipeline as a lightweight inference service exposed through Azure Functions. Running inside Docker and using Azurite to emulate Azure Storage locally, the deployment demonstrates how the analytical workflow can be reproduced in a production-style environment without requiring an Azure subscription.

6.1 Problem Definition

Online retailers receive customer feedback continuously, often accumulating thousands of reviews for a single product. While average star ratings provide a useful summary, they conceal the reasons behind customer satisfaction or dissatisfaction. Two products with similar ratings may generate very different patterns of feedback, making it difficult to identify the issues that deserve attention.

Manual review of this volume of text is rarely practical. As product catalogues expand, organisations need automated methods capable of analysing large collections of reviews consistently while preserving enough linguistic information to distinguish positive, neutral, and negative opinions. Such models can support downstream applications including customer experience monitoring, product quality assessment, and the early detection of emerging issues.

This project addresses that problem by formulating sentiment analysis as a supervised multi-class classification task. Customer ratings are converted into three sentiment categories—negative, neutral, and positive—and the corresponding review text is used to predict the underlying sentiment. The analysis is based on the **Amazon Reviews 2023** dataset, one of the largest publicly available collections of Amazon customer reviews, which combines large-scale review data with detailed product information for research purposes [30].

A practical challenge in review modelling is preventing overly optimistic evaluation caused by information leakage. Reviews of the same product frequently share vocabulary, describe similar characteristics, or even repeat common expressions. If reviews from a single product appear in both the training and test sets, reported performance may reflect memorisation of product-specific language rather than genuine generalisation. To obtain a more realistic assessment, this project partitions the data by product rather than by individual review, ensuring that the model is evaluated on products it has never encountered during training.

The objective is therefore not only to develop an accurate sentiment classifier, but also to demonstrate a workflow that remains reproducible from model development through deployment. The resulting pipeline illustrates how a classical Natural Language Processing approach can be transformed into a lightweight inference service suitable for integration into larger analytical systems while remaining entirely executable on a local machine.

6.2 Modeling Approach

The objective of this project is to develop a sentiment classifier that is both accurate and easily reproducible. Rather than pursuing increasingly complex language models, the emphasis is placed on building a transparent workflow that follows well-established Natural Language Processing practices while remaining suitable for deployment in a lightweight production environment.

The modelling pipeline begins by transforming raw review text into a structured numerical representation. Each review is normalised through a standard text preprocessing stage, including lowercasing, tokenisation, removal of punctuation, stop-word filtering, and lemmatisation. These steps reduce lexical variability while preserving the semantic information required for sentiment classification.

The processed text is then represented using **Term Frequency–Inverse Document Frequency (TF–IDF)**. Unlike simple word counts, TF–IDF assigns greater importance to terms that are characteristic of individual reviews while reducing the influence of words that appear frequently throughout the corpus. This representation has remained a strong baseline for many document classification problems because it combines computational efficiency with good predictive performance on sparse textual data.

The resulting feature vectors are used to train a multinomial Logistic Regression classifier. Despite its conceptual simplicity, Logistic Regression has consistently demonstrated competitive performance for sentiment analysis tasks when paired with appropriate textual representations. The model is also straightforward to interpret, computationally efficient to train, and well suited to deployment scenarios where low inference latency and reproducibility are desirable.

A key aspect of the experimental design is the evaluation strategy. Instead of randomly splitting individual reviews into training and testing sets, the data are partitioned at the **product level**. All reviews associated with a particular product are assigned exclusively to either the training or the testing partition. This prevents the model from learning product-specific vocabulary that could otherwise appear in both sets, providing a more realistic estimate of how well the classifier generalises to previously unseen products.

To improve reproducibility and reduce the risk of implementation inconsistencies, the preprocessing steps, TF–IDF vectoriser, and trained classifier are encapsulated within a single Scikit-Learn pipeline. This design ensures that identical transformations are applied during both training and inference, while also allowing the complete workflow to be exported as a single serialized model artifact.

This packaged pipeline forms the bridge between model development and deployment. Rather than requiring custom inference code, the serialized artifact can be loaded directly by the Azure Functions application, which exposes the trained model through a lightweight HTTP endpoint. Running inside Docker, with Azurite providing a local emulation of Azure Storage, the deployment demonstrates how a traditional machine learning workflow can be packaged as a production-style inference service while remaining fully executable on a local machine.

6.3 Methodology and Implementation

The analytical workflow consists of two complementary stages. The first focuses on developing and evaluating a sentiment classification model from customer reviews, while the second packages the trained model as a lightweight inference service suitable for reproducible local deployment. Although these stages serve different purposes, they share the same Scikit-Learn pipeline, ensuring that identical preprocessing and prediction steps are applied during both model development and inference.

6.3.1 Data Preparation

The analysis begins by extracting the review text, product identifier and customer rating from the Amazon Reviews 2023 `All_Beauty` category. Records containing empty review text are removed, duplicated observations are discarded, and the remaining data are validated before model development. The original five-star rating is then converted into three sentiment categories, grouping one- and two-star reviews as **negative**, three-star reviews as **neutral**, and four- and five-star reviews as **positive**.

To obtain a realistic assessment of predictive performance, the data are partitioned at the **product level** rather than at the review level. All reviews associated with a given product are assigned exclusively to either the training or the testing partition. This prevents product-specific vocabulary from appearing in both sets and provides a more reliable estimate of how well the classifier generalises to previously unseen products.

6.3.2 Text Representation and Sentiment Classification

Customer reviews are first normalised through a standard text preprocessing workflow, including lowercasing, punctuation removal, stop-word filtering and lemmatisation. The cleaned text is subsequently transformed into numerical feature vectors using TF–IDF, allowing each review to be represented according to the relative importance of its terms within the corpus.

These feature vectors are used to train a multinomial Logistic Regression classifier. Rather than implementing each processing step independently, the preprocessing operations, TF–IDF vectoriser and classifier are encapsulated within a single Scikit-Learn pipeline. This design simplifies experimentation while ensuring that exactly the same sequence of transformations is applied whenever new reviews are analysed.

6.3.3 Topic Discovery

While sentiment classification identifies whether customers express positive or negative opinions, it does not explain the underlying reasons for those opinions. To complement the classifier, the project applies topic modelling to the subset of negative reviews in order to identify recurring themes associated with customer dissatisfaction.

Candidate topic models with different numbers of topics are compared using quantitative and qualitative criteria, balancing reconstruction quality with the interpretability of the resulting topics. The selected model provides a concise summary of the most common issues reported by customers, supporting subsequent business interpretation.

6.3.4 Packaging for Local Deployment

Once the sentiment classifier has been trained, the complete Scikit-Learn pipeline is exported as a serialized model artifact. This packaged pipeline is then exposed through an Azure Functions HTTP endpoint, allowing new customer reviews to be submitted for inference without requiring the original training environment.

The inference service runs inside Docker, while Azurite provides a local emulation of Azure Storage, enabling the complete deployment to be executed on a standard workstation without an Azure subscription. Rather than introducing additional machine learning functionality, this deployment layer demonstrates how an analytical model can be packaged into a lightweight, reproducible service using tools commonly encountered in production environments.

6.4 Results, Discussion, and Limitations

6.4.1 Data Preparation

The data preparation pipeline retained nearly the entire analytical sample while removing duplicated observations and records containing unusable review text. As summarised in Table 6.1, more than 99% of the original observations were preserved after cleaning, indicating that the source data were of high quality and required only minimal preprocessing before model development.

Table 6.1: Summary of data quality checks performed before model development. The cleaning pipeline retained almost all observations while removing duplicates and unusable records.

Metric	Value
Input rows	50,000
Whole-row duplicates in raw sample	45
Invalid ratings	0
Empty review text	47
Missing product identifiers	0
Analytical duplicates removed	46
Prepared rows	49,907
Retention rate	99.81%

The small number of discarded records suggests that missing information and duplicated reviews were unlikely to influence the subsequent modelling process. Consequently, the resulting dataset provides a reliable foundation for sentiment classification while avoiding unnecessary sources of bias introduced by incomplete or redundant observations.

6.4.2 Rating Distribution

Customer ratings exhibit the characteristic imbalance commonly observed in online retail platforms, where positive reviews substantially outnumber negative ones. As shown in Table 6.2, almost sixty percent of the reviews awarded the maximum five-star rating, whereas low ratings represent a considerably smaller proportion of the dataset.

Table 6.2: Distribution of customer ratings in the analytical dataset.

Rating	Reviews	Median words	Review share
1	7,304	20	14.64%
2	3,017	24	6.05%
3	4,019	24	8.05%
4	5,635	25	11.29%
5	29,932	17	59.98%

Rather than artificially balancing the classes, the original distribution was preserved throughout the analysis. This decision produces a more realistic evaluation setting by reflecting the conditions under which sentiment classification models would typically operate in production. At the same time, it highlights the importance of evaluating model performance using class-specific metrics in addition to overall accuracy, since aggregate measures alone may overestimate performance on underrepresented sentiment classes.

6.4.3 Sentiment Classification Performance

The TF–IDF Logistic Regression model substantially outperformed the most-frequent baseline across all evaluation metrics, demonstrating that the classifier learned meaningful linguistic patterns rather than simply exploiting the predominance of positive reviews. As shown in Table 6.3, overall accuracy increased from 71.1% to 81.3%, while the macro-averaged F1-score more than doubled, providing a more balanced assessment of performance across all sentiment classes.

Table 6.3: Comparison between the baseline classifier and the TF–IDF logistic regression model.

Model	Macro F1	Weighted F1	Accuracy
Most-frequent baseline	0.277	0.591	0.711
TF–IDF Logistic Regression	**0.674**	**0.830**	**0.813**

Model	Macro F1	Weighted F1	Accuracy

The improvement in macro F1 is particularly important because it gives equal weight to each sentiment category regardless of its frequency. Since positive reviews dominate the dataset, accuracy alone would provide an incomplete picture of model performance. The increase in macro F1 therefore indicates that the classifier captures meaningful distinctions across all three sentiment classes rather than relying primarily on the majority class.

A more detailed breakdown is presented in Table 6.4. The classifier achieved its strongest performance for positive reviews, with an F1-score of 0.906, reflecting the relatively consistent language used in favourable customer feedback. Negative reviews were also identified reliably, achieving an F1-score of 0.755. As expected, the neutral class proved considerably more challenging, exhibiting lower precision and recall because reviews with intermediate ratings often combine both positive and negative opinions within the same text.

Table 6.4: Precision, recall and F1-score for each sentiment class on the test set.

Sentiment	Precision	Recall	F1-score	Reviews
Negative	0.738	0.774	0.755	1,729
Neutral	0.283	0.491	0.359	678
Positive	0.955	0.861	0.906	5,927

These class-level metrics are further illustrated by the confusion matrix shown in Figure 6.1. Most classification errors occur between neighbouring sentiment categories rather than between strongly positive and strongly negative reviews. This behaviour is consistent with the subjective nature of customer feedback, where moderately positive or moderately negative reviews frequently contain mixed opinions that are difficult to assign to a single sentiment category.

6.4.4 Topic Modelling

Although sentiment classification identifies whether a review expresses a favourable or unfavourable opinion, it does not reveal the underlying reasons behind customer dissatisfaction. To complement the classifier, topic modelling

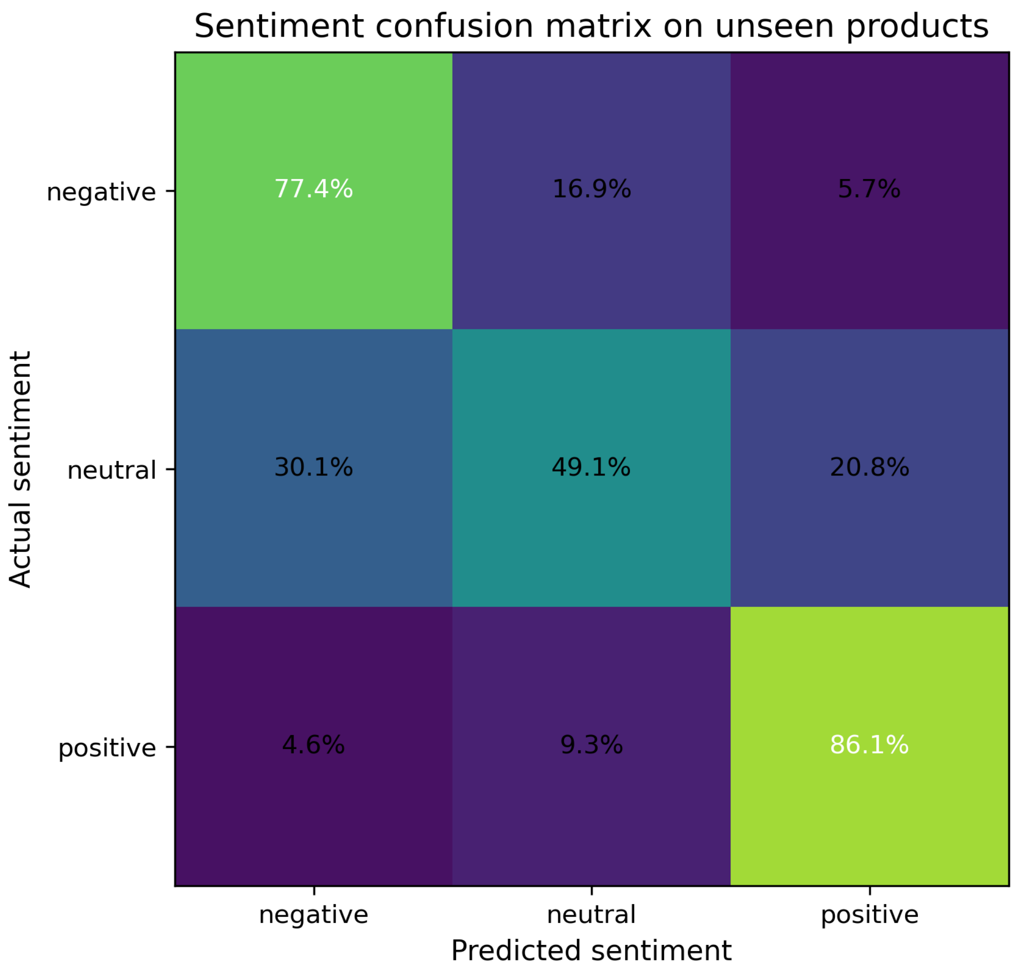

Figure 6.1: Confusion matrix for sentiment classification on previously unseen products. The figure shows that most prediction errors occur around the neutral class, while positive and negative reviews are generally well separated.

was applied to the subset of negative reviews with the objective of identifying the most frequently discussed issues across products.

Several candidate models were evaluated by varying the number of topics. As summarised in Table 6.5, the final configuration of five topics was selected as the best compromise between reconstruction quality, topic coherence, stability and interpretability. While models with a larger number of topics achieved marginal improvements in reconstruction error, these gains came at the expense of producing more fragmented and less meaningful themes.

Table 6.5: Quantitative comparison of candidate topic models. The five-topic solution was selected as the best compromise between coherence, stability and interpretability.

Topics	Reconstruction error	Stability	Term diversity	NPMI coherence	Selected
4	87.632	0.897	0.875	0.155	No
5	**87.393**	**0.892**	**0.850**	**0.174**	**Yes**
6	87.200	0.877	0.847	0.174	No
7	87.015	0.907	0.869	0.172	No
8	86.846	0.878	0.833	0.161	No

The resulting topics correspond to recurring product issues that can be readily interpreted from a business perspective. Rather than representing abstract statistical groupings, they describe concrete themes such as products failing to perform as expected, concerns about durability, perceived poor value for money and quality issues affecting specific product categories.

Table 6.6: Summary of the topics identified in customer reviews and their associated business relevance.

Topic	Topic label	Actionability	Reviews	Products	Negative share
0	Product did not work as expected	High	1,243	1,117	75.70%
3	Limited durability or staying power	High	1,122	1,010	73.08%

Topic	Topic label	Actionability	Reviews	Products	Negative share
2	Poor value for money	High	705	661	94.75%
4	Hair-product quality concerns	Medium	1,739	1,471	70.79%
1	General dissatisfaction (residual)	Low	8,176	6,259	68.71%

From an operational perspective, these topics are considerably more actionable than sentiment labels alone. A retailer can readily identify which complaints are most prevalent, prioritise investigations into specific product defects, monitor whether recurring issues persist over time and evaluate the effectiveness of corrective actions. In this way, topic modelling complements sentiment classification by transforming large volumes of unstructured customer feedback into information that can directly support product management and customer experience decisions.

6.4.5 Discussion and Limitations

The results demonstrate that a classical Natural Language Processing pipeline can provide reliable sentiment classification while remaining computationally efficient and straightforward to deploy. By combining TF–IDF features with Logistic Regression, the proposed workflow achieves strong predictive performance without relying on computationally intensive deep learning architectures. This balance between accuracy, interpretability and reproducibility makes the approach well suited to many retail analytics applications, particularly where transparency and ease of maintenance are important considerations.

The integration of topic modelling further extends the analytical value of the workflow. Rather than treating sentiment prediction as an end in itself, the project illustrates how customer feedback can be transformed into structured business intelligence. Identifying recurring themes behind negative reviews provides actionable insights that can support product quality monitoring, customer experience initiatives and prioritisation of corrective actions.

Several limitations should nevertheless be acknowledged. First, the analysis focuses exclusively on the `All_Beauty` product category. Although the modelling framework is readily transferable to other retail domains, vocabulary and customer behaviour may differ substantially across product categories, making additional training necessary before deployment in a different context.

Second, the sentiment labels are derived directly from customer ratings rather than manual annotation of review text. While this strategy enables the construction of large training datasets, it assumes that numerical ratings accurately reflect the sentiment expressed in the written review. In practice, some customers assign moderate ratings despite expressing strongly positive or negative opinions, introducing a degree of label uncertainty that cannot be eliminated entirely.

Finally, the project deliberately adopts a classical machine learning approach instead of recent transformer-based language models. Large language models can often achieve higher predictive performance, particularly for nuanced linguistic phenomena such as sarcasm or complex contextual relationships. However, they also require substantially greater computational resources, more complex deployment pipelines and longer inference times. Within the objectives of this project, the selected approach provides an effective compromise between predictive performance, computational efficiency and reproducibility while naturally supporting deployment as a lightweight inference service.

A complete Python implementation of the workflow presented in this chapter is available in the project repository.

An Azure-compatible deployment implementing the inference service presented in this chapter is available in the project repository.

6.5 Technical Appendix

The workflow presented in this chapter follows a classical Natural Language Processing (NLP) pipeline for supervised document classification. Although transformer-based language models have become increasingly popular, statistical text representations combined with linear classifiers remain highly competitive for many industrial applications owing to their computational efficiency, interpretability and ease of deployment. Comprehensive treatments of sentiment analysis and opinion mining are provided by Pang and Lee [31] and Liu [32].

6.5.1 Text Preprocessing

Customer reviews are first transformed into a standardised textual representation before feature extraction. The objective of preprocessing is to reduce lexical variability while preserving the semantic information required for sentiment classification.

The preprocessing pipeline consists of the following sequential operations:

1. **Lowercasing**, ensuring that words differing only by capitalisation are treated as identical tokens.
2. **Tokenisation**, which decomposes each review into individual lexical units.
3. **Removal of punctuation and non-alphabetic characters**, reducing noise introduced by formatting symbols.
4. **Stop-word filtering**, eliminating extremely frequent words (e.g., *the*, *and*, *is*) that contribute little discriminative information.
5. **Lemmatisation**, mapping inflected words to their canonical dictionary form. For example, *running*, *runs* and *ran* are all reduced to the lemma *run*.

Collectively, these transformations reduce the dimensionality of the vocabulary and improve the consistency of the subsequent feature representation. The resulting corpus is therefore less sparse and more suitable for statistical learning algorithms such as TF–IDF vectorisation followed by Logistic Regression.

6.5.2 TF–IDF Feature Representation

After preprocessing, each review is transformed into a numerical feature vector using the **Term Frequency–Inverse Document Frequency (TF–IDF)** weighting scheme. Unlike a simple bag-of-words representation, TF–IDF assigns higher importance to terms that are frequent within an individual document but relatively uncommon across the entire corpus. This weighting emphasises words that are more informative for distinguishing between sentiment classes while reducing the influence of ubiquitous vocabulary.

For a term t appearing in document d, the TF–IDF weight is computed as

$$\text{TFIDF}(t, d) = \text{TF}(t, d) \cdot \log\left(\frac{N}{DF(t)}\right), \tag{6.1}$$

where:

- $\text{TF}(t, d)$ is the **term frequency**, measuring how often term t appears in document d;
- $DF(t)$ is the **document frequency**, corresponding to the number of documents containing term t;
- N is the total number of documents in the corpus.

The first component measures the local importance of a term within a review, whereas the inverse document frequency discounts words that occur in many documents. As a result, highly informative terms receive greater weights than common vocabulary shared across most reviews.

Applying Equation 6.1 to every term in the vocabulary produces a sparse document-term matrix in which each review is represented as a high-dimensional feature vector. Although the resulting feature space may contain thousands of dimensions, most entries are zero because any individual review contains only a small subset of the overall vocabulary. Sparse representations of this kind are computationally efficient and are particularly well suited to linear classifiers such as Logistic Regression.

6.5.3 Multinomial Logistic Regression

The sentiment classifier is based on **multinomial Logistic Regression**, a probabilistic linear model designed for multi-class classification problems. Given a TF–IDF feature vector **x** representing a customer review, the classifier estimates the probability that the review belongs to each sentiment class and assigns the class with the highest predicted probability.

For a problem with K sentiment classes, the probability of assigning observation **x** to class k is computed using the softmax function,

$$P(y = k \mid \mathbf{x}) = \frac{\exp\left(\theta_k^\top \mathbf{x}\right)}{\sum_{j=1}^{K} \exp\left(\theta_j^\top \mathbf{x}\right)}, \tag{6.2}$$

where:

- **x** denotes the TF–IDF feature vector associated with a review;
- θ_k is the parameter vector corresponding to sentiment class k;

- $K = 3$ represents the negative, neutral and positive sentiment categories.

The softmax transformation ensures that the predicted probabilities form a valid probability distribution,

$$\sum_{k=1}^{K} P(y = k \mid \mathbf{x}) = 1. \tag{6.3}$$

Model parameters are estimated by solving the regularised optimisation problem

$$\hat{\theta} = \arg\min_{\theta} \left\{ -\mathcal{L}(\theta) + \lambda \left\| \theta \right\|_2^2 \right\}. \tag{6.4}$$

where $\mathcal{L}(\theta)$ denotes the log-likelihood of the training data, λ is the regularisation parameter, and $|\theta|_2^2$ is the squared Euclidean norm of the parameter vector.

The L2 penalty discourages unnecessarily large parameter estimates, reducing the risk of overfitting while improving generalisation to previously unseen products. This regularisation strategy is particularly effective for TF–IDF representations, where the number of features is typically large and the document-term matrix is highly sparse.

To ensure reproducibility, the preprocessing operations, TF–IDF vectoriser and Logistic Regression classifier are encapsulated within a single Scikit-Learn pipeline. Consequently, exactly the same sequence of transformations is applied during both model training and inference, eliminating inconsistencies between development and deployment environments.

6.5.4 Topic Modelling

While sentiment classification determines whether a review expresses a positive or negative opinion, it does not explain the underlying reasons behind customer satisfaction or dissatisfaction. Topic modelling addresses this limitation by identifying latent semantic themes shared across groups of reviews, allowing large collections of customer feedback to be summarised into a small number of interpretable topics.

The analysis is performed exclusively on the subset of reviews classified as negative. Restricting the corpus to dissatisfied customers increases the semantic coherence of the extracted topics and produces themes that are directly relevant for product quality assessment and customer experience management.

Selecting an appropriate number of topics is a critical modelling decision. A model with too few topics tends to merge distinct customer concerns into overly broad themes, whereas an excessively large number of topics often fragments coherent issues into several overlapping clusters that are difficult to interpret. Consequently, the optimal solution is not necessarily the one that minimises reconstruction error alone.

To balance statistical performance and practical interpretability, several candidate models were evaluated using multiple complementary criteria:

- **Reconstruction error**, measuring how accurately the latent representation reproduces the original document-term matrix.
- **Topic coherence**, evaluating the semantic relatedness of the highest-weighted words within each topic.
- **Model stability**, assessing the consistency of the extracted topics across repeated model fits.
- **Term diversity**, quantifying the extent to which different topics capture distinct vocabularies rather than repeatedly selecting the same high-frequency terms.

The final model was selected by jointly considering these quantitative measures together with qualitative inspection of the resulting topics. This multi-criteria evaluation avoids selecting a statistically optimal solution that lacks practical interpretability and instead favours a compact set of topics that can be readily translated into actionable business insights.

Once the optimal model has been identified, each review is assigned to its dominant topic according to the highest topic probability. The resulting topic assignments provide an interpretable representation of the principal sources of customer dissatisfaction, complementing the sentiment classifier by explaining *why* customers express negative opinions rather than simply identifying *whether* those opinions are positive or negative.

6.5.5 Deployment Notes

Although the primary objective of this project is the development of a sentiment classification model, the complete workflow also incorporates a lightweight deployment layer that demonstrates how the trained model can be packaged for inference. Rather than treating deployment as an independent cloud-computing exercise, the implementation focuses on preserving consistency between model development and production-style execution.

The Scikit-Learn preprocessing pipeline, TF–IDF vectoriser and Logistic Regression classifier are encapsulated within a single pipeline object. Once training is complete, this pipeline is serialized using `joblib`, producing a self-contained artifact that includes every transformation required to process new customer reviews. Consequently, no additional preprocessing logic is required during inference, eliminating the risk of discrepancies between training and deployment environments.

The serialized model is loaded by an Azure Functions application, which exposes a RESTful HTTP endpoint for sentiment prediction. Incoming requests containing raw review text are processed using exactly the same pipeline employed during model development, ensuring that every prediction is generated under identical preprocessing and modelling assumptions.

To facilitate reproducibility, the inference service executes inside a Docker container that encapsulates the Python runtime together with all project dependencies. Containerisation guarantees that the application behaves consistently across different operating systems while simplifying installation and deployment.

Although Azure Functions typically interact with Azure Storage services, this project relies on **Azurite**, Microsoft's local emulator for Azure Storage. Azurite reproduces the behaviour of Azure Blob Storage within a local development environment, allowing the complete inference workflow to execute without requiring an active Azure subscription or cloud resources.

This architecture illustrates a common deployment pattern for classical machine learning models. Model development, serialization and inference remain clearly separated, while the prediction service preserves exactly the same preprocessing pipeline validated during experimentation. As a result, the project demonstrates not only how a statistical learning model can be developed and evaluated, but also how it can be packaged into a reproducible, Azure-compatible inference service suitable for subsequent integration into larger analytical systems.

Chapter 7

Sediment Source Modeling

Sediment source apportionment aims to estimate how much each potential source contributes to the sediment collected at a downstream location. The problem arises in watershed management, erosion assessment, and river restoration, where understanding the origin of transported sediments is often more important than simply measuring their volume.

This project compares three complementary source apportionment approaches using laboratory mixtures with known compositions and synthetic mixtures generated from the same experimental dataset. Alongside a probabilistic fingerprinting model, two deterministic geochemical deconvolution algorithms originally developed for petroleum production allocation are adapted to sediment source apportionment, allowing their performance to be evaluated under controlled conditions.

7.1 Problem Definition

Sediments transported through a watershed are typically generated by multiple erosion processes acting simultaneously. Agricultural fields, forested areas, channel banks, and other landscape units may all contribute material to the sediment ultimately deposited downstream. Although the resulting mixture can be sampled directly, the relative contribution of each source cannot be measured independently, making source apportionment an inverse problem.

A common strategy is to characterize each potential source using a set of geochemical tracers and compare these fingerprints with those measured in the collected sediment. If the selected tracers are sufficiently discriminative, the observed composition can be interpreted as the result of combining material from multiple sources in unknown proportions. The objective is therefore to estimate a vector of source proportions that reproduces the measured composition while satisfying the physical constraints of the mixing process.

The overall source apportionment workflow is illustrated in Figure 7.1. Representative geochemical signatures are first collected from potential sediment sources, which subsequently mix during erosion, transport, and deposition. After sampling and laboratory analysis, the measured composition is used to estimate the relative contribution of each source to the final mixture.

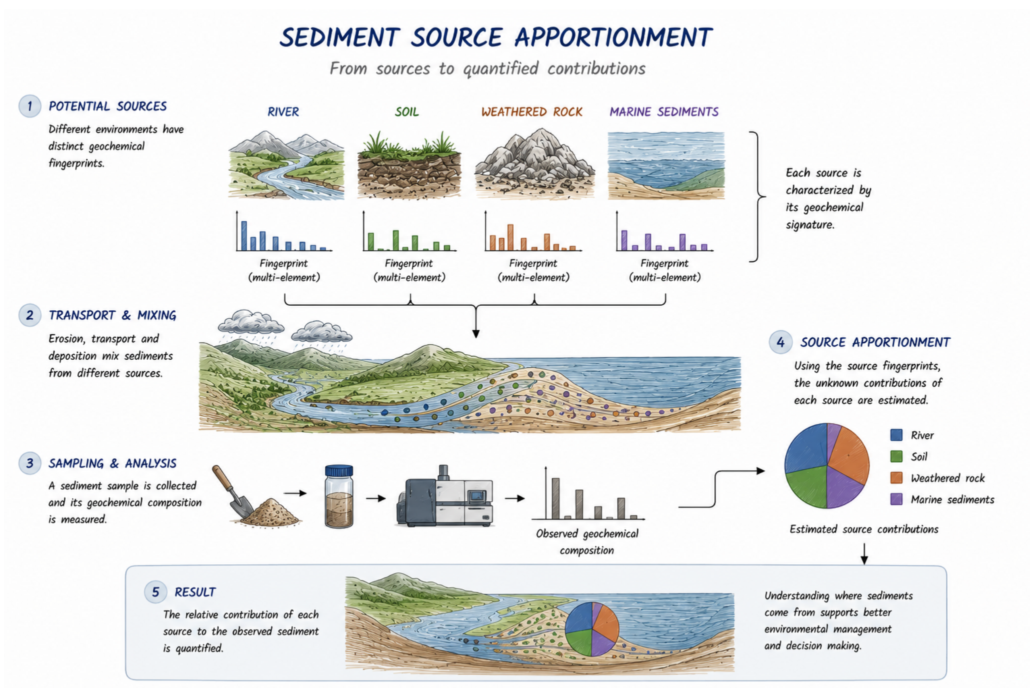

Figure 7.1: Conceptual workflow of sediment source apportionment. Representative geochemical fingerprints are measured for potential sediment sources, transported and mixed within the watershed, and subsequently used to estimate the relative contribution of each source to an observed sediment sample.

Several factors make this estimation challenging. Geochemical signatures from different sources often overlap, laboratory measurements are subject to analytical uncertainty, and the number of available tracers is usually limited. Moreover, the estimated source proportions must remain non-negative and sum to one, introducing additional constraints that distinguish the problem from an unconstrained regression task.

One of the main difficulties when assessing source apportionment methods is that the true source proportions are rarely known for field observations. As a result, performance is often evaluated indirectly through sensitivity analyses or expert interpretation. To overcome this limitation, Batista, Laceby, and Evrard [33] proposed an experimental framework based on laboratory-prepared sediment mixtures with known source proportions. Because the true composition of each mixture is available, different modelling approaches can be evaluated objectively using quantitative performance metrics.

This project adopts that experimental framework and compares three complementary source apportionment methods using the same benchmark dataset. In addition to the laboratory mixtures, the evaluation also includes a large collection of synthetic mixtures generated from the experimental data, allowing model behaviour to be examined over a much broader range of source combinations while preserving complete knowledge of the ground truth.

7.2 Modeling Approach

At first glance, sediment fingerprinting and petroleum production allocation appear to address entirely different scientific problems. The former seeks to determine the origin of sediments transported through a watershed, whereas the latter estimates the contribution of individual reservoirs to a commingled production stream. Despite these differences, both can be formulated as the same inverse problem: recovering the proportions of multiple sources from a compositional mixture subject to physical constraints.

This shared mathematical structure motivated the comparison presented in this chapter. Rather than proposing a new source apportionment algorithm, the objective is to evaluate whether deconvolution methods that have been successfully applied in petroleum geochemistry can also recover sediment source proportions from environmental fingerprinting data. Laboratory-prepared mixtures with known compositions provide an ideal benchmark because every prediction can be compared directly with the true proportions.

The first method considered is the Bootstrapped Mixing Model (BMM) described by Batista, Laceby, and Evrard [33]. This probabilistic framework repeatedly resamples the geochemical observations to propagate measurement variability through the mixing model, producing a distribution of plausible source proportions instead of a single deterministic estimate. Because it explicitly represents uncertainty, the BMM serves as a natural reference against which alternative approaches can be evaluated.

The remaining methods originate from petroleum geochemistry. McCaffrey et al. (2011) [34] formulated the allocation problem as a constrained least-squares optimization, estimating the unknown source proportions directly from elemental concentrations while enforcing non-negativity and mass conservation. Sandoval et al. (2022) [35] later extended this idea by performing the estimation in log-ratio space, an approach specifically designed for compositional data that mitigates numerical issues associated with the constant-sum constraint and improves the treatment of relative geochemical information.

Although neither deterministic model was originally developed for sediment fingerprinting, both rely on assumptions that remain valid for conservative geochemical tracers. Consequently, applying them to sediment source apportionment requires no modification of their mathematical formulation, only a reinterpretation of what constitutes a "source." In petroleum applications, the sources correspond to producing reservoirs, whereas in this study they represent sediment-producing landscape units.

To ensure that differences in performance arise from the modelling strategies rather than the data themselves, all three methods are evaluated using exactly the same benchmark datasets. Their estimates are then compared against the known source proportions, providing a controlled assessment of accuracy, robustness, and the practical advantages and limitations of each formulation.

7.3 Methodology and Implementation

The implementation was organized as three independent workflows, one for each modelling approach. This separation was intentional: each model uses a different representation of the mixing problem, different assumptions about uncertainty, and different diagnostic outputs. Keeping the workflows independent makes the comparison easier to audit and avoids forcing the three methods into a single artificial pipeline.

All methods are evaluated using the same sediment fingerprinting dataset and the same known source proportions. Consequently, any differences in performance arise from the modelling formulation rather than from changes in the input data. This distinction is particularly important because the objective is not only to obtain accurate source apportionments, but also to understand how each formulation behaves when applied to the same controlled mixtures.

7.3.1 Bootstrapped Mixing Model

The Bootstrapped Mixing Model (BMM) serves as the probabilistic baseline throughout this study. Unlike deterministic mixing models that produce a single estimate of the source proportions, the BMM propagates measurement uncertainty through repeated bootstrap resampling. The result is not only an estimate for each sediment source, but a distribution of plausible solutions for every mixture.

The implementation follows the evaluation framework proposed by Batista, Laceby, and Evrard [33]. The workflow begins by examining the available source fingerprints, then selects the tracers with the strongest discriminatory power, and finally estimates the source proportions for both laboratory-prepared and synthetically generated benchmark mixtures.

A complete Python implementation of the Bootstrapped Mixing Model presented in this section is available in the project repository.

7.3.1.1 Exploratory Analysis

Before fitting the mixing model, the source data were inspected to verify that each sediment class was adequately represented and that the candidate geochemical tracers carried sufficient discriminatory information. This step is important because source apportionment depends on separating source signatures rather than simply reproducing a numerical mixture.

Figure 7.2 summarizes the number of samples available for each sediment source. Although the sampling effort is not perfectly balanced, every class contains enough observations to characterize source variability and support the bootstrap resampling procedure.

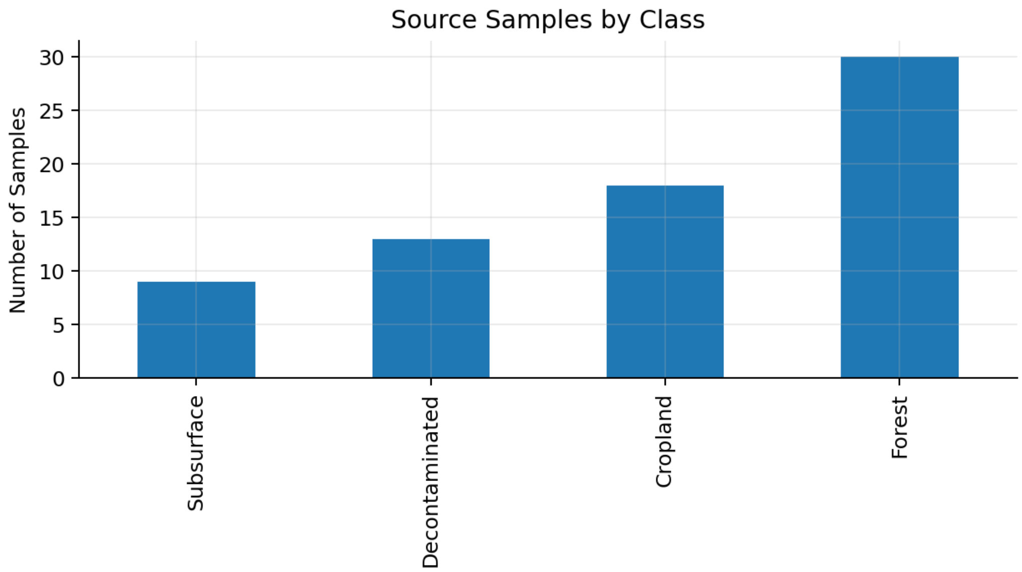

Figure 7.2: Number of samples available for each sediment source.

Candidate tracers were screened using the non-parametric Kruskal–Wallis test. The retained variables, listed in Table 7.1, show statistically significant differences among the four source classes and were therefore selected for the subsequent source apportionment analysis.

Table 7.1: Geochemical tracers retained after Kruskal–Wallis screening.

Tracer	H Statistic	p-value
L	51.647	3.56e-11
h	45.480	7.32e-10
Si	43.607	1.83e-09
C*	34.421	1.61e-07
K	33.015	3.20e-07
Y	23.563	3.08e-05
Mn	13.278	4.07e-03
Zn	12.499	5.86e-03
Ti	9.213	2.66e-02

The selected tracer set was also examined through Linear Discriminant Analysis. As shown in Figure 7.3, most source classes occupy distinct regions of the projected feature space, although some overlap remains. This behaviour is expected in environmental fingerprinting studies and further supports the use of a probabilistic framework rather than a purely classificatory approach.

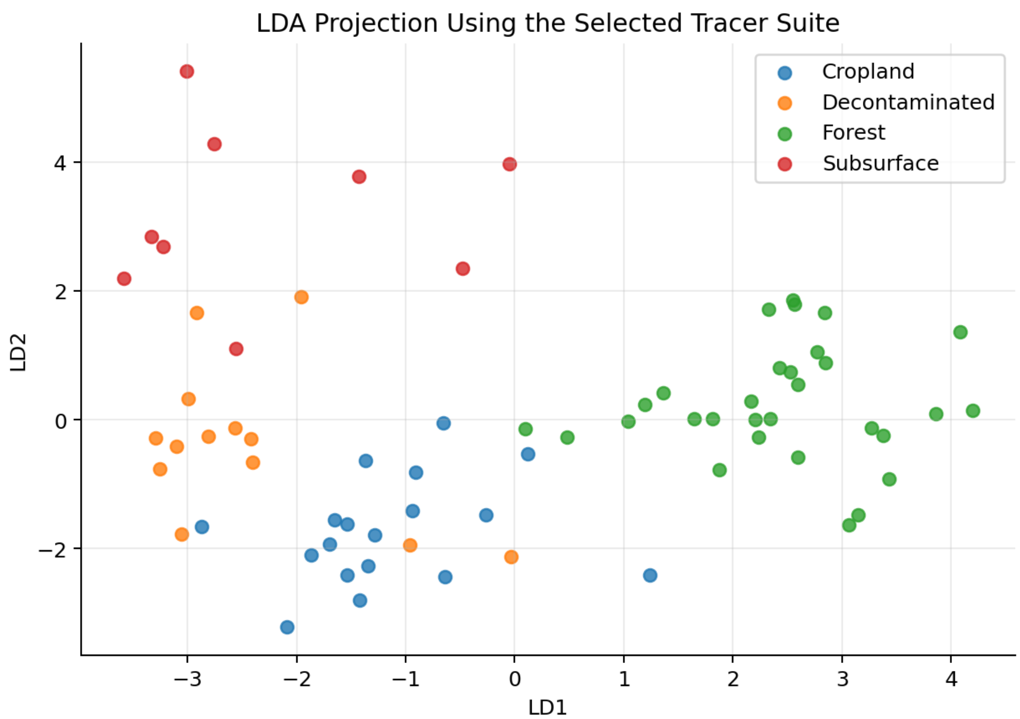

Figure 7.3: Linear Discriminant Analysis projection using the selected geochemical tracers.

7.3.1.2 Source Apportionment

The selected tracers were then used to estimate the source proportions for both the laboratory-prepared and synthetic benchmark mixtures. For each sample, the BMM returns predictive intervals that quantify the uncertainty associated with every estimated proportion.

The laboratory results are presented in Figure 7.4. In most cases, the known source proportions fall within the predicted intervals, indicating that the model captures the main structure of the mixing process while appropriately representing the uncertainty arising from overlapping source fingerprints.

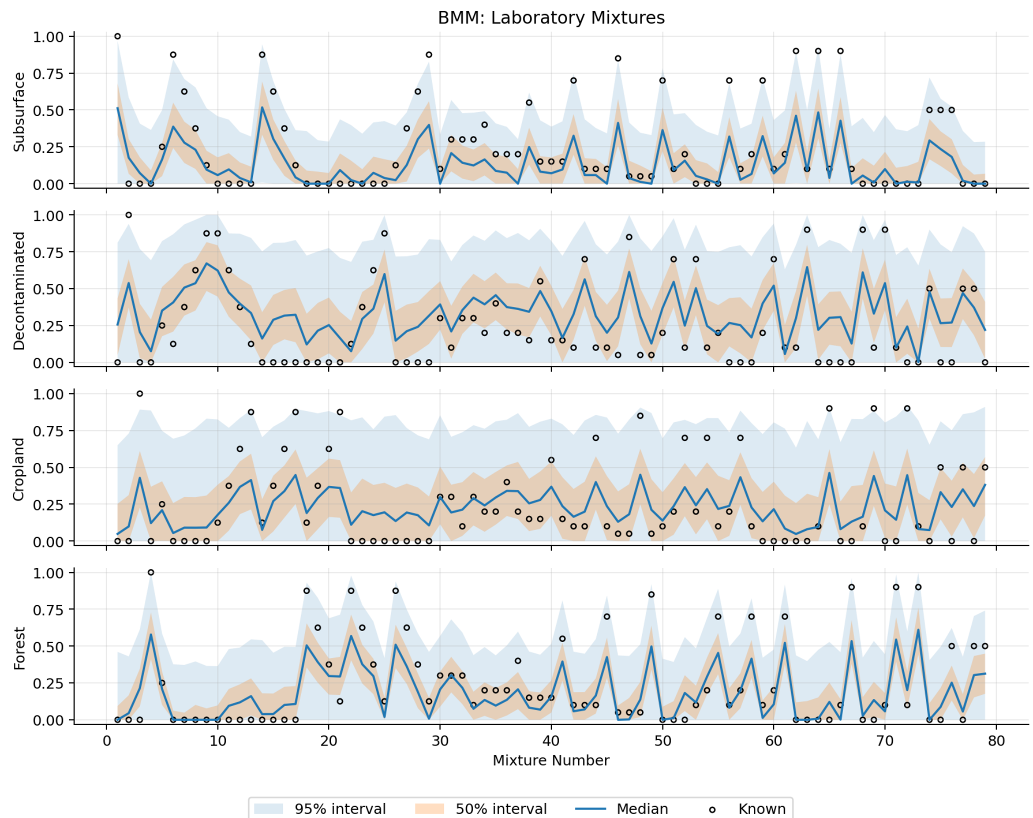

Figure 7.4: Predicted source contributions for the laboratory-prepared mixtures using the Bootstrapped Mixing Model.

The same procedure was subsequently applied to the synthetic benchmark. As shown in Figure 7.5, predictive performance improves substantially when evaluated on these mixtures, which provide a more systematic coverage of the compositional space than the smaller laboratory dataset.

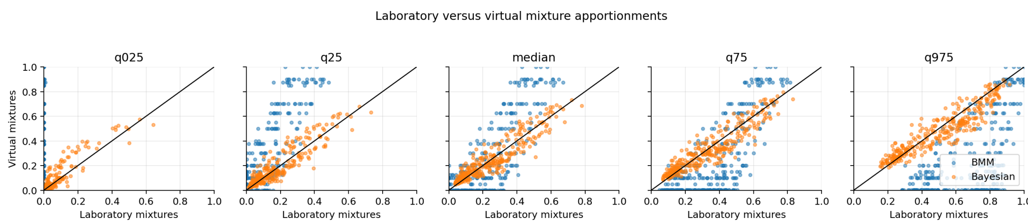

Figure 7.5: Comparison of predictive intervals obtained for the laboratory and synthetic benchmark mixtures.

The overall performance metrics reported in Table 7.2 confirm this pattern. The laboratory mixtures exhibit wider predictive intervals and larger reconstruction errors, whereas the synthetic benchmark is reconstructed almost exactly. This difference should not be interpreted as a limitation of the method, but rather as a consequence of the evaluation protocol: the laboratory mixtures retain the full variability of experimentally measured samples, while the synthetic dataset provides a more regular and controlled benchmark.

Table 7.2: Overall performance of the Bootstrapped Mixing Model.

Dataset	**P50**	**P95**	**W50**	**W95**	**MAE50**	**MAE95**	**CRPS**	**HR95**
Laboratory	0.424	0.962	0.278	0.676	0.046	0.001	0.103	1.000
Synthetic	0.978	1.000	0.0001	0.0013	0.0000	0.0000	0.00003	1.000

The source-level metrics reported in Table 7.3 provide a more detailed view of where uncertainty remains. Cropland and Decontaminated areas exhibit wider 95% predictive intervals than Forest and Subsurface soils, suggesting that their geochemical signatures are more difficult to distinguish. Even so, coverage remains consistently high across all four source classes, making the BMM an effective probabilistic reference against which the deterministic approaches can be compared.

Table 7.3: Source-level performance metrics of the Bootstrapped Mixing Model.

Source	Mean CRPS	Median Absolute Error	95% Coverage	95% Width
Cropland	0.115	0.118	0.962	0.801
Decontaminated	0.118	0.194	0.987	0.825
Forest	0.081	0.088	0.987	0.584
Subsurface	0.098	0.082	0.911	0.494

7.3.2 McCaffrey Constrained Least-Squares Model

The second approach adapts the constrained least-squares formulation proposed by McCaffrey et al. [34] for production allocation in commingled oil and gas reservoirs. Rather than estimating a distribution of plausible source proportions, this method seeks a single solution that best reproduces the measured geochemical composition while satisfying the physical constraints of the mixing process. The estimated proportions are therefore constrained to remain non-negative and to sum to one, ensuring that every solution is physically interpretable.

Although originally developed for petroleum production allocation, the mathematical formulation is directly applicable to sediment fingerprinting. In both cases, the observed sample represents a mixture of multiple sources with unknown proportions, and the objective is to recover those proportions from a set of conservative geochemical tracers. This shared mathematical structure makes the McCaffrey formulation a natural deterministic alternative to the probabilistic Bootstrapped Mixing Model.

A complete Python implementation of the McCaffrey constrained least-squares model presented in this section is available in the project repository.

7.3.2.1 Representative Source Signatures

Unlike the Bootstrapped Mixing Model, which explicitly propagates measurement uncertainty through repeated resampling, the McCaffrey formulation requires a single representative geochemical signature for each sediment source. These signatures were obtained by aggregating the available observations from each source and subsequently scaling the retained tracers to comparable numerical ranges before solving the constrained optimization problem.

The resulting scaling factors are summarized in Table 7.4.

Table 7.4: Scaling factors applied to the selected geochemical tracers prior to optimization.

Tracer	**Scaling Factor**
L	3.940
h	4.311
Si	12.271
Zn	4.902
Mn	6.847
Ti	7.614
K	8.520
C*	5.178
Y	5.096

After estimating the source proportions, the representative compositions can be reconstructed and compared with their known values. As shown in Figure 7.6, the reconstructed signatures closely reproduce the observed geochemical fingerprints, indicating that the optimization successfully captures the dominant characteristics of each sediment source.

7.3.2.2 Source Apportionment

The constrained least-squares model was subsequently applied to every laboratory mixture included in the benchmark dataset. The estimated source proportions are compared with the known values in Figure 7.7. Most predictions lie close to the one-to-one reference line, suggesting that the deterministic optimization captures the overall mixing behaviour despite the absence of an explicit uncertainty model.

Overall performance is summarized in Table 7.5. The model achieves a mean absolute error of approximately 0.08, an RMSE of 0.11, and a Nash–Sutcliffe efficiency close to 0.87. In addition, the dominant sediment source is correctly identified for more than 83% of the evaluated mixtures, demonstrating that the deterministic formulation preserves most of the information required for practical source attribution.

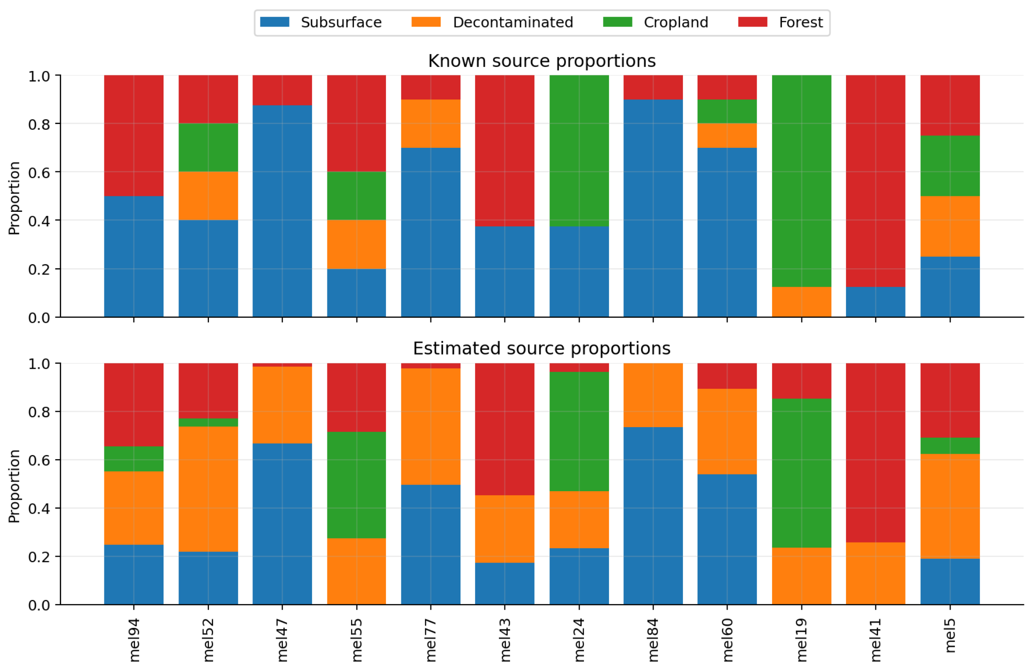

Figure 7.6: Known and reconstructed representative source compositions obtained with the McCaffrey constrained least-squares model.

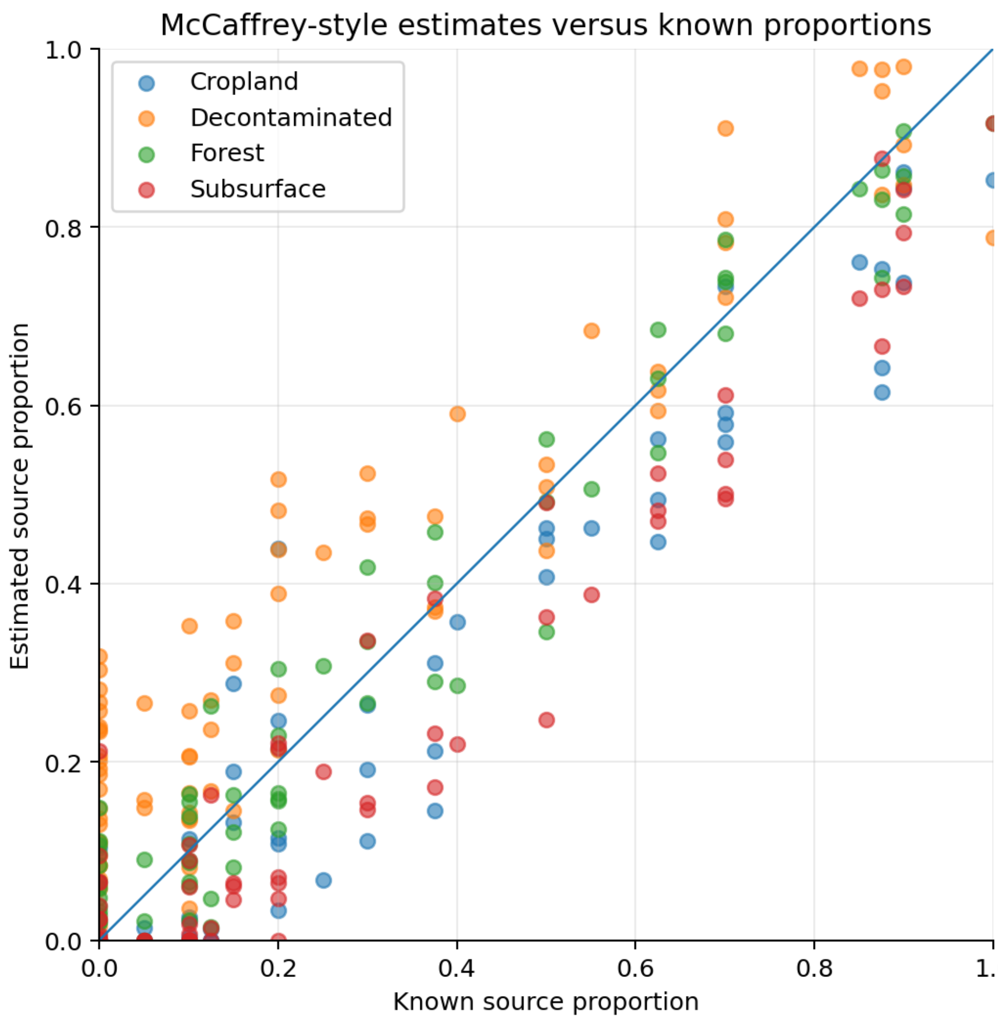

Figure 7.7: Estimated versus true source contributions obtained with the McCaffrey constrained least-squares model.

Table 7.5: Overall performance of the McCaffrey constrained least-squares model.

Metric	Value
Mean Absolute Error (MAE)	0.0793
Root Mean Squared Error (RMSE)	0.1086
Nash–Sutcliffe Efficiency (NSE)	0.8683
Dominant Source Accuracy	0.8354
Mean Aitchison Distance	16.6797

The aggregated metrics conceal some differences between sediment sources. As shown in Table 7.6, Forest and Cropland exhibit the smallest prediction errors, whereas the Decontaminated source shows the largest bias. This behaviour is consistent with the greater overlap between some geochemical fingerprints observed during the exploratory analysis.

Table 7.6: Source-level performance metrics for the McCaffrey constrained least-squares model.

Source	MAE	RMSE	Mean Error
Subsurface	0.0768	0.1037	-0.0601
Decontaminated	0.1234	0.1517	0.1106
Cropland	0.0693	0.0975	-0.0532
Forest	0.0478	0.0625	0.0028

Compared with the probabilistic BMM, the McCaffrey model sacrifices uncertainty quantification in exchange for a considerably simpler optimization procedure and a single deterministic estimate for each mixture. This trade-off does not necessarily represent a limitation; in many industrial applications, a stable point estimate is preferable when uncertainty intervals are not required for operational decision-making.

7.3.3 Sandoval Ratio-Based Mixing Model

The final approach implements the ratio-based constrained deconvolution model proposed by Sandoval et al. [35]. Although it addresses the same source apportionment problem as the McCaffrey model, it adopts a different representation of the geochemical data. Instead of comparing elemental concentrations directly, the optimization is performed using log-ratios between tracers, reducing the influence of the constant-sum constraint that characterizes compositional datasets.

This formulation is particularly attractive for geochemical applications because many analytical measurements are inherently relative rather than absolute. By expressing the observations as ratios, the optimization focuses on the proportional relationships between tracers instead of their individual magnitudes, making the solution less sensitive to scaling effects and compositional closure.

A complete Python implementation of the Sandoval ratio-based mixing model presented in this section is available in the project repository.

7.3.3.1 Representative Source Signatures

As in the McCaffrey formulation, a representative fingerprint was first constructed for each sediment source. These reference signatures were subsequently transformed into a log-ratio representation before solving the constrained optimization problem.

The tracer-specific scaling factors used during preprocessing are summarized in Table 7.7.

Table 7.7: Scaling factors applied to the selected geochemical tracers prior to log-ratio transformation.

Tracer	**Scaling Factor**
L	51.728
h	74.501
Si	214014.572
Zn	135.606
Mn	972.318
Ti	4696.537
K	2784.897
C*	3.657
Y	5.830

The reconstructed representative compositions are compared with the observed source fingerprints in Figure 7.8. Despite operating in log-ratio space, the model accurately reproduces the characteristic geochemical signatures of the four sediment sources, indicating that the transformation preserves the relevant compositional information.

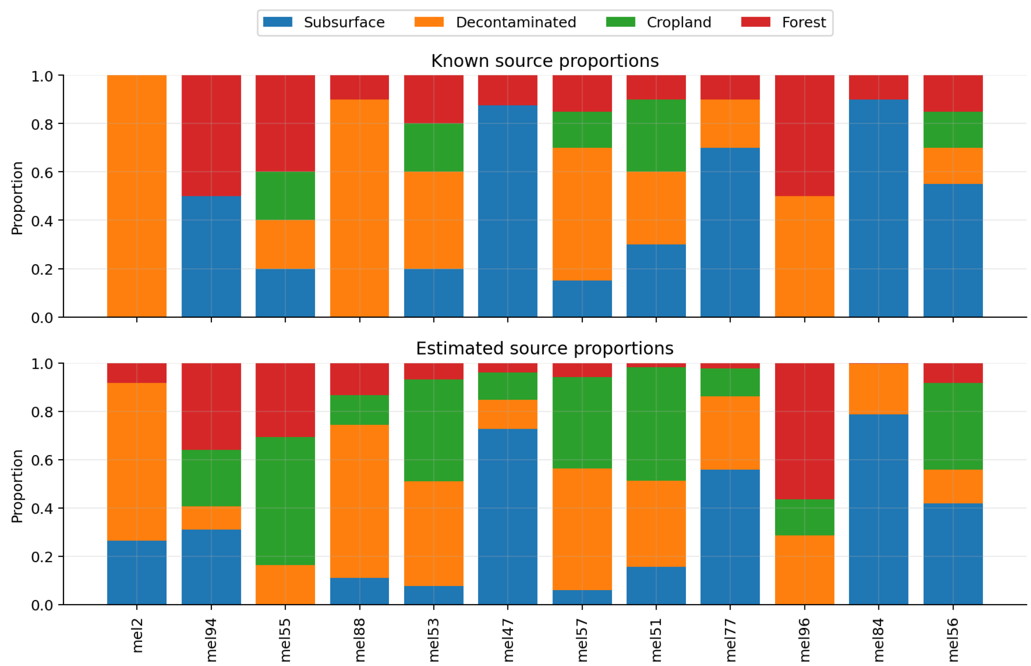

Figure 7.8: Known and reconstructed representative source compositions obtained with the Sandoval ratio-based model.

7.3.3.2 Source Apportionment

After calibration, the ratio-based model was applied to the laboratory mixtures using the same benchmark dataset employed for the previous methods. The estimated source proportions are compared with the known values in Figure 7.9. Most predictions remain close to the one-to-one reference line, indicating that the log-ratio formulation successfully recovers the dominant source proportions while maintaining numerical stability throughout the optimization.

The overall performance metrics are summarized in Table 7.8. Compared with the McCaffrey formulation, the ratio-based model achieves lower prediction errors, a higher Nash–Sutcliffe efficiency, and a modest improvement in dominant

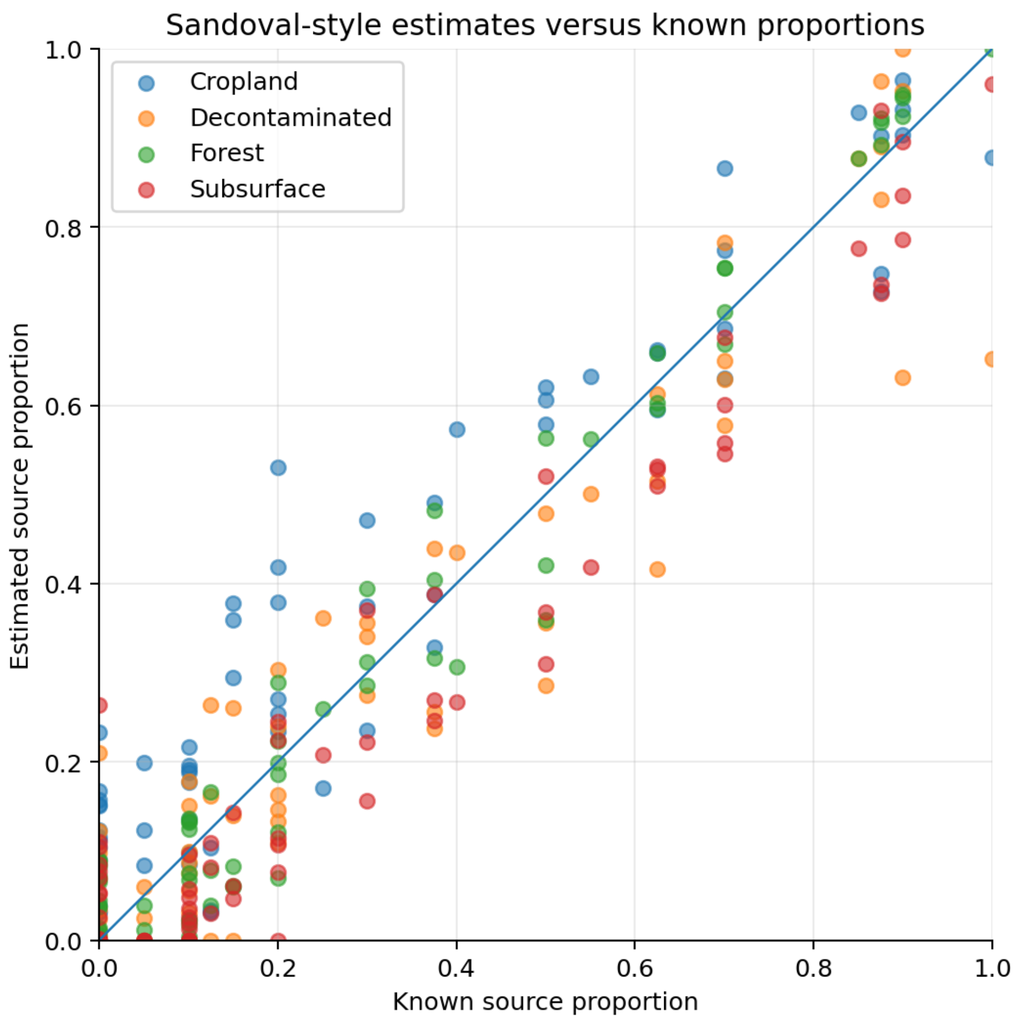

Figure 7.9: Estimated versus true source contributions obtained with the Sandoval ratio-based model.

source identification. The average Aitchison distance also decreases, suggesting that representing the mixtures in log-ratio space provides a more appropriate description of compositional differences.

Table 7.8: Overall performance of the Sandoval ratio-based mixing model.

Metric	Value
Mean Absolute Error (MAE)	0.0620
Root Mean Squared Error (RMSE)	0.0864
Nash–Sutcliffe Efficiency (NSE)	0.9166
Dominant Source Accuracy	0.8481
Mean Aitchison Distance	14.3363
Mean Ratio Objective	0.00143

The source-level results presented in Table 7.9 provide a more detailed perspective on model performance. Forest exhibits the smallest reconstruction error, whereas Cropland remains the most challenging source to estimate accurately, consistent with the overlap observed during the exploratory analysis.

Table 7.9: Source-level performance metrics for the Sandoval ratio-based mixing model.

Source	MAE	RMSE	Mean Error
Subsurface	0.0656	0.0866	-0.0384
Decontaminated	0.0609	0.0909	-0.0151
Cropland	0.0799	0.1055	0.0515
Forest	0.0415	0.0545	0.0019

One practical advantage of the Sandoval formulation is that it naturally accommodates the relative nature of geochemical measurements without increasing the complexity of the optimization procedure. While both deterministic approaches produce comparable source apportionments, the log-ratio representation provides a measurable improvement in reconstruction accuracy and compositional consistency, making it the strongest deterministic performer among the three methods evaluated in this study.

7.4 Results, Discussion, and Limitations

Although the three source apportionment methods were developed from different modelling philosophies, all successfully recovered the source proportions of the benchmark mixtures. This agreement is perhaps the most important outcome of the study. Rather than demonstrating the superiority of a particular algorithm, it shows that the underlying inverse problem is sufficiently well posed to admit multiple valid formulations. Whether approached through probabilistic resampling, constrained least-squares optimization, or log-ratio deconvolution, the essential objective remains the same: identifying the combination of source proportions that best explains the measured geochemical composition while satisfying the physical constraints of the mixing process.

The Bootstrapped Mixing Model provides the richest representation of the problem by explicitly accounting for uncertainty. Instead of producing a single estimate, it generates confidence intervals that reflect both analytical variability and the natural overlap between source fingerprints. This additional information is particularly valuable when the estimated proportions are intended to support environmental interpretation, where understanding the confidence associated with a prediction is often as important as the prediction itself. The price for this richer representation is a more computationally intensive workflow that requires repeated resampling and optimization.

The deterministic formulations approach the problem from a different perspective. By directly solving a constrained optimization problem, they produce a single physically consistent estimate for each sediment mixture without explicitly modelling uncertainty. This considerably simplifies the computational workflow while preserving the physical constraints that define a valid compositional mixture.

Between the two deterministic approaches, the ratio-based formulation proposed by Sandoval et al. consistently achieved the strongest overall performance. Although both methods rely on the same optimization principles, representing the observations in log-ratio space reduces the influence of compositional closure and places greater emphasis on the relative relationships between tracers. The resulting improvements in reconstruction accuracy are modest but consistent across the evaluation metrics, suggesting that explicitly accounting for the compositional nature of the data provides a measurable practical advantage.

Perhaps the most interesting outcome of the comparison lies beyond the numerical results themselves. The two deterministic methods were originally developed for petroleum production allocation, yet they required remarkably few conceptual modifications before being applied to sediment fingerprinting. The optimization problem, the governing constraints, and the mathematical formulation remain essentially unchanged; only the interpretation of the sources differs. Reservoirs contributing to a commingled production stream become sediment-producing landscape units, while reservoir geochemical fingerprints become sediment fingerprints. This demonstrates that many inverse problems share a common mathematical structure despite arising in entirely different scientific domains.

Several limitations should nevertheless be acknowledged. The evaluation is based on laboratory-prepared mixtures and synthetically generated benchmark datasets, where the true source proportions are known exactly. While these datasets provide an objective basis for comparing competing methods, natural catchments often exhibit greater spatial variability, temporal changes in source composition, and additional measurement uncertainty that cannot be fully reproduced under controlled experimental conditions. Furthermore, all three methods assume that the selected tracers behave conservatively during erosion, transport, and deposition. Processes that preferentially alter or fractionate individual tracers could violate this assumption and reduce the reliability of the estimated proportions.

Overall, the results indicate that no single method is universally preferable. The Bootstrapped Mixing Model is the natural choice when uncertainty quantification is an essential component of the analysis, whereas the deterministic formulations provide simpler and computationally efficient alternatives when a single robust estimate is sufficient. Among these, the ratio-based model offers the best balance between numerical stability and predictive accuracy.

Beyond the comparison itself, this project illustrates a broader principle that appears repeatedly throughout this portfolio. Effective data science is often less about developing entirely new algorithms than about recognizing when an existing mathematical formulation can be transferred to a different domain. Once the underlying structure of a problem has been identified, methodologies developed for one application can often provide effective solutions to another with only minor adaptations. In this study, techniques originally designed for petroleum production allocation proved equally capable of solving a sediment source apportionment problem, highlighting the value of focusing on mathematical structure rather than disciplinary boundaries.

7.5 Technical Appendix (Optional)

The three source apportionment methods presented in this chapter originate from different scientific disciplines and employ different estimation strategies. Nevertheless, they all solve the same underlying inverse problem: recovering the relative proportions of multiple sources whose combined geochemical signatures reproduce an observed sediment mixture.

Recognizing this common mathematical structure is often more valuable than studying each algorithm in isolation. Once the inverse mixing problem has been defined, the differences between the Bootstrapped Mixing Model, the McCaffrey formulation, and the Sandoval formulation reduce primarily to how the solution is estimated and how uncertainty is represented.

Rather than reviewing the implementation details presented earlier in the chapter, this appendix focuses on the mathematical concepts shared by all three approaches before introducing the ideas that distinguish them. The discussion begins with the linear mixing model, proceeds to the inverse problem and its physical constraints, and then examines how probabilistic and deterministic formulations provide alternative solutions to the same estimation problem.

The conceptual relationships among these ideas are summarized in Figure 7.10. Although the three methods differ in how they estimate the unknown source proportions, they all originate from the same physical mixing model, share the same constrained inverse formulation, and differ primarily in how uncertainty is represented and how the objective function is defined.

7.5.1 Linear Mixing Models

At the core of sediment fingerprinting lies a simple physical assumption: the geochemical composition measured in a sediment sample results from mixing material originating from several upstream sources. If the selected tracers behave conservatively—that is, they are neither created nor destroyed during erosion, transport, and deposition—the composition of the mixture can be approximated as a weighted combination of the compositions of its constituent sources.

This assumption is known as the **linear mixing model** and forms the mathemati-

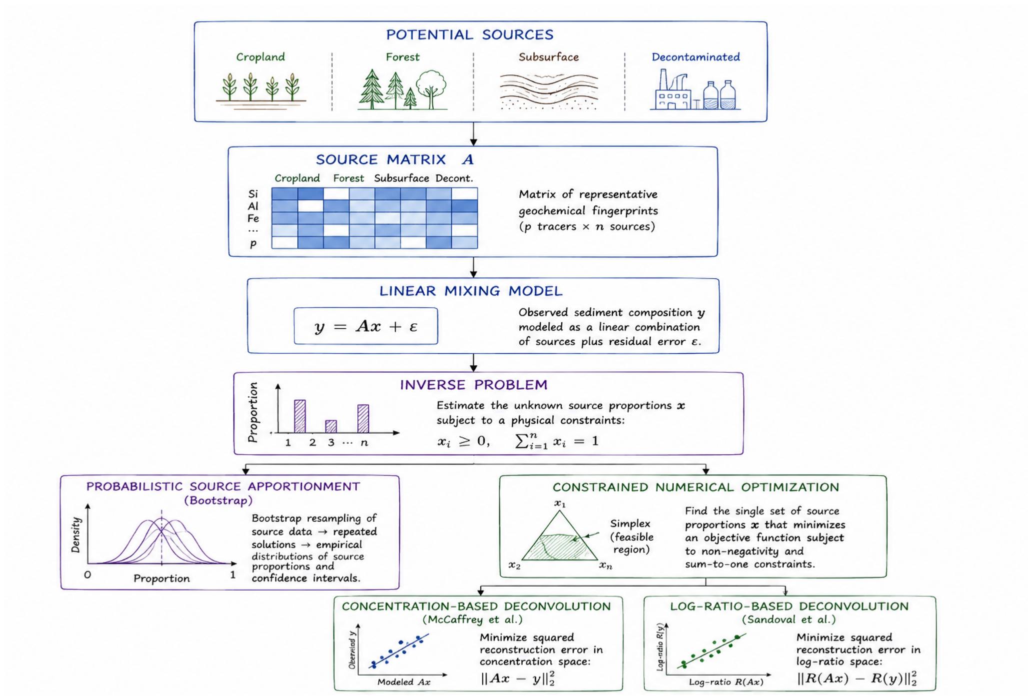

Figure 7.10: Conceptual framework of the source apportionment problem. All three methods considered in this chapter solve the same inverse mixing problem under identical physical constraints, differing only in how uncertainty is represented and how the objective function is formulated.

cal foundation of every source apportionment method presented in this chapter. Although the three approaches differ in how they estimate the unknown source proportions, they all rely on the same physical description of the mixing process.

Suppose that m geochemical tracers are measured for n potential sediment sources. The corresponding source fingerprints can then be arranged into the matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \tag{7.1}$$

where each column corresponds to one sediment source and each row contains the concentration of a particular tracer measured for that source.

The observed sediment mixture is represented by

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \tag{7.2}$$

while the unknown source proportions are collected in

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \tag{7.3}$$

where each component represents the fraction of sediment contributed by one source.

Under the assumption of conservative mixing, the relationship between the source fingerprints and the measured sediment composition is expressed as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \varepsilon, \tag{7.4}$$

where ε represents measurement uncertainty together with any discrepancy between the simplified mixing model and the observations.

Although deceptively compact, Equation 7.4 encapsulates the entire source apportionment problem. The matrix $\mathbf{A}$ is assumed to be known from measurements collected at the potential source areas, while the mixture vector $\mathbf{y}$ is obtained from downstream samples. The only unknown quantity is the vector of source proportions $\mathbf{x}$, which becomes the target of the estimation process.

The challenge, however, is that estimating $\mathbf{x}$ is considerably more difficult than evaluating Equation 7.4 in the forward direction. Recovering the source proportions from an observed mixture requires solving an inverse problem, where multiple solutions may reproduce the measured composition with similar accuracy.

7.5.2 The Inverse Mixing Problem

If the source proportions are known, predicting the composition of the resulting sediment mixture is straightforward: the linear mixing model introduced in Equation 7.4 can be evaluated directly by multiplying the source matrix by the vector of source proportions. This is commonly referred to as the **forward problem**.

Sediment source apportionment, however, requires solving the opposite task. The measured sediment composition is available, but the proportions contributed by each source are unknown. The objective is therefore to estimate the vector $\mathbf{x}$ from the mixture vector $\mathbf{y}$ and the known source fingerprints contained in $\mathbf{A}$. This inverse formulation is considerably more challenging because the solution cannot generally be obtained by simply rearranging Equation 7.4.

Several factors contribute to this difficulty. First, laboratory measurements inevitably contain analytical uncertainty, meaning that the measured composition is only an approximation of the true mixture. Second, different sediment sources often exhibit similar geochemical fingerprints, making it difficult to distinguish their individual contributions. Finally, environmental datasets typically contain only a limited number of informative tracers, reducing the amount of independent information available for estimating the unknown source proportions.

These factors imply that multiple combinations of source proportions may reproduce the measured composition with nearly identical accuracy. Consequently, source apportionment should not be viewed as the search for a unique exact solution, but rather as the identification of the most plausible solution that remains consistent with both the observations and the underlying physics of the mixing process.

Physical considerations impose two additional constraints on every admissible solution. Because the unknown variables represent proportions, they cannot be negative,

$$x_i \geq 0, \qquad i = 1, \dots, n, \tag{7.5}$$

and, because the entire sediment sample must originate from the available sources, the proportions must sum to one,

$$\sum_{i=1}^{n} x_i = 1. \tag{7.6}$$

These constraints are not merely mathematical conveniences; they reflect the conservation of mass and ensure that every estimated source apportionment has a physically meaningful interpretation. Any solution violating either condition would imply either negative sediment contributions or the artificial creation (or loss) of material during the mixing process.

Together, Equation 7.5 and Equation 7.6 define the feasible solution space explored by all three methods presented in this chapter. The principal difference between the Bootstrapped Mixing Model, the McCaffrey formulation, and the Sandoval formulation lies not in these physical constraints, which are common to all approaches, but in how each method explores this feasible space and identifies the solution that best explains the measured composition.

7.5.3 Constrained Numerical Optimization

Once the inverse mixing problem has been formulated, the next question is how the unknown source proportions should be estimated. In principle, one might attempt to solve Equation 7.4 analytically using ordinary least squares. Such an approach minimizes the discrepancy between the measured composition and the reconstructed mixture, producing the solution that best fits the available observations in a purely algebraic sense.

In practice, however, unconstrained least-squares solutions are not guaranteed to satisfy the physical requirements established in Equation 7.5 and Equation 7.6. Small measurement errors, correlated tracers, or poorly conditioned source matrices may produce negative proportions or values exceeding one. Although these solutions minimize the residual error, they cannot represent physically meaningful sediment mixtures.

For this reason, the deterministic methods implemented in this project were reformulated as constrained numerical optimization problems. Rather than solving the system analytically and verifying the constraints afterwards, the optimization searches directly within the physically admissible solution space. Every candidate solution therefore satisfies the non-negativity and mass conservation constraints throughout the optimization process.

The general optimization problem can be written as

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & x_i \geq 0, \qquad i = 1, \dots, n, \\ & \sum_{i=1}^{n} x_i = 1, \end{aligned} \tag{7.7}$$

where the objective function $f(\mathbf{x})$ depends on the particular source apportionment method being used. The optimization variables are always the unknown source proportions, while the physical constraints remain unchanged.

Reformulating the estimation in this way offers several practical advantages. First, every solution returned by the optimizer is physically interpretable, eliminating the need for post-processing corrections to enforce valid proportions. Second, the optimization framework is independent of the specific objective function. This makes it possible to compare different source apportionment formulations while using exactly the same numerical solution strategy.

This separation between the optimization algorithm and the objective function is one of the central design principles of the implementation developed for this project. The McCaffrey and Sandoval methods differ primarily in the quantity they seek to minimize, yet both can be solved using the same constrained optimization framework introduced in Equation 7.7. As a result, differences in performance can be attributed to the modelling formulation itself rather than to changes in the numerical solution procedure.

The Bootstrapped Mixing Model follows a different estimation strategy. Instead of searching for a single optimal solution, it repeatedly perturbs the source fingerprints through bootstrap resampling and solves the mixing problem many times. Nevertheless, each realization remains subject to the same physical constraints, ensuring that every sampled source apportionment represents a feasible sediment mixture. The principal distinction between the probabilistic and deterministic approaches therefore lies not in the underlying physics, but in how uncertainty is represented and propagated through the estimation process.

7.5.4 Probabilistic Source Apportionment

The deterministic optimization framework described in the previous section assumes that the source fingerprints are known exactly. In practice, however, geochemical measurements are subject to analytical uncertainty, natural spatial variability, and sampling error. As a result, the representative fingerprint assigned to each sediment source should be regarded as an estimate rather than as a fixed quantity.

The Bootstrapped Mixing Model (BMM) addresses this uncertainty by repeatedly resampling the available source observations before solving the mixing problem. Instead of producing a single estimate of the source proportions, the model generates a large collection of equally plausible solutions, each corresponding to a different realization of the source fingerprints. The variability among these solutions provides a direct estimate of the uncertainty associated with the predicted proportions.

The procedure can be summarized as four consecutive steps:

1. A bootstrap sample is generated independently for each sediment source by randomly resampling the available geochemical observations with replacement.
2. Representative source fingerprints are computed from the resampled datasets.
3. The source apportionment problem is solved using the reconstructed fingerprints.
4. The process is repeated many times to obtain an empirical distribution of the estimated source proportions.

Because each bootstrap realization is treated as an equally plausible representation of the original dataset, the collection of estimated source proportions approximates the sampling distribution of the unknown solution. Confidence intervals can then be obtained directly from the empirical quantiles without requiring any assumption about the analytical form of the underlying probability distribution.

Conceptually, the procedure can be written as

$$\mathbf{x}^{(b)} = \arg\min_{\mathbf{x}} f_b(\mathbf{x}), \qquad b = 1, \dots, B, \tag{7.8}$$

where B denotes the number of bootstrap realizations and $f_b(\mathbf{x})$ is the objective function associated with the b-th resampled dataset. Each optimization is solved independently, producing a collection of estimated source proportions,

$$\left\{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(B)}\right\}, \tag{7.9}$$

from which summary statistics such as the mean, median, prediction intervals, and confidence intervals can be computed.

An important consequence of this formulation is that uncertainty arises naturally from the variability of the source fingerprints rather than being imposed through an explicit probabilistic model. The bootstrap therefore provides a practical way to propagate measurement uncertainty through the source apportionment process while preserving the physical constraints introduced in Equation 7.7 for every realization.

Unlike the deterministic methods discussed later in this appendix, the objective of the Bootstrapped Mixing Model is not to identify a single optimal solution. Instead, it characterizes the range of solutions that remain compatible with the observations. This distinction explains why the BMM produces confidence intervals for the estimated source proportions, whereas the deterministic formulations return only a single point estimate for each sediment source.

7.5.5 Concentration-Based Deconvolution

The Bootstrapped Mixing Model characterizes uncertainty by repeatedly perturbing the source fingerprints and solving the inverse problem many times. The deterministic approaches follow a different philosophy. Rather than estimating a distribution of plausible solutions, they seek a single set of source proportions that best reproduces the measured composition.

The first deterministic formulation considered in this chapter is based on the constrained least-squares approach proposed by McCaffrey et al. [34]. Originally developed for geochemical allocation of commingled oil and gas production, the method assumes that the measured composition of a mixture can be reconstructed by combining the representative fingerprints of the contributing sources.

Within the optimization framework introduced in Equation 7.7, the McCaffrey formulation defines the objective function as the squared Euclidean distance between the measured composition and the composition reconstructed from the estimated source proportions,

$$f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 . \tag{7.10}$$

The optimization therefore seeks the physically admissible source proportions that minimize the overall reconstruction error while satisfying the non-negativity and mass conservation constraints.

The least-squares criterion has an intuitive interpretation. If the estimated source proportions perfectly reproduce the measured composition, every residual becomes zero and the objective function reaches its minimum possible value. In practice, measurement uncertainty and natural variability prevent an exact reconstruction, so the optimization instead identifies the solution with the smallest overall discrepancy.

An important characteristic of this formulation is that all tracers contribute simultaneously to the objective function. Rather than attempting to match each geochemical variable independently, the optimization balances the residuals across the complete fingerprint, producing the set of source proportions that provides the best overall agreement with the observations.

The constrained optimization framework adopted in this project differs slightly from the original analytical formulation presented by McCaffrey et al. Instead of deriving the solution directly through matrix algebra, the unknown source proportions are estimated numerically under the physical constraints defined by Equation 7.5 and Equation 7.6. Although both approaches solve the same mathematical problem, the numerical formulation guarantees physically admissible solutions while providing a common framework that can easily be extended to alternative objective functions.

This flexibility becomes particularly valuable when the assumptions underlying ordinary least squares are no longer appropriate. Geochemical measurements are compositional by nature: only the relative relationships between tracers are physically meaningful, while their absolute magnitudes are affected by the constant-sum constraint. The next section introduces the basic concepts of compositional data analysis before presenting the log-ratio formulation proposed by Sandoval et al., which explicitly accounts for these characteristics.

7.5.6 Compositional Data Analysis

The least-squares formulation presented in the previous section implicitly treats each geochemical tracer as an independent continuous variable. While this assumption is reasonable in many regression problems, it becomes problematic for compositional data, where the individual components cannot vary independently.

Geochemical fingerprints are often expressed as concentrations or proportions whose total is constrained by the composition of the sample. Increasing the concentration of one tracer necessarily changes the relative abundance of the remaining tracers, even if their absolute amounts remain unchanged. This dependency is known as the **closure problem** and is one of the defining characteristics of compositional data.

The consequence of closure is that conventional Euclidean geometry no longer provides an entirely appropriate representation of the data. Distances between samples, correlations among variables, and residual errors may all be influenced by the constant-sum constraint rather than by genuine geochemical differences. As a result, statistical analyses performed directly on the original concentrations may identify relationships that arise from the mathematical structure of the data rather than from the underlying physical processes.

To illustrate this idea, consider a sediment mixture composed of four tracers whose concentrations sum to 100%. If the concentration of one tracer increases while the total remains fixed, the remaining three concentrations must decrease, even when no physical relationship exists between them. Apparent negative correlations therefore emerge naturally from the compositional constraint.

A widely adopted solution is to analyze **ratios** rather than the original concentrations. Ratios preserve the relative information contained in the composition while eliminating the dependence on the arbitrary total. Because multiplicative relationships become additive after applying a logarithmic transformation, the analysis is typically performed using log-ratios instead of raw ratios.

For two tracers i and j, the simplest log-ratio transformation is

$$z_{ij} = \log\left(\frac{x_i}{x_j}\right), \tag{7.11}$$

which expresses the abundance of one tracer relative to another. Since the logarithm converts multiplicative differences into additive ones, many statistical techniques developed for unconstrained Euclidean spaces become applicable once the data have been transformed.

The central idea is therefore not that the absolute concentration of a tracer carries the relevant information, but that its abundance relative to the remaining tracers is often more informative. This perspective forms the foundation of **Compositional Data Analysis (CoDA)** and motivates the log-ratio source apportionment model proposed by Sandoval et al.

Although several log-ratio transformations have been developed—including additive, centered, and isometric log-ratios—the implementation presented in this chapter follows the formulation introduced by Sandoval et al. [35], which was specifically designed for geochemical source allocation problems.

7.5.7 Log-Ratio-Based Deconvolution

The ratio-based deconvolution model proposed by Sandoval et al. [35] builds directly upon the concepts introduced in the previous sections. Like the McCaffrey formulation, it seeks the source proportions that best reproduce the measured geochemical composition while satisfying the physical constraints of the mixing process. The fundamental difference lies not in the optimization itself, but in how the geochemical information is represented.

Instead of comparing elemental concentrations directly, the model first transforms the source fingerprints and the observed mixture into a set of log-ratios. The optimization is then performed in this transformed space, where the relative relationships between tracers are preserved and the effects of compositional closure are substantially reduced.

Within the constrained optimization framework introduced in Equation 7.7, the objective function can be expressed as

$$f(\mathbf{x}) = \|\mathbf{R}(\mathbf{A}\mathbf{x}) - \mathbf{R}(\mathbf{y})\|_2^2, \tag{7.12}$$

where $\mathbf{R}(\cdot)$ denotes the log-ratio transformation applied to the compositional data. Rather than minimizing differences between the original concentrations, the optimization minimizes the discrepancy between the transformed fingerprints.

This apparently simple modification has important practical consequences. Because the optimization operates on relative rather than absolute information, the estimated source proportions become less sensitive to scaling effects and to the constant-sum constraint inherent in compositional datasets. Consequently, the reconstructed mixtures better preserve the geochemical relationships that distinguish one source from another.

From an optimization perspective, the log-ratio formulation remains remarkably similar to the least-squares approach described previously. The optimization variables are still the unknown source proportions, the non-negativity and mass conservation constraints remain unchanged, and the same numerical optimization framework can be employed without modification. Only the objective function differs, allowing both formulations to be implemented within a common computational framework.

This separation between the optimization algorithm and the objective function is one of the main advantages of the implementation developed for this project. Once the constrained optimization problem has been defined, alternative source apportionment models can be incorporated simply by redefining the objective function, without altering the numerical solution procedure itself.

The experimental results presented in this chapter indicate that the log-ratio formulation consistently produces lower reconstruction errors than the concentration-based least-squares model. Although the numerical improvements are moderate, they are observed across multiple evaluation metrics and are consistent with the theoretical expectations of Compositional Data Analysis.

By respecting the relative nature of geochemical measurements, the optimization recovers the source proportions more accurately while maintaining the same computational simplicity as the original deterministic formulation.

Taken together, the three methods discussed in this appendix illustrate complementary ways of solving the same inverse mixing problem. The Bootstrapped Mixing Model emphasizes uncertainty quantification through repeated resampling, the McCaffrey formulation seeks the best deterministic reconstruction in the original concentration space, and the Sandoval formulation extends this deterministic approach to log-ratio space to account explicitly for the compositional nature of geochemical data. Although these methods differ in their estimation strategies, they all rely on the same physical model of sediment mixing, the same conservation constraints, and the same constrained optimization framework. Once the inverse problem has been formulated, the choice of method primarily reflects how uncertainty is represented and how the discrepancy between the measured and reconstructed compositions is quantified, rather than a change in the underlying physics.

More broadly, this shared mathematical foundation illustrates an idea that extends well beyond sediment fingerprinting. Scientific problems that appear unrelated at first glance are often governed by remarkably similar mathematical structures. Recognizing these common patterns makes it possible to transfer methodologies across disciplines with only minor adaptations, allowing existing models to address new challenges without fundamentally changing their underlying principles. In this chapter, techniques originally developed for petroleum production allocation proved equally applicable to sediment source apportionment, illustrating that effective solutions often emerge not from developing entirely new algorithms, but from recognizing the mathematical structures that different scientific problems have in common.

Chapter 8

Skin Lesion Triage

The ability to prioritise patients who may require earlier clinical attention is an important aspect of many healthcare workflows. In dermatology, dermoscopic imaging has become a valuable tool for assessing pigmented skin lesions, but interpreting these images remains a specialised task that depends on clinical expertise and available resources.

This project explores whether deep learning can support that prioritisation process by distinguishing lesions that warrant earlier dermatological review from those that can be assigned a lower priority. Rather than attempting to automate diagnosis, the objective is to investigate how image classification models can contribute to a cautious triage workflow while preserving clear boundaries between decision support and clinical judgement.

The chapter follows the complete modelling process, from data preparation and model development to performance evaluation and interpretation, with particular attention to the methodological decisions required to obtain a reliable assessment of model performance.

8.1 Problem Definition

Dermoscopic image classification is commonly framed as a diagnostic task in which each image is assigned to a specific disease category. While this formulation is appropriate for benchmarking machine learning models, it does not necessarily reflect how such models are most likely to be used in clinical practice. In many situations, the immediate objective is not to establish a definitive diagnosis but to identify which lesions should be prioritised for specialist assessment.

This project therefore approaches skin lesion analysis as a binary triage problem. Rather than predicting one of the seven diagnostic categories provided by the original dataset, the objective is to distinguish lesions that warrant **priority dermatological review** from those assigned a lower review priority. This reformulation shifts the focus from diagnostic classification towards clinical prioritisation, where the cost of overlooking a potentially serious lesion is considerably higher than recommending an unnecessary specialist examination.

The binary target is created by grouping melanoma (`mel`), basal cell carcinoma (`bcc`), and actinic keratosis or intraepithelial carcinoma (`akiec`) into the priority-review class. Melanocytic naevi (`nv`), benign keratoses (`bkl`), dermatofibroma (`df`), and vascular lesions (`vasc`) constitute the non-priority class. This grouping is introduced solely for the purposes of this study and should not be interpreted as a validated clinical referral protocol. Instead, it provides a practical framework for evaluating whether deep learning models can effectively rank lesions according to their potential clinical urgency.

This formulation introduces several modelling challenges. The priority-review class represents a relatively small proportion of the available observations, creating a naturally imbalanced classification problem. In addition, multiple dermoscopic images may correspond to the same underlying lesion, making careful data partitioning essential to avoid information leakage between training and evaluation. These characteristics influence both the modelling strategy and the interpretation of the resulting performance.

The objective is therefore not simply to maximise predictive accuracy, but to develop a reproducible workflow capable of supporting image-based lesion prioritisation while providing a realistic assessment of its strengths, limitations, and potential role within a clinical decision-support setting.

8.2 Modeling Approach

The objective of this project is not to identify the most sophisticated image classification architecture, but to evaluate whether deep learning can provide effective support for prioritising dermoscopic lesions that may require earlier clinical attention. Rather than comparing a large number of competing models, the analysis focuses on two complementary modelling strategies that represent different levels of complexity while sharing the same evaluation framework.

The first approach consists of a compact convolutional neural network (CNN) developed specifically for this study. Training a model from scratch establishes a transparent baseline and provides a useful reference for assessing the extent to which predictive performance can be achieved without relying on external image representations. Although relatively simple, this architecture allows the complete learning process to be observed directly and serves as a benchmark against which more advanced approaches can be evaluated.

The second approach adopts transfer learning using ResNet-18, a convolutional neural network pretrained on the ImageNet dataset. Instead of learning visual features entirely from the available dermoscopic images, the model starts from representations acquired from a large-scale natural image dataset and is subsequently fine-tuned for the binary lesion triage task. Transfer learning has become a common strategy in medical image analysis because it often improves performance when labelled clinical datasets are relatively limited.

Both models address the same binary classification problem, are trained using the same lesion-level data partitions, and are evaluated under an identical validation and testing protocol. This common framework allows differences in predictive performance to be attributed primarily to the modelling strategy rather than to variations in data preparation or evaluation methodology.

The overall objective is therefore not only to determine which model achieves superior predictive performance, but also to assess the practical trade-offs between a lightweight architecture trained from scratch and a widely adopted transfer learning approach when applied to automated skin lesion triage.

8.3 Methodology and Implementation

The modelling approach described in the previous section was implemented through a structured workflow covering data preparation, model development, evaluation, and interpretation. Particular attention was given to preserving the independence of the evaluation data, mitigating class imbalance, and ensuring that model selection was based exclusively on validation performance. The implementation follows established practices in deep learning for medical image analysis while adapting them to the specific objective of binary skin lesion triage.

Throughout the workflow, methodological decisions prioritise reproducibility and a realistic assessment of predictive performance rather than maximising results on a single dataset.

8.3.1 Dataset

The implementation is based on the **HAM10000** (*Human Against Machine with 10,000 Training Images*) dataset, introduced by Tschandl et al. [36] and released as the principal dataset for the ISIC 2018 Skin Lesion Analysis Challenge [37]. With 10,015 dermoscopic images spanning seven diagnostic categories, HAM10000 has become one of the most widely used public benchmarks for developing and evaluating machine learning methods for skin lesion analysis.

The dataset combines images collected from different clinical sources and represents a broad spectrum of pigmented skin lesions encountered in dermatological practice. Diagnostic labels were established through histopathology where available, together with follow-up examinations, expert consensus, or confocal microscopy, resulting in a carefully curated dataset suitable for supervised learning.

Although HAM10000 was originally developed as a seven-class diagnostic dataset, this project reformulates the task as a binary lesion triage problem. The original diagnostic categories were grouped according to whether they require priority dermatological assessment, as introduced in the Problem Definition section. The mapping between the original labels and the binary target used throughout the analysis is summarised in Table 8.1.

Table 8.1: Mapping from the original HAM10000 diagnostic categories to the binary triage target used in this study.

Original diagnosis	Label	Project class
Melanoma	`mel`	Priority review
Basal cell carcinoma	`bcc`	Priority review
Actinic keratosis / Bowen's disease	`akiec`	Priority review
Melanocytic naevus	`nv`	Non-priority
Benign keratosis	`bkl`	Non-priority
Dermatofibroma	`df`	Non-priority
Vascular lesion	`vasc`	Non-priority

After this transformation, the dataset contains **1,954 priority-review images** and **8,061 non-priority images**, reflecting the natural imbalance between potentially malignant and predominantly benign lesions. This imbalance constitutes an important consideration throughout model development and is explicitly addressed during training.

A further characteristic of HAM10000 is that multiple images may correspond to the same underlying lesion. Since these images are not statistically independent, preserving lesion-level separation between the training, validation, and test partitions becomes essential to avoid information leakage. The partitioning strategy adopted in this study is described in the following section.

8.3.2 Exploratory Data Analysis

Before developing the classification models, an exploratory analysis was conducted to characterise the dataset and identify properties that could influence model development and evaluation. Rather than searching for predictive relationships, the objective of this stage was to verify the quality of the available data, understand the distribution of the target classes, and identify potential sources of bias that should be considered throughout the modelling process.

The first step was to examine the distribution of the binary triage target. Figure Figure 8.1 summarises the number of images and unique lesions assigned to each class. As expected, non-priority lesions represent the majority of observations, whereas lesions requiring priority review constitute a substantially smaller proportion of the dataset. This imbalance reflects routine dermatological practice, where benign lesions are considerably more common than lesions requiring immediate clinical attention, and motivates the use of evaluation metrics that extend beyond overall classification accuracy.

A second characteristic of HAM10000 is that multiple dermoscopic images may correspond to the same underlying lesion. Figure Figure 8.2 illustrates the distribution of the number of images available for each lesion. While many lesions are represented by a single image, a substantial proportion includes multiple photographs captured from different viewpoints or magnifications. This characteristic reinforces the importance of performing lesion-level rather than image-level partitioning, ensuring that the evaluation reflects the model's ability to generalise to previously unseen lesions rather than recognising repeated visual appearances of the same case.

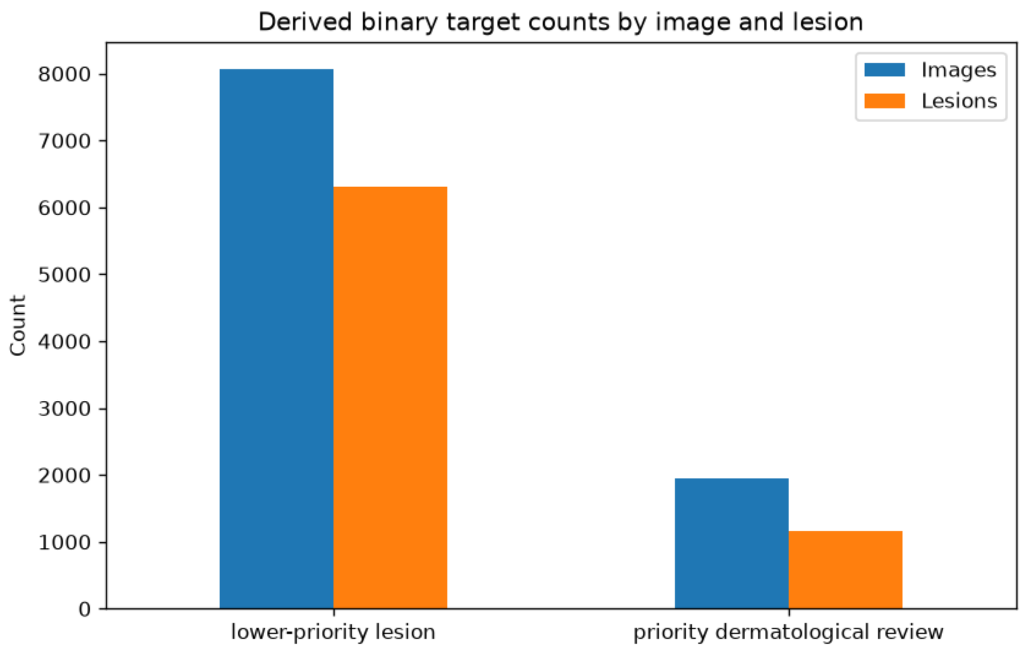

Figure 8.1: Distribution of images and unique lesions across the binary triage classes.

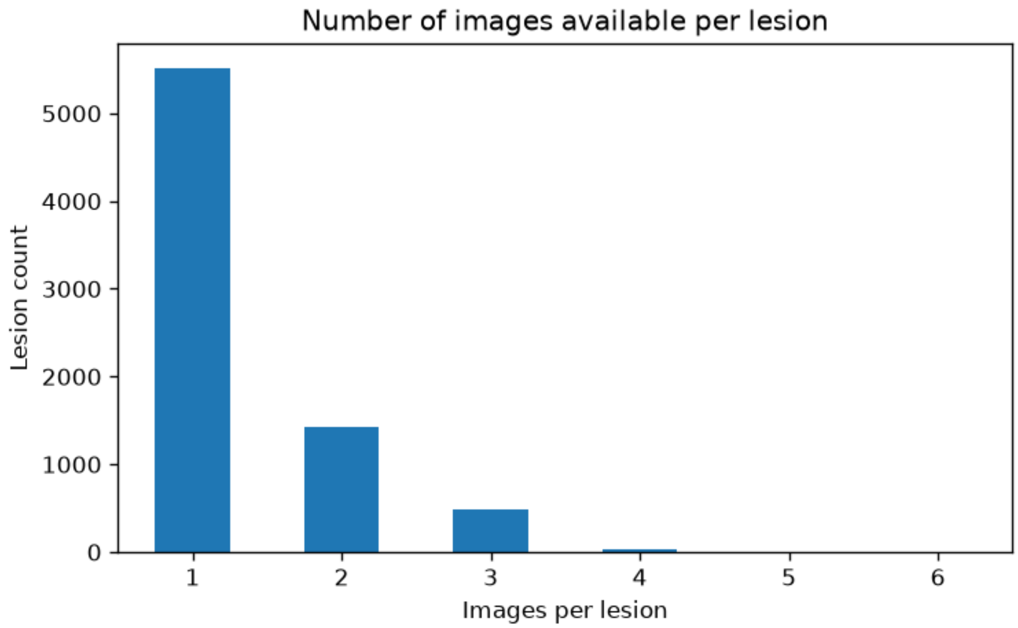

Figure 8.2: Distribution of the number of images available for each unique lesion.

Patient demographics provide additional context regarding the composition of the study population. Figure Figure 8.3 presents the age distribution for the two binary target classes. Most observations correspond to middle-aged and older adults, consistent with the epidemiology of pigmented skin lesions. Although the age distributions overlap considerably, this analysis provides useful context for interpreting the characteristics of the study population.

The distribution of patients by biological sex is shown in Figure Figure 8.4. Male and female patients are both well represented across the two triage categories, reducing the likelihood that the classification models are driven primarily by differences in sex representation within the dataset.

Figure Figure 8.5 summarises the anatomical location of lesions across the binary target classes. Lesions originate from multiple body regions, with the trunk and lower extremities accounting for a substantial proportion of the observations. This diversity contributes to the visual variability of the dataset and reflects the heterogeneous clinical presentation of pigmented skin lesions encountered in routine practice.

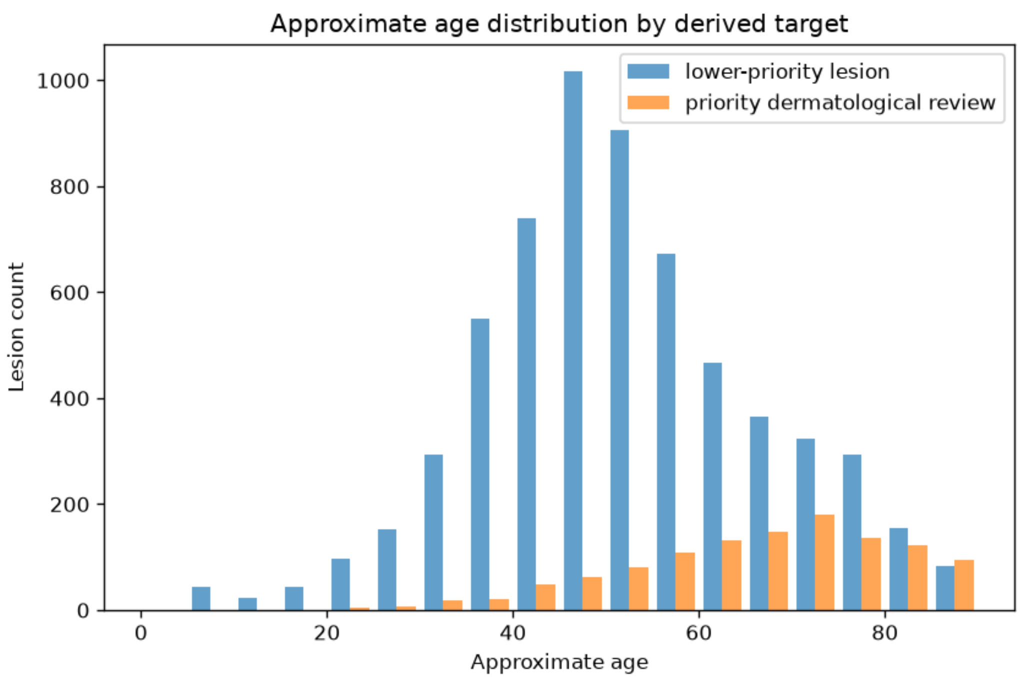

Figure 8.3: Patient age distribution stratified by binary triage class.

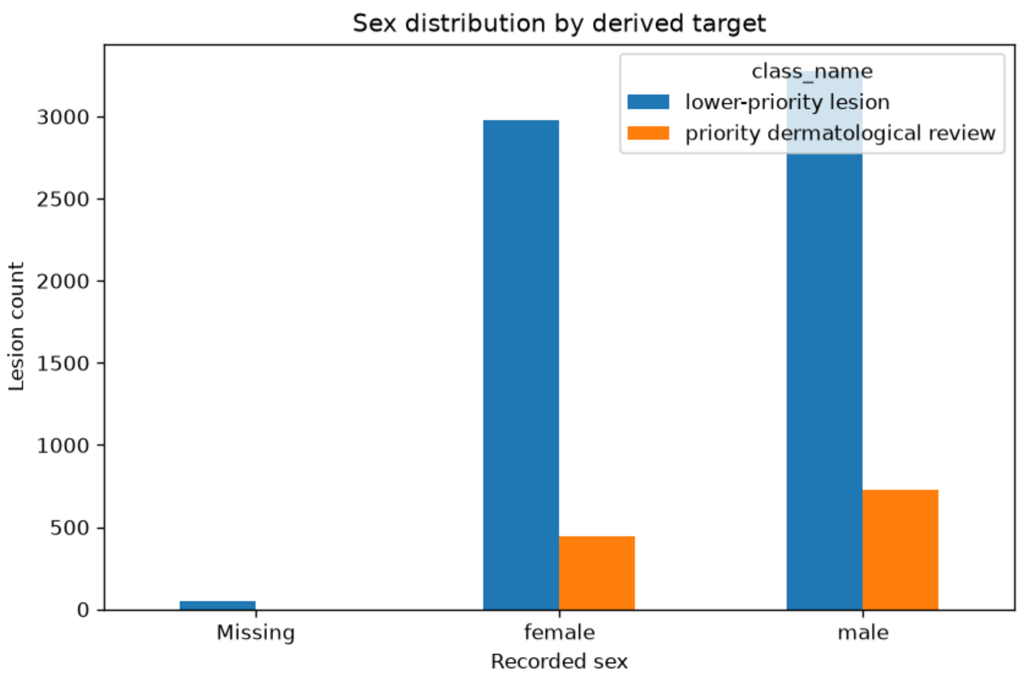

Figure 8.4: Distribution of patient sex stratified by binary triage class.

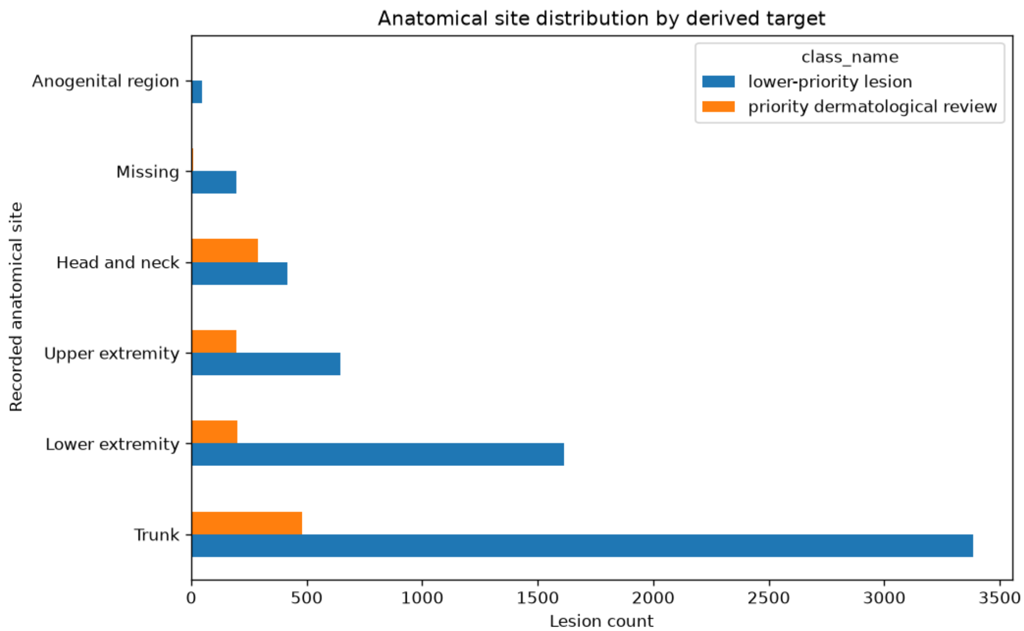

Figure 8.5: Distribution of anatomical sites stratified by binary triage class.

To better understand the composition of the binary target, the original diagnostic categories were also examined. Figure Figure 8.6 shows the number of images and lesions contributed by each diagnostic class before collapsing them into the binary triage formulation. This analysis confirms that the priority-review group combines several clinically important but comparatively infrequent diagnoses, whereas benign melanocytic nevi dominate the dataset.

Finally, Figure Figure 8.7 summarises the methods used to establish the reference diagnosis for each lesion. Histopathology represents the predominant confirmation method for malignant lesions, whereas benign lesions include a mixture of histopathological confirmation, expert consensus, and follow-up examinations. Understanding these differences provides additional context regarding the reliability of the ground-truth labels used throughout model development and evaluation.

Overall, the exploratory analysis confirms several characteristics that must be considered throughout model development, including class imbalance, heterogeneous patient demographics, variation in anatomical location, and the presence of multiple images corresponding to the same lesion. Among these, the repeated

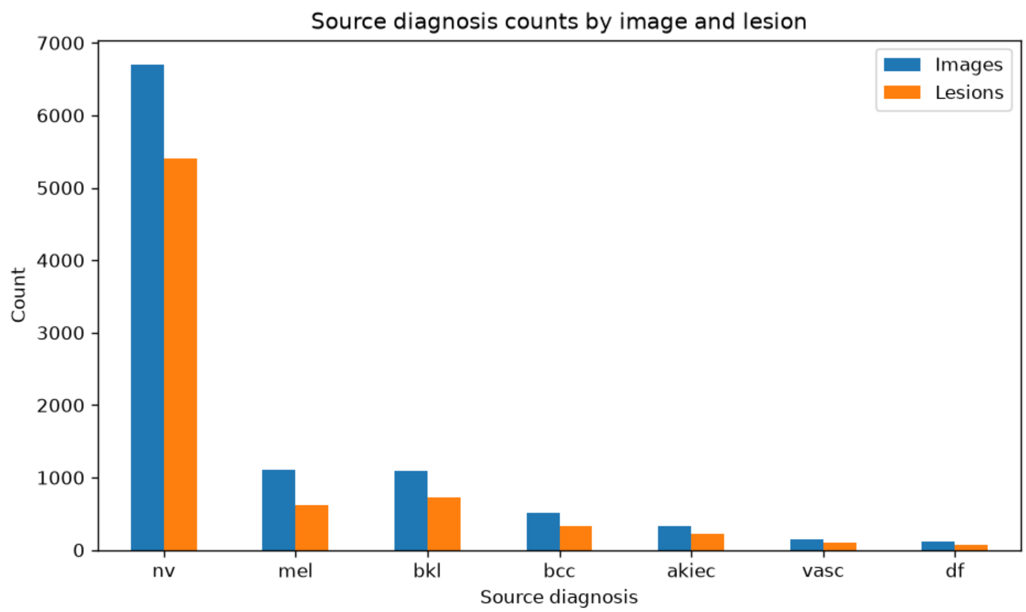

Figure 8.6: Distribution of images and lesions across the original diagnostic categories.

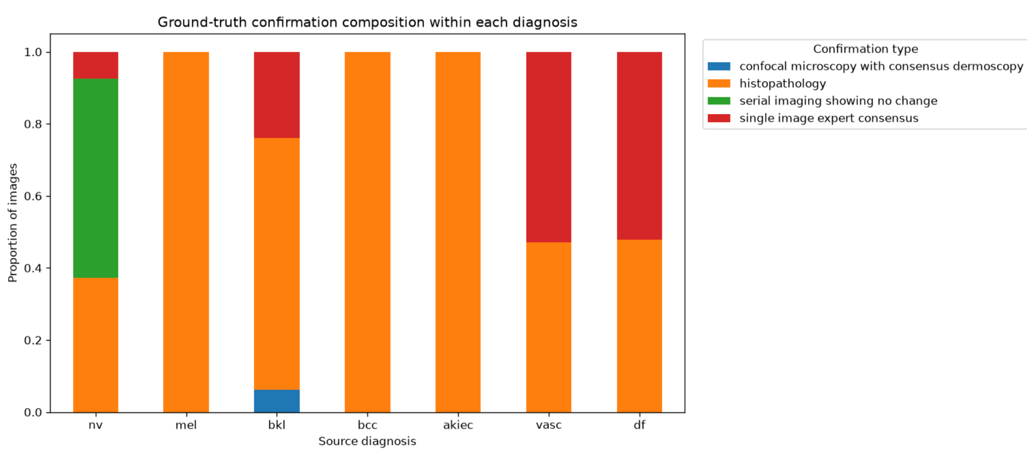

Figure 8.7: Reference diagnosis confirmation methods across the original diagnostic categories.

representation of individual lesions has the greatest methodological implications, directly motivating the lesion-level data partitioning strategy adopted throughout the remainder of this study.

8.3.3 Lesion-Level Data Partitioning

A critical characteristic of the HAM10000 dataset is that multiple dermoscopic images may correspond to the same underlying lesion. Although these images differ in magnification, lighting conditions, orientation, or acquisition time, they remain highly correlated because they depict the same clinical case. Consequently, randomly splitting individual images between training and testing sets would allow visually similar representations of the same lesion to appear in both partitions, leading to overly optimistic estimates of model performance.

To prevent this form of information leakage, the dataset was partitioned at the lesion level rather than at the image level. Each unique lesion identifier was assigned exclusively to either the training, validation, or test set, ensuring that no lesion was represented in more than one partition. This strategy provides a more realistic assessment of the model's ability to generalise to previously unseen lesions, which is ultimately the objective in a clinical decision-support setting.

The resulting partition is summarised in Table 8.2. Approximately 70% of the images were allocated to the training set, while the remaining observations were divided almost equally between the validation and independent test sets. Importantly, the proportion of priority-review lesions remains remarkably consistent across all three partitions, indicating that the lesion-level split preserves the overall class distribution while maintaining complete separation between lesions.

Table 8.2: Summary of the lesion-level data partition used throughout model development and evaluation.

Partition	Images	Unique lesions	Priority-review images	Priority-review proportion
Training	7,014	5,229	1,372	19.6%
Validation	1,506	1,120	299	19.9%
Test	1,495	1,121	283	18.9%

Once established, these partitions remained fixed throughout the entire analysis.

All preprocessing operations, image normalisation, data augmentation, model optimisation, threshold selection, and performance assessment were performed using this predefined split. The validation set was used exclusively for model selection and threshold calibration, whereas the test set remained untouched until the final evaluation.

By separating lesions rather than individual images, the evaluation protocol better reflects the intended clinical application. During deployment, a model is expected to analyse lesions that have never been observed previously, rather than additional photographs of lesions encountered during training. Adopting a lesion-level partition therefore provides a substantially more reliable estimate of the model's ability to generalise to new patients.

With the data partitions established, the next stage of the workflow consists of standardising the input images before they are presented to the neural networks.

8.3.4 Image Preprocessing

Before being presented to the neural networks, all dermoscopic images underwent a standardised preprocessing pipeline to ensure consistent input dimensions and pixel distributions across the dataset. Standardising the input is an essential step in deep learning workflows, as convolutional neural networks require images of fixed size while also benefiting from comparable intensity statistics during optimisation.

All images were resized to 224×224 pixels using bilinear interpolation. This resolution represents a practical compromise between preserving diagnostically relevant visual structures and maintaining computational efficiency during training. Moreover, it matches the input size expected by the pre-trained ResNet18 architecture, allowing both models to be evaluated under identical conditions.

Pixel values were subsequently converted to floating-point tensors and normalised using channel-wise statistics computed exclusively from the training partition. Applying the same transformation to the validation and test sets ensures that no information from unseen data contributes to the preprocessing stage, thereby preventing a subtle but important source of information leakage.

No handcrafted image enhancement techniques were applied. In particular, procedures such as colour correction, contrast enhancement, hair removal, lesion segmentation, or artefact suppression were intentionally omitted. This decision allows the models to learn directly from the original dermoscopic images and avoids introducing additional preprocessing assumptions that may not generalise across acquisition devices or clinical settings.

Consequently, the preprocessing pipeline remains deliberately simple, focusing on producing consistent inputs while preserving the visual characteristics of the original images. This design also facilitates reproducibility and makes the comparison between the baseline convolutional network and the transfer learning approach more transparent, as both models receive exactly the same preprocessed data.

Although image preprocessing standardises the inputs, it does not address the limited variability of the training data. To improve the robustness of the learned representations and reduce the risk of overfitting, a series of data augmentation techniques were applied exclusively to the training partition, as described in the following section.

8.3.5 Data Augmentation

Although the lesion-level partition prevents information leakage, the number of available training images remains relatively limited for training deep convolutional neural networks. Furthermore, dermoscopic images exhibit natural variability arising from differences in patient positioning, camera orientation, magnification, and illumination. Data augmentation was therefore applied to increase the diversity of the training data while preserving the underlying diagnostic characteristics of each lesion.

All augmentation operations were performed online during training, meaning that transformed images were generated dynamically at each training epoch rather than being stored as additional samples. This approach exposes the model to a broader range of image variations while avoiding unnecessary duplication of the dataset.

The augmentation pipeline included random horizontal and vertical flips, small rotations, affine transformations, resized crops, and mild colour perturbations. Together, these operations encourage the networks to learn features that are robust to changes in orientation, scale, and illumination without altering the clinical label associated with each lesion.

Importantly, data augmentation was applied exclusively to the training partition.

Validation and test images were subjected only to the standard preprocessing pipeline described in the previous section. Restricting augmentation to the training data ensures that model selection and final performance evaluation are conducted using images that faithfully represent unseen clinical observations.

This strategy serves two complementary purposes. First, it increases the effective diversity of the training data, helping to reduce overfitting in a dataset of moderate size. Second, it encourages the learned feature representations to become less sensitive to superficial image characteristics, allowing the models to focus on diagnostically meaningful visual patterns.

With the input pipeline fully defined, the following sections describe the two modelling approaches evaluated in this study: a baseline convolutional neural network trained from scratch and a transfer learning model based on a pretrained ResNet18 architecture.

8.3.6 Baseline Convolutional Neural Network

As a benchmark for comparison, a convolutional neural network was trained from scratch using only the dermoscopic images available in the training partition. The purpose of this baseline model was not to maximise predictive performance but to establish a reference against which the benefits of transfer learning could be evaluated under an identical experimental protocol.

The network follows a conventional convolutional architecture composed of successive feature extraction blocks followed by fully connected classification layers. The convolutional layers progressively learn increasingly abstract image representations, beginning with low-level visual features such as edges, colour transitions, and simple textures before combining them into more complex lesion characteristics relevant to the classification task.

To improve optimisation stability and reduce overfitting, the architecture incorporates batch normalisation, non-linear activation functions, and dropout regularisation. Max-pooling operations progressively reduce the spatial resolution of the feature maps while increasing the receptive field of subsequent convolutional layers, allowing the network to capture both local morphological patterns and broader structural characteristics of each lesion.

The network was trained using the Adam optimiser and binary cross-entropy loss, with the objective of distinguishing lesions requiring priority clinical review from those considered lower priority. During optimisation, model parameters were updated exclusively using the training partition, while the validation set was used to monitor learning progress and identify the best-performing model.

Rather than selecting the final model according to classification accuracy alone, the validation Average Precision (AP) score was adopted as the primary selection criterion. Because the dataset exhibits moderate class imbalance and the principal objective is to identify clinically important lesions, Average Precision provides a more informative assessment of model quality than overall accuracy by evaluating performance across all possible decision thresholds.

The weights corresponding to the highest validation Average Precision were retained and subsequently used for the final evaluation on the independent test set. This model therefore represents the strongest version of the baseline architecture obtained without exposure to the test data, providing a fair point of comparison for the transfer learning approach described in the following section.

8.3.7 Transfer Learning with ResNet18

In addition to the baseline convolutional network, a transfer learning approach based on the ResNet18 architecture was evaluated. Transfer learning has become a standard strategy in medical image analysis because it enables models to leverage visual representations learned from large-scale image datasets, reducing the amount of task-specific data required to achieve strong predictive performance.

ResNet18 was originally trained on the ImageNet dataset, which contains millions of natural images spanning a wide range of object categories. Although dermoscopic images differ substantially from natural photographs, the lower layers of deep convolutional networks learn general-purpose visual features such as edges, textures, colour gradients, and geometric patterns that remain useful across many image classification tasks. Initialising the network with pre-trained weights therefore provides a more informative starting point than random parameter initialisation.

To adapt the model to the binary lesion triage task, the original classification layer was replaced with a new fully connected output layer producing a single probability corresponding to the priority-review class. During training, the entire network was fine-tuned using the dermoscopic images rather than restricting optimisation to the final classification layer. This allows the higher-level feature representations to progressively adapt to the visual characteristics of skin lesions while retaining the robust low-level features learned during pre-training.

The transfer learning model was trained using the same lesion-level partitions, preprocessing pipeline, data augmentation strategy, optimiser, loss function, and validation protocol adopted for the baseline convolutional network. Maintaining identical experimental conditions ensures that any observed differences in predictive performance can be attributed primarily to the modelling strategy rather than to differences in data preparation or optimisation.

As with the baseline model, validation Average Precision was used as the primary model selection criterion. The network parameters corresponding to the highest validation AP score were retained and subsequently evaluated on the independent test set. This consistent evaluation protocol enables a direct comparison between a model trained entirely from scratch and one that benefits from prior visual knowledge acquired through large-scale pre-training.

8.3.8 Evaluation Protocol

Both models were evaluated using exactly the same experimental protocol to ensure that any performance differences reflected the modelling approach rather than inconsistencies in data preparation or model selection. Throughout the study, the lesion-level data partition remained fixed, with the training set used for parameter optimisation, the validation set reserved for model selection and threshold calibration, and the independent test set withheld until the final evaluation.

Given the clinical objective of prioritising lesions requiring further assessment, model selection was based on validation Average Precision (AP) rather than classification accuracy. Average Precision summarises the precision–recall relationship across all possible decision thresholds and is particularly informative when class distributions are imbalanced or when correctly identifying the minority class is of greater clinical importance than maximising overall accuracy.

After selecting the best-performing model according to validation AP, an operating threshold was determined using the validation data. This threshold was then fixed and applied without modification to the independent test set, ensuring that the reported performance represents a genuine out-of-sample evaluation rather than an optimistically tuned estimate.

The final assessment included threshold-independent metrics, such as the Area Under the Receiver Operating Characteristic Curve (ROC AUC) and Average Precision, together with threshold-dependent measures including sensitivity, specificity, precision, F1-score, balanced accuracy, and overall classification accuracy. Reporting both groups of metrics provides a more comprehensive characterisation of model behaviour than relying on a single performance measure.

To quantify the uncertainty associated with the reported results, confidence intervals were estimated using bootstrap resampling of the independent test set. This procedure generates repeated samples with replacement and estimates the variability of each performance metric without relying on strong distributional assumptions.

Because the dataset contains multiple images of some lesions, predictions were also aggregated at the lesion level in addition to the image level. Aggregating predictions better reflects the intended clinical application, where the objective is to support the assessment of individual lesions rather than isolated photographs. This complementary analysis allows the robustness of the models to be evaluated under a more clinically meaningful perspective.

Finally, model performance was examined across relevant demographic subgroups, including biological sex and age category. Although these analyses are exploratory, they provide useful insight into the consistency of predictive performance across different patient populations and help identify potential sources of performance variability that may warrant further investigation.

The methodology described in this chapter is implemented in the accompanying Python notebook, which is available in the project repository.

8.4 Results, Discussion, and Limitations

Section.

8.5 Technical Appendix (Optional)

Section.

Part II

Writings & Essays

Overview

The chapters presented in this section complement the applied projects found elsewhere in this portfolio by exploring the ideas, methods, and scientific principles that underpin quantitative modeling and computational problem solving. Some take the form of technical essays, while others investigate computational methods or methodological questions through independent studies and practical implementations.

Rather than focusing on the immediate demands of a particular application domain, these chapters examine the broader concepts that continue to shape modern quantitative practice. Their topics range from scientific methodology and mathematical reasoning to computational techniques with applications that extend across multiple disciplines.

Although the subject matter varies, they share a common objective: to develop a deeper understanding of the assumptions, limitations, and reasoning that lie behind quantitative methods. The emphasis is not only on how particular techniques work, but also on why they work, when they are appropriate, and how they contribute to solving complex problems.

Together, these chapters complement the applied case studies presented throughout the portfolio by highlighting the broader scientific and computational ideas that continue to inform my approach to quantitative problem solving.

Chapter 9

Data Science in the Age of Hype: Why the scientific method still matters

9.1 The Rise of Data Science

Data has become one of the most valuable assets of any modern organization. The unprecedented flow of information—resulting from countless digital interactions such as clicks, e-commerce transactions, mobile communications, and social media activity—reflects its central role in contemporary society. At the same time, the rapid advances in computing that enable large-scale data collection, storage, and processing are radically transforming the way businesses, governments, scientific communities, and institutions make decisions.

Yet, raw data by itself is meaningless. It only acquires value when subjected to systematic and rigorous analysis through methodologies capable of detecting patterns, inferring relationships, predicting trends, and ultimately translating insights into decisions with real-world impact. This is why Clive Humby's now-famous analogy describes data as "the new oil": like crude oil, data must be refined before it can generate value. The crucial element is not the "raw material" itself but the techniques used to extract its potential and the knowledge derived from it. This reflection naturally leads to a fundamental question: what do we really mean by Data Science?

Historically, the term Data Science emerged from an evolutionary process shaped by multiple contributions that helped formalize it as a discipline. Although its roots lie in statistics and computer science, its consolidation as an autonomous field took place over more than half a century of theoretical and technological developments.

The conceptual seed of Data Science can be traced back to 1962, when American statistician John W. Tukey published his influential essay *The Future of Data Analysis*. In it, Tukey argued that statistics should not be confined to hypothesis confirmation alone, but should place equal emphasis on the exploration, diagnosis, and understanding of data [38], [39]. He introduced *data analysis* as a distinct discipline, anticipating the vision of a "science of learning from data"—one in which empirical inquiry and iterative interaction with data occupy the core of quantitative knowledge.

Later, in 1974, Danish computer scientist Peter Naur, in his *Concise Survey of Computer Methods*, explicitly used the terms *Data Science* and *Datalogy* to refer to the systematic study of data [40]. Naur argued that computing should focus on the structure, meaning, and organization of data rather than on hardware or programming. Although his proposal went largely unnoticed for years, it laid conceptual foundations that would be revisited decades later.

In 1997, statistician C.F. Jeff Wu, during his inaugural lecture at the University of Michigan titled *Statistics = Data Science?*, proposed renaming statistics as *Data Science* and calling modern statisticians *data scientists* [41]. His goal was to signal a new era in which statistical inference should integrate with computation and empirical analysis to move beyond mere data description and outdated stereotypes.

This perspective was further developed in 2001 by William S. Cleveland in his seminal article *Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics* [42]. Cleveland proposed a systematic expansion of statistics by identifying six core technical areas of Data Science: multidisciplinary investigations, models and methods for data, computing with data, pedagogy, tool evaluation, and theory. Rather than allowing statistics to contract into a narrowly mathematical discipline, he emphasized the central role of computation, real-world data analysis, education, and the development and evaluation of tools. His framework laid the conceptual groundwork for understanding Data Science as a comprehensive, data-centered field—deeply rooted in statistics, yet necessarily broader in scope and practice.

That same year, Leo Breiman, a statistician at the University of California, Berkeley, published his influential essay *Statistical Modeling: The Two Cultures*, in which he argued that statistical modeling serves two fundamental purposes: **information**—extracting insight into how nature associates response variables with input variables—and **prediction**—the ability to accurately anticipate future responses for new observations [43]. Breiman critically observed that traditional statistical practice had largely privileged the informational goal through highly structured, assumption-driven models, often at the expense of predictive performance. He advocated for a broader methodological perspective in which algorithmic and data-driven approaches play a central role, emphasizing prediction as an essential complement rather than a secondary objective. This distinction, and the tension it revealed between explanation and prediction, foreshadowed core principles of modern Data Science, where understanding and predictive accuracy are treated as intertwined but distinct aims.

By the late 2000s, the rise of Big Data and the growing need for professionals capable of combining statistics, programming, and domain expertise fueled the popularization of both the term Data Science and the role of the data scientist. Media consolidation came in 2012, when Thomas H. Davenport and DJ Patil published *Data Scientist: The Sexiest Job of the 21st Century* in the *Harvard Business Review* [44]. Since then, Data Science has become a strategic discipline across both academia and industry.

Today, Data Science is recognized as an interdisciplinary field that integrates statistical, computational, and conceptual methods to extract knowledge, infer patterns, and generate predictions from data. In many ways, it represents the natural evolution of statistics a discipline now equipped to address complex phenomena in fields as diverse as economics, biology, marketing, and physics.

However, despite its relevance and expansion, the term *Data Science* has become somewhat diffuse, often distorted by media overexposure. This ambiguity calls for a deeper examination of its epistemological foundations: if we speak of a **Science** of Data, we must ask what makes a discipline a science?

From this perspective, Data Science can indeed be considered a science. Its scientific character lies in the systematic application of the scientific method to the study of data. The data scientist formulates hypotheses, analyzes observations, and tests results just as any other scientist does. The difference lies in the object of study: data as a quantitative representation of reality.

In this sense, Data Science can be viewed as a generalization of statistics extending its focus from **inference** (understanding how the world behaves based on observed data) to **prediction** (anticipating how it might behave under new conditions). This predictive capability, powered by machine learning algorithms, enables the effective application of the **Hypothetical-Deductive Method**: formulating hypotheses about future phenomena and testing them through data and predictive models. In other words, machine learning acts as a technological enabler that materializes Breiman's inference–prediction duality, providing tools to model complex relationships directly from data and to test hypotheses that were previously unreachable with traditional statistical methods. It is this integration of inference, prediction, and empirical validation that solidifies the scientific nature of the discipline.

Understanding these foundations is essential for interpreting the true scope of Data Science. While the discipline draws from statistics, computer science, and domain expertise, its ultimate objective is not merely the manipulation of data but the extraction of reliable knowledge about the world. In this sense, Data Science should not be understood as a collection of algorithms or software tools, but as a methodological framework for reasoning under uncertainty.

However, the rapid growth of the field has also brought an unintended consequence: a proliferation of narratives that portray Data Science primarily as a technological revolution driven by ever more sophisticated algorithms. In many contexts, the emphasis has shifted toward tools, automation, and predictive performance, sometimes overshadowing the epistemological principles that historically guided scientific inquiry.

This tension raises a critical question for modern practitioners: if Data Science claims to be a science, what role does the scientific method play in its practice today? Addressing this question requires revisiting the principles that define scientific reasoning and examining how they apply to the analysis and interpretation of data in an era increasingly shaped by computational power and algorithmic models.

9.2 The Nature of Science and the Scientific Method

The very semantics of the term *Data Science* make it impossible to detach it from the concept of *science*. This forces us to pause and reflect: can Data Science truly be regarded as a scientific discipline, or is it merely a fashionable commercial label? A comprehensive analysis of the nature of science lies far beyond the scope of this article; however, for the sake of intellectual honesty, we should not evade the issue. At the very least, we ought to clarify what makes an activity genuinely *scientific*.

From an epistemological perspective, science cannot be defined merely as any endeavor that leads to new discoveries or expands knowledge. Scientific knowledge evolves what was once considered true has often been revised, refined, or discarded. It is therefore reasonable to assume that part of what we regard today as scientifically valid will eventually be replaced, even if we cannot foresee which part. In other words, science does not guarantee truth. What distinguishes science from other forms of inquiry is not the *results* it produces, but the *method* it employs to generate them [45]. Science is better understood as an iterative mechanism—almost algorithmic in nature—that produces, tests, refines, and occasionally rejects knowledge. This is a central idea, because in essence, a data scientist does exactly that: applies the scientific method as a repeatable process of inquiry. We will return to this point shortly when discussing the *Hypothetico-Deductive Method*.

With this foundation in place, it becomes easier to see why Data Science can be viewed as a generalization of Statistics. This is not a trivial claim. Statistics is undoubtedly a science specifically, a formal science, alongside logic and computer science. Formal sciences study abstract systems through reasoning rather than direct experimentation, distinguishing them from natural, social, and applied sciences. Yet classical statistics has been historically grounded in *inductive inference*: reasoning from the particular to the general. Induction allows us to infer broader conclusions from observed cases, but it carries a fundamental limitation: inductive reasoning can *support* a conclusion, but never *prove* it. This problem, famously articulated by David Hume in the 18th century and known as the *problem of induction*, arises from the logical gap between past observations and future expectations.

Bertrand Russell illustrated this issue through the well-known "inductivist turkey". A turkey observes that every morning at 9 a.m. the farmer brings food. After hundreds of consistent observations—rainy days, sunny days, warm and cold alike—the turkey confidently concludes that it will always be fed at 9 a.m. Yet on Christmas Eve, instead of being fed, it is slaughtered. A single contradictory event collapses a conclusion seemingly supported by overwhelming evidence. The lesson is clear: no amount of past observations can logically ensure the truth of a universal claim about the future. Karl Popper pushed this idea further: induction can never confirm a theory with certainty, whereas a single counterexample can refute it. The turkey's mistake—rooted in the "turkey illusion"—was assuming regularity without understanding the underlying mechanism (the farmer's intention).

None of this implies that induction is a "bad" method. It remains essential as the generator of hypotheses and conjectures. As Klimovsky notes [46], induction may be seen as a probabilistic justification: it evaluates how strongly observations support a hypothesis, rather than certifying its truth. Put simply, science advances by combining induction (to propose hypotheses) and deduction (to test them). Deduction enables us to assess whether predictions derived from hypotheses correspond to reality. Verification, however, does not grant absolute truth; it merely confirms that a hypothesis has survived a test under certain conditions.

This tension lies at the heart of Leo Breiman's critique of classical statistics. While statistics provided the mathematical language to quantify uncertainty and draw inferences about populations from limited data, it remained fundamentally bound to inductive reasoning: no matter how large the sample or how robust the model, a future observation could always contradict it. Machine Learning transformed this landscape by not only inferring patterns from data but also validating predictive performance empirically. A model is no longer judged solely by how well it explains past data, but by how accurately it predicts *unseen* data. This represents a qualitative shift that operationalizes the hypothetico-deductive paradigm: generating hypotheses (models) and testing them empirically against new observations in a reproducible and quantifiable way.

To illustrate this in a Data Science context, consider a credit-scoring model. Classical statistics may estimate relationships between customer features and default risk through inductive inference on historical samples. A machine learning model, however, must *demonstrate its predictive power* on new customers those the model has never encountered. If the predictions fail, the model is rejected or refined. This is falsification in action, applied to data-driven decision-making.

In this sense, Data Science can be expressed as **Inference + Prediction**, or equivalently, as **Statistics + Machine Learning**. It unites the inductive strength of statistics with the deductive rigor of model testing, fulfilling Breiman's vision of a discipline that bridges explanation and prediction a modern science of learning from data. The next section will examine this duality through the lens of the *Hypothetico-Deductive Method*, the framework that ultimately solidifies the scientific character of Data Science.

9.3 The Scientific Mindset of a Data Scientist

At the core of scientific inquiry lies a logical structure that transcends individual disciplines: the **Hypothetico-Deductive Method**. It provides the conceptual backbone of the scientific method, linking theoretical conjecture with empirical observation. The process begins with a hypothesis—a tentative model proposed to explain observed behavior or predict its future course. or predict its future course. From this hypothesis, one deduces empirically testable consequences. These predictions are then confronted with real data through experimentation or systematic observation. When evidence contradicts the hypothesis, it must be revised or discarded; when the evidence supports it, the hypothesis gains provisional credibility, though never definitive truth.

This dynamic interplay between theoretical formulation, deduction, and empirical scrutiny defines the essence of scientific work. What grants scientific legitimacy to a model is not its mathematical elegance or computational sophistication, but its **capacity to generate falsifiable predictions**. The scientific method is therefore inherently self-correcting: each cycle of hypothesizing, testing, and refining moves knowledge closer to truth while acknowledging that truth remains an ever-distant horizon.

Traditional inferential statistics implements this logic only partially, primarily through hypothesis testing. A null hypothesis is proposed, its probabilistic implications are derived, and sample data are used to evaluate whether they justify its rejection. Yet this remains predominantly an inductive process, generalizing from finite observations into probabilistic claims about broader populations. Moreover, statistical validation is typically static once the data are gathered, the hypothesis is evaluated solely in that context, rather than through iterative confrontation with new evidence.

Machine learning, by contrast, operationalizes the deductive stage of scientific reasoning through **predictive validation**. Each model can be viewed as a hypothesis about the functional relationship between predictors and outcomes. Training a model corresponds to formulating the hypothesis; evaluating it on unseen data corresponds to testing its empirical consequences. Performance metrics quantify how well the model withstands empirical scrutiny, and failures become informative, guiding the reformulation or replacement of the underlying hypothesis.

This perspective turns Data Science into a **practical enactment of the Hypothetico-Deductive Method**. It combines inductive discovery of structure within data with deductive testing of the predictive consequences of that structure. More importantly, it enables Data Science to progress from mere correlation to **genuine explanation**, seeking models that capture mechanisms rather than patterns alone.

Consider the problem of **customer churn** in retail. A predictive model might accurately identify customers at risk of leaving based solely on behavioral correlations a purely inductive exercise. But if the goal is to determine whether a discount campaign will *reduce* churn, prediction alone is insufficient. The data scientist must investigate—and, when possible, formalize—**causal models** capturing how interventions alter outcomes. Prediction answers *who* is likely to churn; causality answers *why* and *what we can do about it*.

Under the Hypothetico-Deductive paradigm, this iterative cycle can be summarized visually, as shown in Figure 9.1, and is typically composed of the following stages:

- **Systematic observation** of a phenomenon
- **Formulation of hypotheses** or models to explain observed patterns
- **Deduction of testable predictions** grounded in those hypotheses
- **Empirical evaluation** of predictions via new data
- **Revision or replacement** of hypotheses based on the evidence collected

Viewed from this angle, the role of a data scientist extends far beyond executing algorithms or statistical routines. It is the **iterative application of the scientific method** across diverse business contexts generating hypotheses, building predictive and causal models, confronting them with evidence, and allowing data to arbitrate which explanations survive. Data Science thus emerges not merely as a technical discipline, but as the **contemporary expression of scientific thinking** applied to data-rich environments, where distinguishing correlation from causation and prediction from explanation defines true professional excellence.

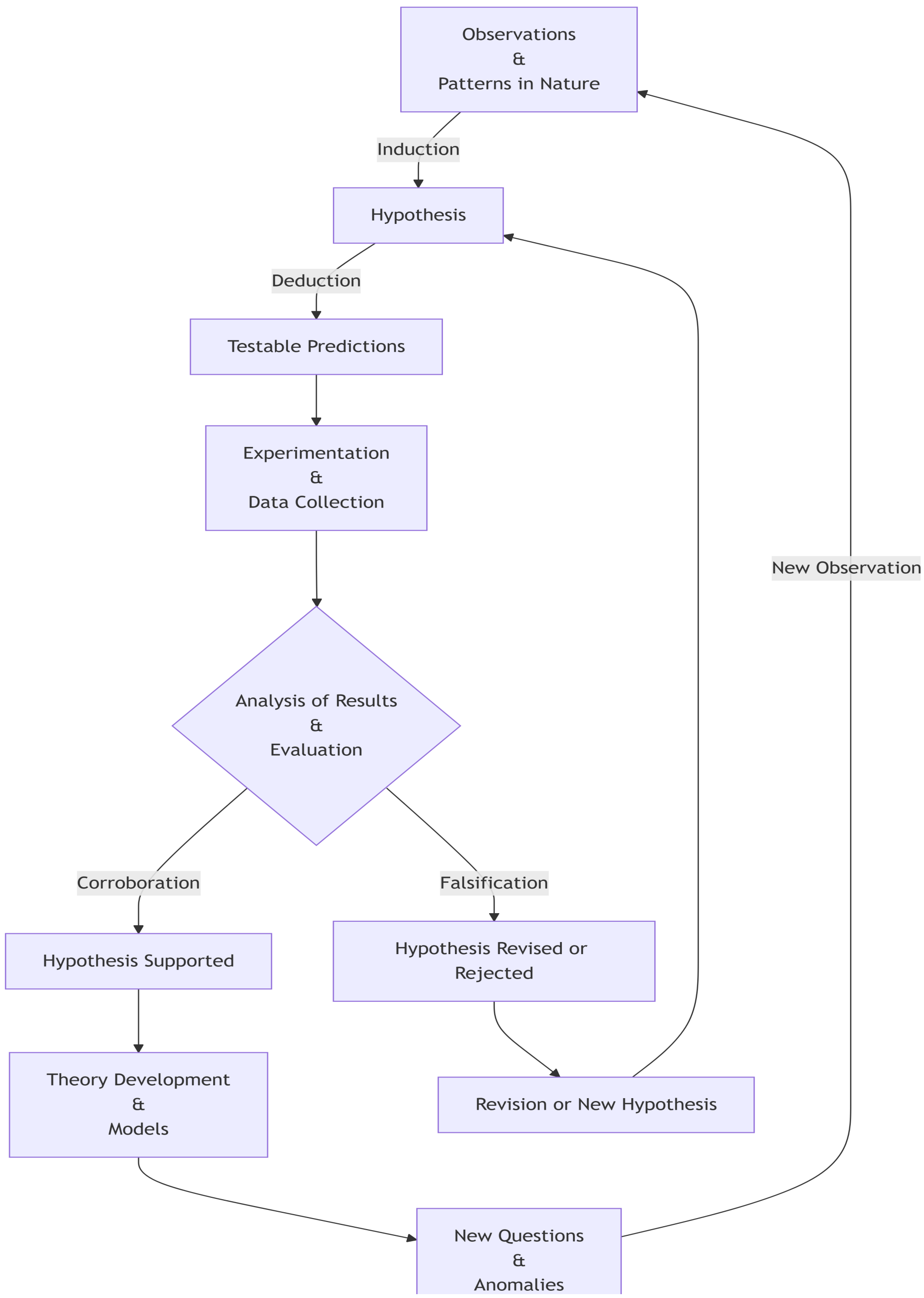

Figure 9.1: The hypothetico-deductive method as an iterative loop: observations motivate hypotheses (induction); hypotheses yield testable predictions (deduction); empirical results either support or challenge the hypothesis, leading to theory formation or revision and ultimately to new questions that restart the process.

9.4 The Fundamental Constituents of Data Science

Despite its rapid expansion and increasing visibility, Data Science is often portrayed as a discipline defined by an ever-growing collection of tools, frameworks, and technological trends. New libraries, platforms, and modeling techniques appear constantly, creating the impression that the field evolves primarily through technological novelty. This perception, however, can be misleading. While modern practice indeed relies on a rich computational ecosystem, the scientific core of Data Science is not defined by software stacks or fashionable methodologies. Rather, it rests on a far more compact and principled foundation: **probabilistic reasoning and statistical inference**.

From a first-principles perspective, Data Science is best understood as a systematic framework for reasoning under uncertainty using data. Probability provides the language to model randomness and variability, while Statistics supplies the methodological machinery for learning from data and evaluating the reliability of the conclusions we draw from it. Together, they form the conceptual backbone that allows data-driven reasoning to remain coherent, interpretable, and scientifically grounded.

Computation plays a crucial—but often misunderstood—role in this framework. The term *computation* derives from the Latin *computatio*, formed by the prefix *com* (“together” or “with”) and *putatio*, related to *putare*, meaning “to calculate” or “to reckon.” In its original sense, the word conveys the idea of **calculating together**. In this light, computation can be understood not as an autonomous source of knowledge, but as a process through which machines assist human reasoning by carrying out calculations at scales and speeds beyond our natural capacity. Algorithms, programming languages, and computational environments therefore act as instruments that enable probabilistic models to be simulated, statistical procedures to be implemented, and empirical hypotheses to be tested. Computation does not define Data Science; it provides the **operational medium** through which probabilistic and statistical reasoning becomes executable in practice.

9.5 Data Science as a Strategic Value Engine

At this point, it is crucial to emphasize a key principle that often gets overlooked: **Data Science is not a goal in itself**. No matter how sophisticated the methodologies or how advanced the machine learning models, they are worthless if they do not generate real and measurable impact. The true purpose of Data Science is to inform and improve strategic decisions—transforming data into outcomes that advance organizational goals.

Data-driven decision making empowers organizations to optimize operations, personalize customer experiences, identify risks proactively, and maintain a competitive advantage. As Kozyrkov highlights, outstanding data analysts ensure that their work is focused on solving the **right business problems**, not simply producing technically impressive outputs [47].

This underscores a critical truth: technical excellence without business alignment leads to solutions that may appear successful internally but **fail to influence decisions or create value**. The impact of a model is not measured solely by statistical accuracy or computational ingenuity, but by the extent to which it **changes what the organization does**.

From a decision-theoretic perspective, integrating Data Science into strategy enables measurable improvements not only in profitability or utility, but also in **decision velocity**. Automated analytical pipelines allow organizations to deploy models directly into operational environments continuously monitoring performance, adapting to new data, and discarding approaches that no longer deliver value. This creates a self-correcting loop where models are evaluated and refined over time, accelerating the pathway from insight to action.

Consider, for example, the case of **customer churn** in retail. Without Data Science, retention strategies typically rely on intuition or broad marketing campaigns, often costly and ineffective. In contrast, a churn prediction model can **identify customers most at risk**, enabling targeted interventions such as personalized offers or loyalty programs. If the strategy does not reduce churn among the targeted segment, performance monitoring mechanisms reveal the failure, prompting refinement or alternative plans. Here, data does not merely describe behavior—it actively shapes business outcomes.

Viewed through this lens, Data Science becomes a **strategic value engine**: a discipline that leverages scientific reasoning, computational tools, and domain expertise to drive decisions that matter. This synthesis—model-driven insight aligned with real-world objectives—ensures that Data Science fulfills its true mission: transforming information into impact.

9.6 Why the Scientific Method Still Matters

The strategic impact of Data Science ultimately stems from something deeper than predictive models or analytical pipelines. Its real strength lies in the intellectual framework that guides how data is interpreted, hypotheses are formulated, and decisions are evaluated. The ability of Data Science to generate meaningful impact in organizations is therefore inseparable from the scientific principles that structure its practice.

In an era defined by rapid technological change and the constant emergence of new tools, it is tempting to define Data Science by its most visible artifacts: programming languages, machine learning libraries, cloud platforms, and ever-evolving frameworks. Yet these elements, while powerful, are ultimately transient. Technologies evolve, tools are replaced, and methodologies shift as the field continues to mature.

What endures is something far more fundamental: the scientific mindset that underlies the discipline. At its core, Data Science is not about algorithms but about **reasoning under uncertainty**, formulating hypotheses, confronting them with data, and refining our understanding of complex phenomena through systematic evidence. Probability provides the language for uncertainty, statistics supplies the methods for inference, and computation serves as the medium through which these ideas become operational.

When practiced in this way, Data Science becomes more than a collection of analytical techniques—it becomes a modern expression of the scientific method applied to data-rich environments. Its true power lies not in automation or predictive accuracy alone, but in its capacity to transform data into knowledge and knowledge into better decisions. In the end, the enduring value of Data Science does not come from the sophistication of its tools, but from the rigor of the thinking that guides their use.

The original version is available on Medium here.

Chapter 10

Conformal Geometry and Electrostatics in Three Dimensions

Header.

10.1 Problem Definition

Section.

10.2 Modeling Approach

Section.

10.3 Methodology and Implementation

Section.

10.4 Results, Discussion, and Limitations

Section.

10.5 Technical Appendix (Optional)

Section.

10.6 Laplace's equation

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0 \tag{10.1}$$

10.7 Conformal Mappings

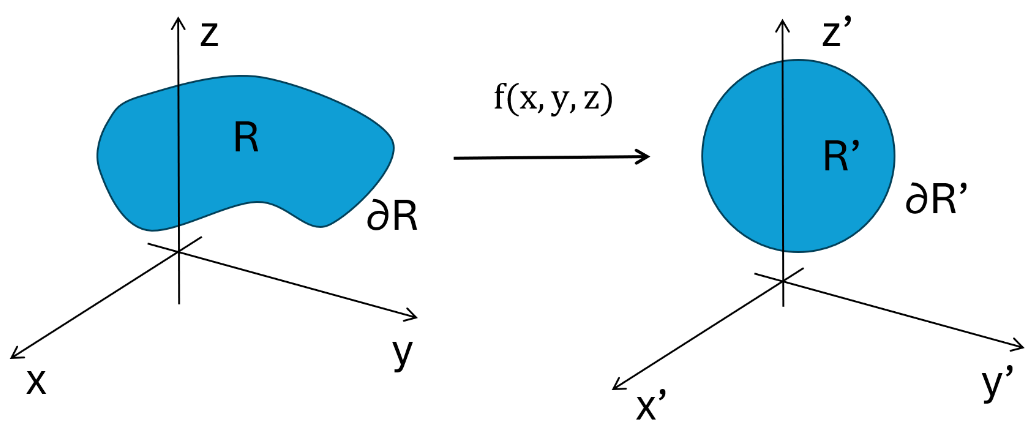

Figure 10.1: Schematic representation of a conformal transformation $f(x, y, z)$ mapping a region R with boundary ∂R in the (x, y, z) coordinate system into a transformed region R' with boundary $\partial R'$ in the (x', y', z') coordinate system.

As shown in Figure 10.1, ...

Suppose we have a region R in xyz space in which a scalar field ϕ satisfies Laplace's equation Equation 10.1. We want to determine what conditions a transformation $f : R \to R'$ must satisfy such that Laplace's equation remains invariant under this transformation. For simplicity, let's consider the simplest case in $\mathbb{R}^2$.

Applying the chain rule we have that:

$$
\begin{aligned}
\frac{\partial \phi}{\partial x} &= \frac{\partial \phi}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial \phi}{\partial y'}\frac{\partial y'}{\partial x} \\
\frac{\partial \phi}{\partial y} &= \frac{\partial \phi}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial \phi}{\partial y'}\frac{\partial y'}{\partial y}
\end{aligned}
\tag{10.2}
$$

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